



CAPSTONE PROJECT REPORT

Report 3 – Software Requirement Specification

– Can Tho, July 2026 –

Table of Contents

I. Project Report	3
1. Status Report	3
2. Team Involvements.....	3
3. Issues/Suggestions	3
II. Software Requirement Specification	4
1. Product Overview	4
2. User Requirements	5
2.1 Actors	5
2.2 Use Cases	6
3. Functional Requirements.....	21
3.1 System Functional Overview.....	21
3.2 Authentication	67
3.4 Clinic Management	78
3.5 Schedule Management	81
3.6 Patient Management	84
3.7 Appointment Management	88
3.8 Service Catalog Management	98
3.9 Clinical Examination	100
3.10 Dental Imaging	106
3.11 Performance Management.....	108
4. Non-Functional Requirements.....	111
4.1 External Interfaces	111
4.2 Quality Attributes.....	112
5. Requirements Appendix.....	115
5.1 Business Rules	115
5.1 Appendix1 - Messages List.....	116

I. Project Report

1. Status Report

#	Work Item	Status	Notes (Work Item in Details)
1		Pending	
2		In Progress	
3		Completed	

2. Team Involvements

#	Task	Member	Notes (Task Details, etc.)
1		KienNT	
2		TuanTV	
3		AnhLM	

3. Issues/Suggestions

#	Issue	Status	Notes (Solution, Suggestion, etc.)
1		Pending	
2		In Progress	
3		Completed	

II. Software Requirement Specification

1. Product Overview

S.M.I.L.E (Smart Dental Management and Diagnosis System) is a comprehensive dental healthcare management platform implementing medical record integrity with AI-powered diagnostic capabilities.

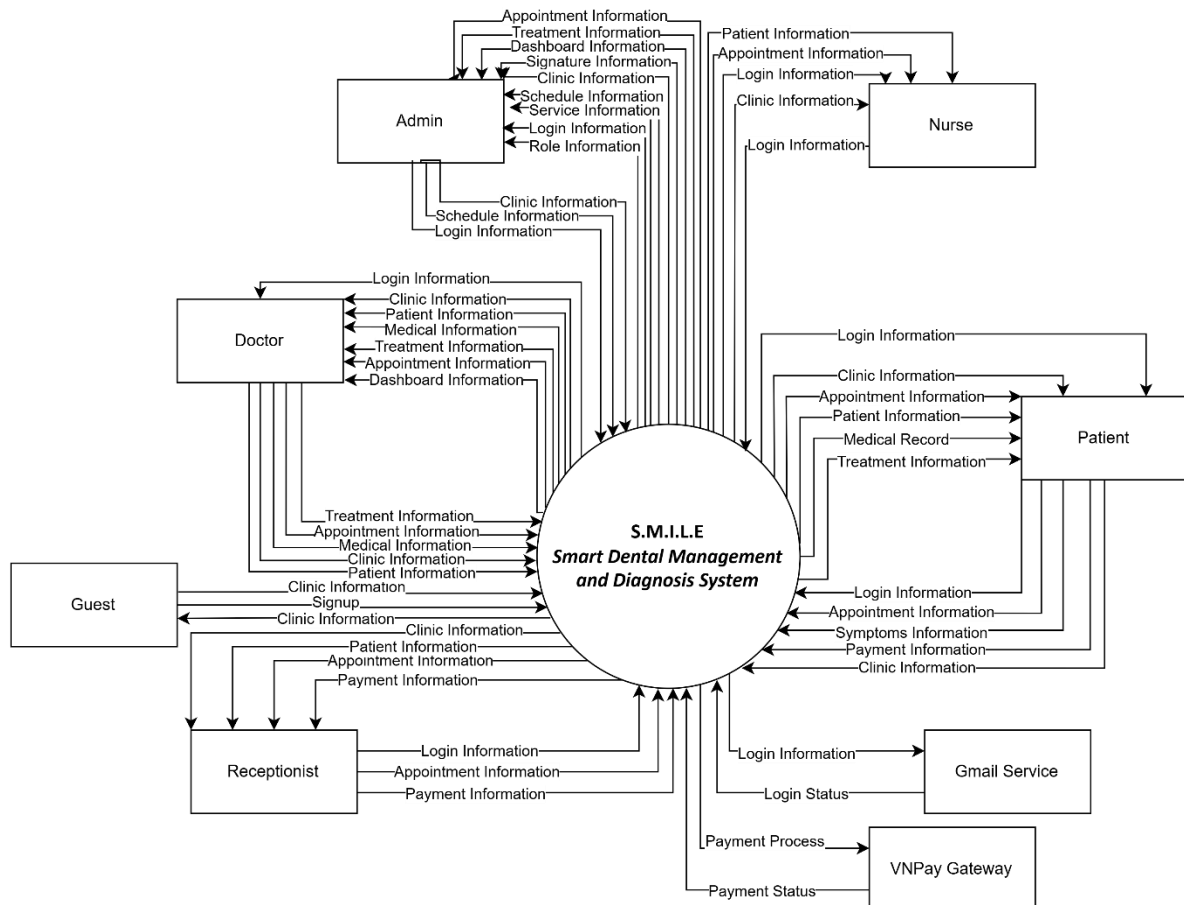


Figure 1. System diagram

2. User Requirements

2.1 Actors

#	Actor	Description
1	Admin	System owner with full access to all modules and permissions. Admin manages users, roles, clinics, rooms, schedules, and system configuration, and can lock/unlock accounts and review audit/security activities.
2	Doctor	Clinical care provider responsible for diagnosis, treatment, and medical documentation. Doctor manages assigned appointments, creates and updates medical records, treatment plans, prescriptions, and clinical notes, and can access patient data needed for treatment.
2	Nurse	Coordinates treatment session logistics and room status; views shift schedules and basic patient info, no access to clinical records or financial/administrative actions.
4	Receptionist	Front-desk coordinator focused on patient flow and operations. Receptionist handles patient registration, appointment booking/updates, check-in confirmation, basic profile updates, and payment support, with limited access to clinical data (mostly read-only).
5	Patient	End user receiving care with self-service access to personal information. Patient can view and update own profile, book/manage own appointments, view own medical records and treatment plans, and provide digital consent/signatures.
6	Guest	Unauthenticated visitor with public access only. Guest can browse clinic information and starts onboarding actions such as account registration before becoming a registered patient.

2.2 Use Cases

a. Use Case Diagram

1. Guest

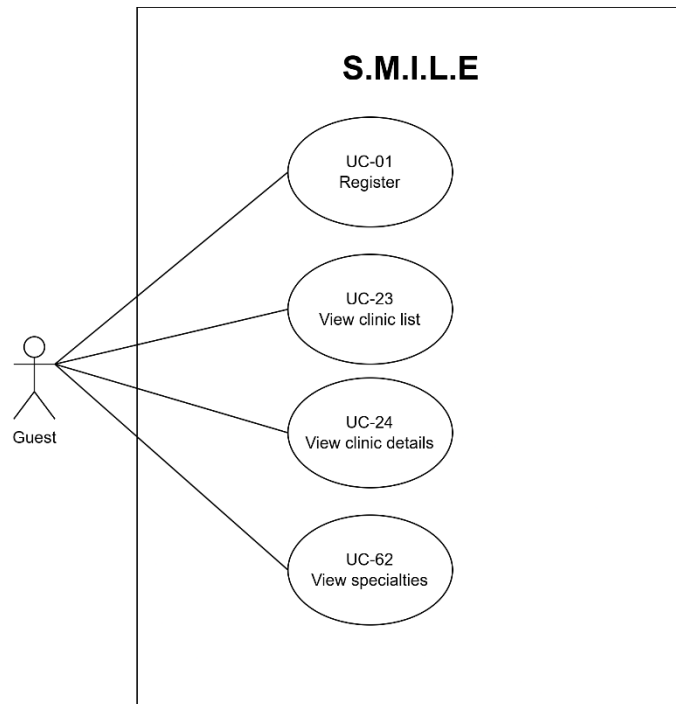


Figure 2. Use case diagram of Guest

2. Patient

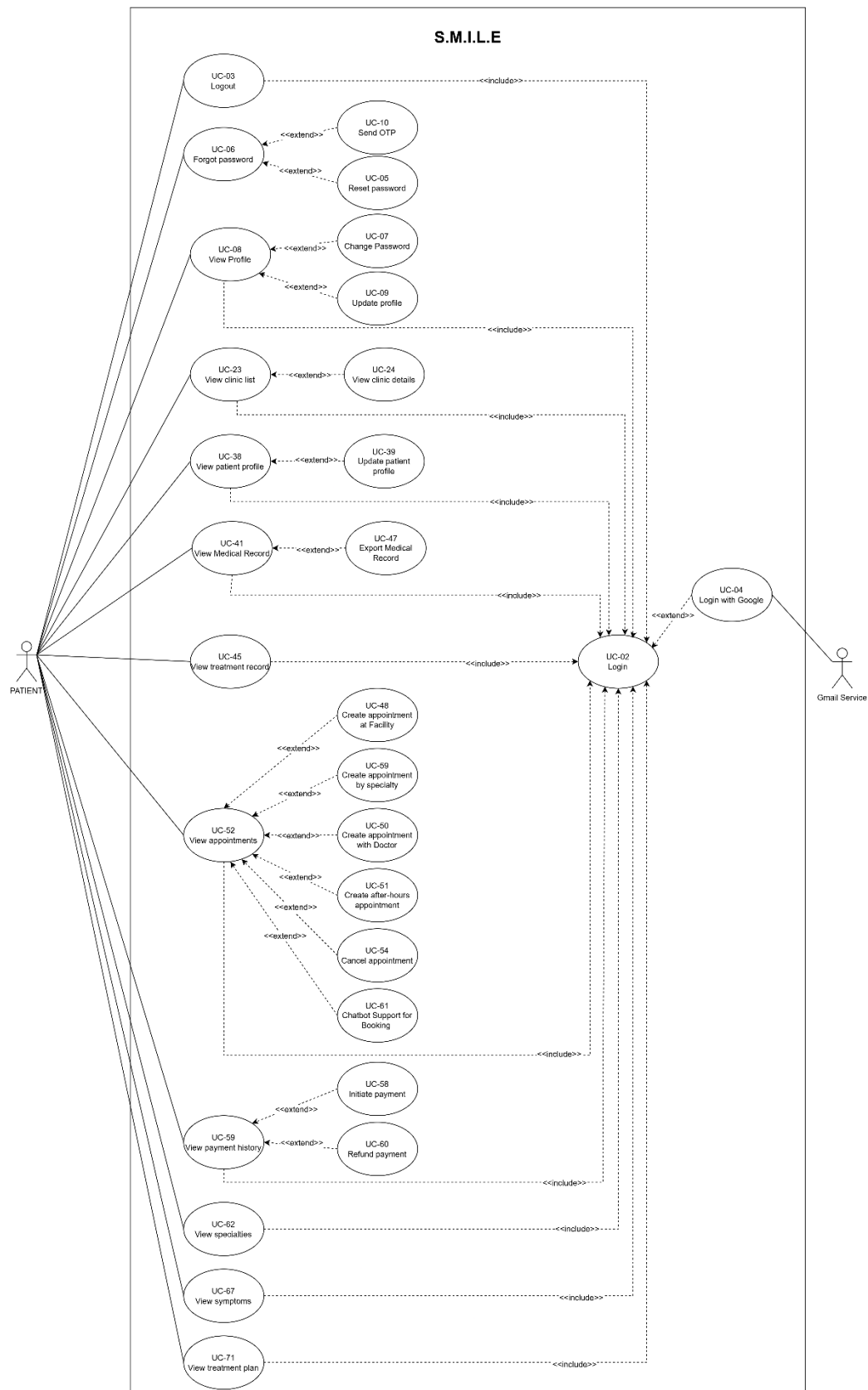


Figure 3. Use case diagram of Patient

3. Nurse

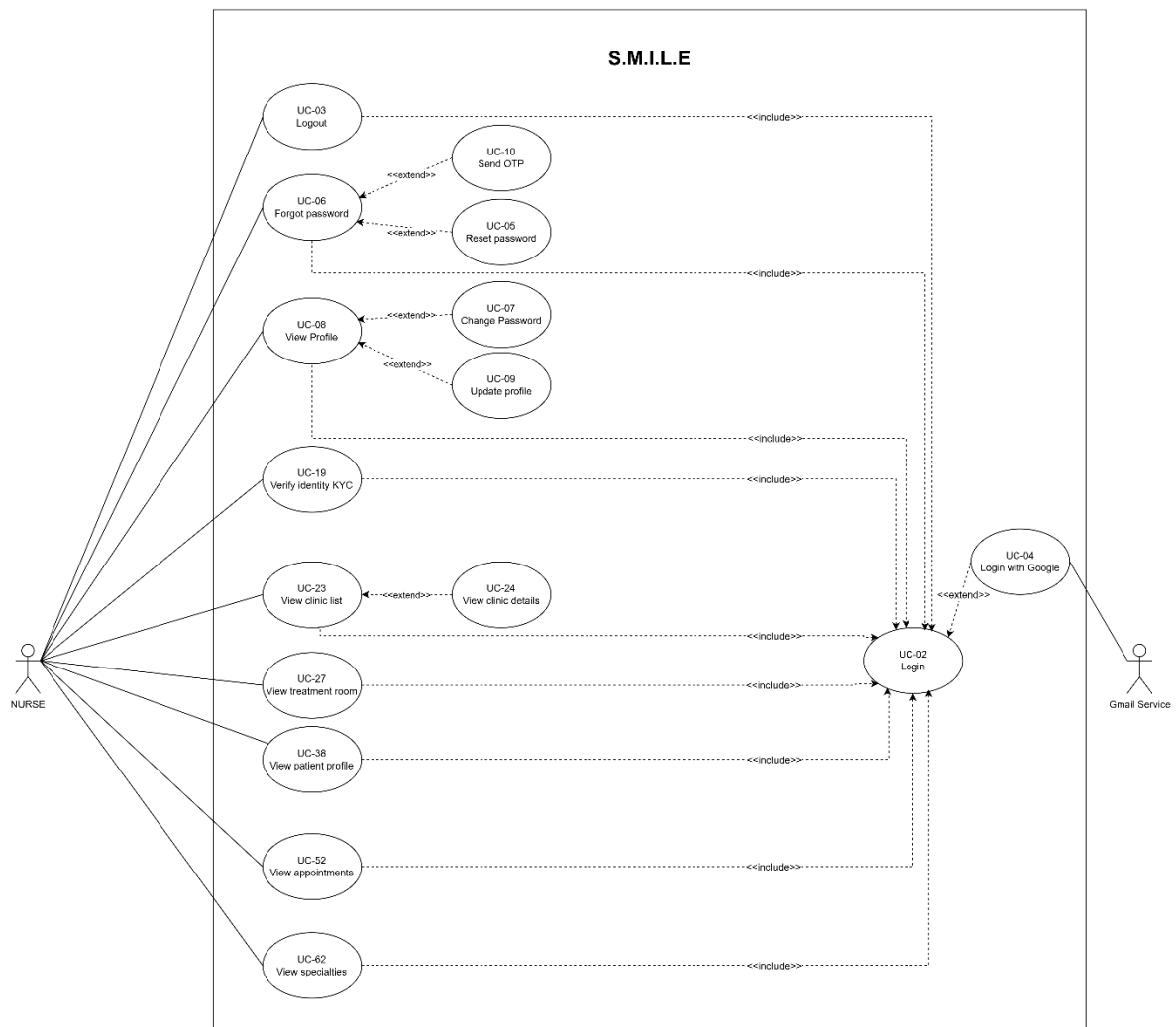


Figure 4. Use case diagram of Nurse

4. Doctor

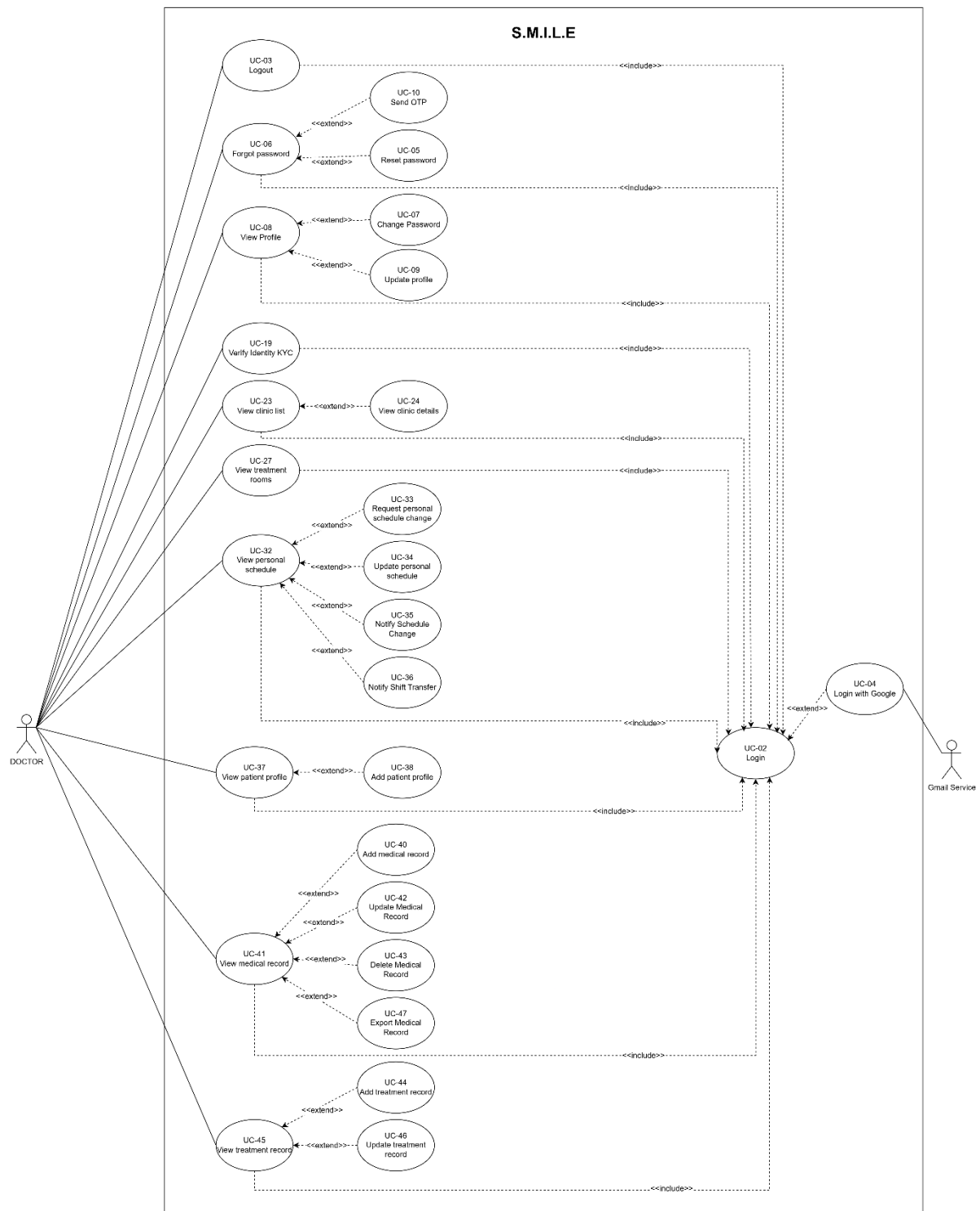


Figure 5. Use case diagram of Doctor (1)

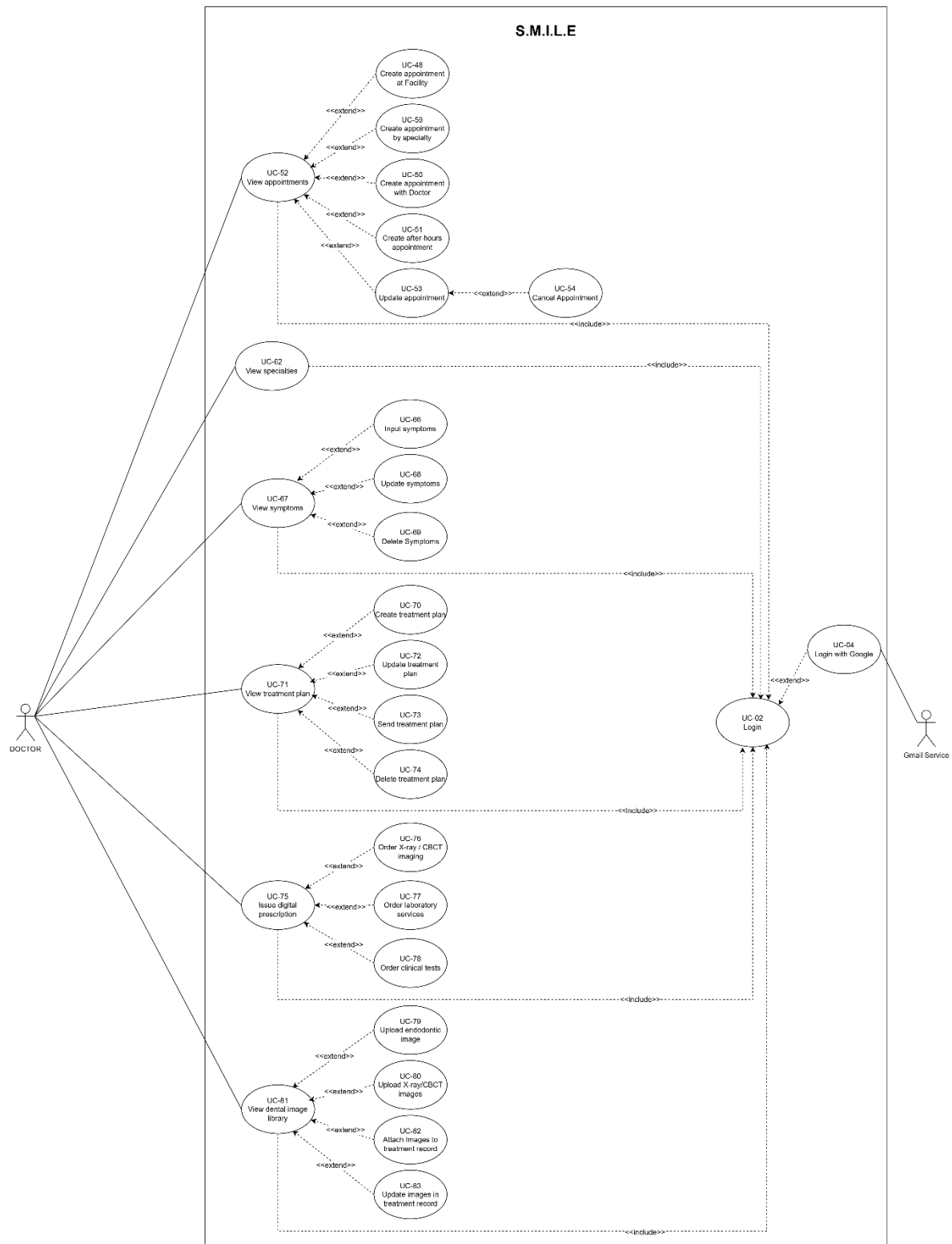


Figure 6. Use case diagram of Doctor (2)

5. Admin

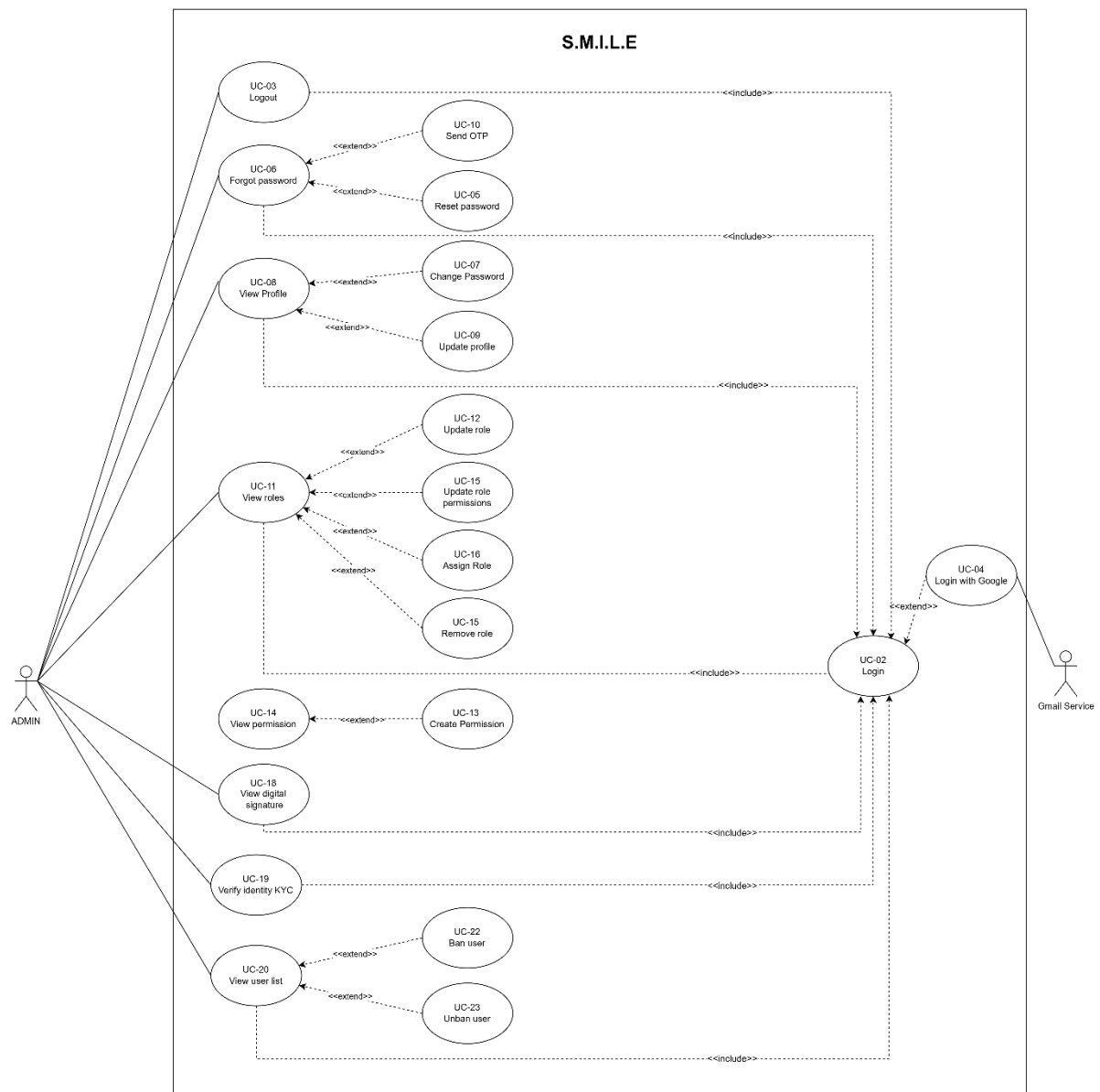


Figure 7. Use case diagram of Admin (1)

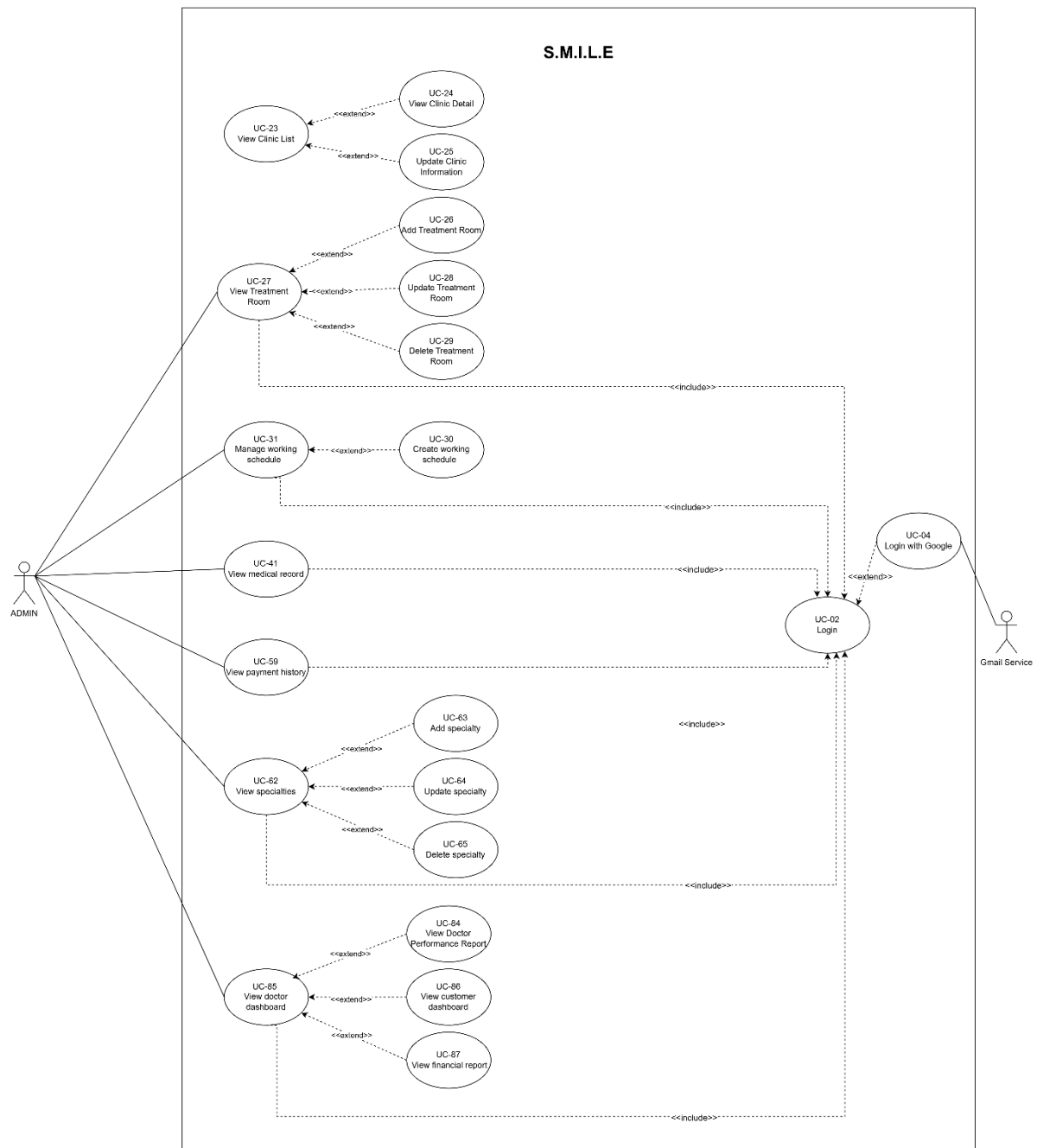


Figure 8. Use case diagram of Admin (2)

6. Receptionist

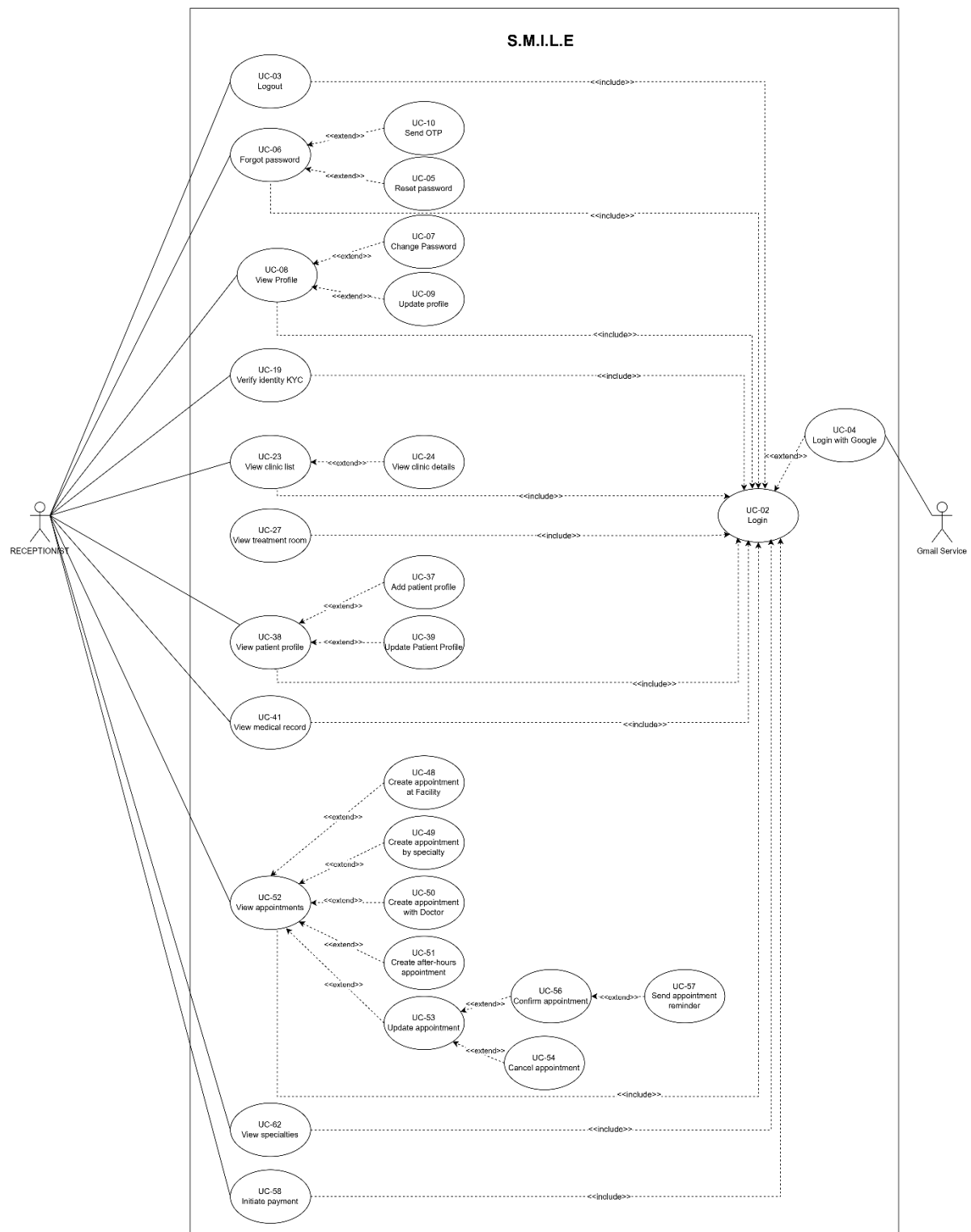


Figure 9. Use case diagram of Receptionist

c. Use Cases Description

ID	Use Case	Primary Actors	Secondary Actors	Use Case Description
1	Register	Guest		Guest submits email, password and profile details to create a patient account; the system validates that the email, phone and username are unique, creates the account and returns access and refresh tokens.
2	Login	Admin, Nurse, Doctor Receptionist, Patient		User submits email or phone plus password; the system verifies the credentials and issues a JWT access token and a refresh token.
3	Logout	Admin, Nurse, Doctor Receptionist, Patient		User ends the session; the system adds the token's jti to the Redis blacklist so the token is rejected for the remainder of its lifetime.
4	Login with Google	Admin, Nurse, Doctor Receptionist, Patient	Gmail Service	User authenticates through Google OAuth 2.0; the system verifies the Google token, links or creates the local account and issues its own tokens.
5	Reset password	Admin, Nurse, Doctor, Receptionist, Patient		User supplies a valid reset token and a new password; the system verifies the token and replaces the stored password hash.
6	Forget password	Admin, Nurse, Doctor, Receptionist, Patient		User requests a reset for a registered email; the system generates a time-limited reset token.
7	Change password	Admin, Nurse, Doctor, Receptionist, Patient		Signed-in user supplies the current and new password; the system verifies the old password before storing the new hash, rejecting the request if it does not match.
8	View profile	Admin, Nurse, Doctor, Receptionist, Patient		User opens their own profile; the system returns the account and profile record for the authenticated user.
9	Update profile	Admin, Nurse, Doctor, Receptionist, Patient		User edits name, date of birth, gender, avatar or contact details; the system validates and saves the changes.
10	Send OTP	Admin, Doctor, Receptionist, Patient		User requests a one-time code for phone verification; the system generates the OTP, caches it in Redis with an expiry and dispatches it.
11	View roles	Admin		Admin lists the defined roles; the system returns each role with its description.

12	Update role	Admin		Admin edits a role's name or description; the system validates uniqueness, saves the change and writes an audit-log entry.
13	Create permission	Admin		Admin defines a new permission from a resource and an action; the system validates that the permission name is unique and stores it.
14	View permissions	Admin		Admin lists all permissions, optionally filtered by role; the system returns the matching permission records.
15	Update role permission	Admin		Admin attaches or detaches permissions on a role; the system updates the role-permission mapping and writes an audit-log entry.
16	Assign Role	Admin		Admin grants a role to a user; the system creates the user-role link and writes an audit-log entry.
17	Remove role	Admin		Admin revokes a role from a user; the system deletes the user-role link and writes an audit-log entry.
18	Access log	Admin		Admin reviews recorded security and administrative activity; the system returns the audit-log entries with actor, action, target and timestamp.
19	Verify Identity KYC	Admin, Nurse, Doctor, Receptionist,		User submits identity documents and verifies a phone OTP; the system stores the files AES-encrypted, optionally extracts fields via the OCR service, and moves the verification to its next status.
20	View user list	Admin		Admin browses registered users with filters and pagination; the system returns the matching user profiles.
21	Ban user	Admin		Admin locks an account that violates policy; the system sets the account status, force-logs-out active sessions and writes an audit-log entry.
22	Unban user	Admin		Admin restores a locked account; the system reactivates it and writes an audit-log entry.
23	View clinics list	Admin, Nurse, Doctor, Receptionist, Patient, Guest		Actor browses the clinics in the network; the system returns the clinic list, publicly readable without authentication.

24	View clinic details	Admin, Nurse, Doctor, Receptionist, Patient, Guest		Actor opens one clinic; the system returns its address, contact details, opening hours and offered services.
25	Update clinic information	Admin		Admin edits a clinic's details; the system validates and saves the record.
26	Add treatment room	Admin		Admin registers a new room for a clinic; the system validates that the room code is unique within the clinic and stores it.
27	View treatment rooms	Admin, Doctor, Nurse, Receptionist		Actor lists the rooms of a clinic; the system returns each room with its type and status.
28	Update treatment room	Admin		Admin edits a room's name, type or availability; the system saves the change.
29	Delete treatment room	Admin		Admin removes a room no longer in use; the system deletes it unless it is still referenced by a schedule or appointment.
30	Create working schedule	Admin		Admin creates a doctor's shift for a clinic, date and room; the system rejects a duplicate of an existing doctor/date/shift combination.
31	Manage working schedule	Admin		Admin edits shift, room, capacity, status or notes on an existing schedule; the system checks the schedule is still mutable, requires a changed_by actor, applies the change and records old and new values in the schedule-change log.
32	View personal schedule	Doctor		Doctor views their own assigned shifts; the system returns the schedules for that doctor over the selected period.
33	Request personal schedule change	Doctor		Doctor submits a leave or change request with a date range and reason; the system validates the range, rejects it if it overlaps a pending or approved request, and stores it with status PENDING for approval.
34	Update personal schedule	Doctor		Doctor amends their own schedule entry; the system applies the change and records it in the schedule-change log.
35	Notify Schedule Change	Doctor		The system informs affected staff when a schedule changes, by creating a notification through the IAM notification service.

36	Notify Shift Transfer	Doctor		Doctor hands a shift to a colleague; the system reassigns the shift and notifies the receiving doctor.
37	Add patient profile	Doctor, Receptionist		Staff registers a new patient with demographics, allergies and insurance details; the system creates the patient master record.
38	View patient profile	Doctor, Receptionist, Nurse, Patient		Actor opens a patient profile; the system returns it, restricting patients to their own record and those they represent.
39	Update patient profile	Receptionist, Patient		Actor edits patient demographic or contact details; the system validates ownership and saves the change.
40	Add medical record	Doctor		Doctor creates a visit record linked to a patient, clinic and appointment; the system forces the new record to status draft and rejects any attempt to create it already finalized.
41	View medical record	Admin, Doctor, Receptionist, Patient		Actor opens a patient's medical record; the system returns the record and its version history, limiting patients to their own.
42	Update medical record	Doctor		Doctor edits complaint, diagnosis, plan or notes; the system saves a new version snapshot with the editing actor and reason, and blocks edits once the record is finalized.
43	Delete medical record	Doctor		Doctor removes a record created in error; the system deletes it and the deletion is captured in the audit trail.
44	Add treatment record	Doctor		Doctor records a treatment performed on a tooth or region; the system links it to the patient and the medical record.
45	View treatment record	Doctor, Patient		Actor reviews treatment history by patient, record or tooth number; the system returns the matching entries.
46	Update treatment record	Doctor		Doctor corrects or completes a treatment entry; the system validates and saves it.
47	Export medical record	Doctor, Patient		Actor requests an export of a medical record; the system creates an export job and returns the generated document when it is ready.

48	Create appointment at Facility	Receptionist, Patient		Actor books at a chosen clinic and time slot; the system checks availability and rejects overlaps for the doctor, room or patient via database exclusion constraints.
49	Create appointment at Specialty	Receptionist, Patient		Actor books by specialty rather than a named doctor; the system allocates an available doctor in that specialty and applies the same conflict checks.
50	Create appointment with Doctor	Receptionist, Patient		Actor books a specific doctor; the system verifies that doctor's schedule and availability before creating the appointment.
51	Create after-hours appointment	Receptionist, Patient		Actor requests a slot outside published working hours; the system flags it as an exception and records it for approval.
52	View appointments	Doctor, Receptionist, Nurse, Patient		Actor lists appointments filtered by date, doctor, patient or status; the system returns the matching appointments and their current state.
53	Update appointment	Doctor, Receptionist		Staff reschedules or amends an appointment; the system re-checks slot availability and moves the appointment through its status state machine.
54	Cancel appointment	Doctor, Receptionist, Patient		Actor cancels an appointment with a reason; the system records who cancelled it and when, releases the slot and sets the status to cancelled.
55	Confirm appointment	Receptionist		Receptionist confirms a booked appointment; the system transitions its status to confirmed.
56	Send appointment confirmation	Receptionist		Receptionist sends the confirmation notice for an appointment; the system dispatches the notification and logs the delivery attempt.
57	Send appointment reminder	Receptionist		Receptionist sends a reminder ahead of an appointment; the system dispatches it, honours the patient's reminder preference and logs the outcome, with retry available.
58	Initiate payment	Receptionist, Patient		Actor starts payment for an appointment; the system creates a pending payment and returns a signed VNPay checkout URL.
59	View payment history	Admin, Patient		Actor reviews past payments, optionally filtered by status; the

				system returns the payment records with amount, status, provider and refund state.
60	Refund payment	Patient		Patient requests a refund on a paid transaction; the system opens a refund request that an admin then approves or rejects, moving it through the refund state machine.
61	Chatbot Support for Booking	Patient		Patient describes a booking need in natural language; the conversational agent extracts intent and details and calls the appointment endpoints on the patient's behalf.
62	View specialties	Admin, Doctor, Nurse, Receptionist, Patient, Guest		Actor browses the dental specialties offered; the system returns the specialty catalogue.
63	Add specialty	Admin		Admin adds a specialty to the catalogue; the system validates that the name is unique and stores it.
64	Update specialty	Admin		Admin edits a specialty's name or description; the system saves the change.
65	Delete specialty	Admin		Admin removes a specialty; the system deletes it unless doctors or services still reference it.
66	Input symptoms	Doctor		Doctor records the symptoms reported during an examination session; the system stores them against the session and patient.
67	View symptoms	Doctor, Patient		Actor reviews recorded symptoms by session or patient; the system returns the matching entries.
68	Update symptoms	Doctor		Doctor corrects or expands a symptom entry; the system saves the change.
69	Delete symptoms	Doctor		Doctor removes a symptom entered in error; the system deletes the entry.
70	Create treatment plan	Doctor		Doctor drafts a treatment plan for a patient, linked to an examination session or medical record; the system stores it in draft state.
71	View treatment plan	Doctor, Patient		Actor opens a treatment plan; the system returns its steps, estimated cost and current status.
72	Update treatment plan	Doctor		Doctor revises the plan's steps or details; the system saves the change while the plan is still editable.

73	Send treatment plan	Doctor		Doctor proposes the finished plan to the patient; the system moves the plan to proposed state so the patient can accept or decline it.
74	Delete treatment plan	Doctor		Doctor discards a plan that is no longer needed; the system deletes it.
75	Issue digital prescription	Doctor		Doctor issues a prescription from a draft; the system validates the dosing of every item, captures a consent snapshot for paediatric cases and transitions the prescription to issued.
76	Order X-ray / CBCT imaging	Doctor		Doctor raises a diagnostic imaging order for a patient; the system creates the order with a tracking code and pending status.
77	Order laboratory services	Doctor		Doctor orders laboratory work; the system creates the clinical order and later accepts the returned results, flagging abnormal values.
78	Order clinical tests	Doctor		Doctor orders an in-clinic clinical test; the system records the order against the examination session and tracks its status.
79	Upload endodontic images	Doctor		Doctor uploads endodontic images for a patient; the system stores the image metadata with category and tooth numbers.
80	Upload X-ray / CBCT images	Doctor		Doctor uploads radiographic images; the system stores the metadata and links it to the patient and record.
81	View dental image library	Doctor		Doctor browses stored images filtered by patient, record, category or uploader; the system returns the matching image entries.
82	Attach images to treatment record	Doctor		Doctor links one or more images to a treatment record; the system creates the association so the images appear with that record.
83	Update images in treatment record	Doctor		Doctor edits image metadata or annotations, or archives an image; the system saves the change.
84	View doctor performance report	Admin		Admin reviews per-doctor activity over a period; the system aggregates appointment and treatment counts into the report.
85	View doctor dashboard	Admin		Admin opens the doctor-oriented dashboard; the system returns the

				aggregated workload and schedule metrics.
86	View customer dashboard	Admin		Admin opens the patient-oriented dashboard; the system returns patient volume and visit metrics.
87	View financial report	Admin		Admin reviews revenue over a period; the system aggregates paid amounts and subtracts refunds to produce gross and net revenue.

3. Functional Requirements

3.1 System Functional Overview

a. Screen Flow

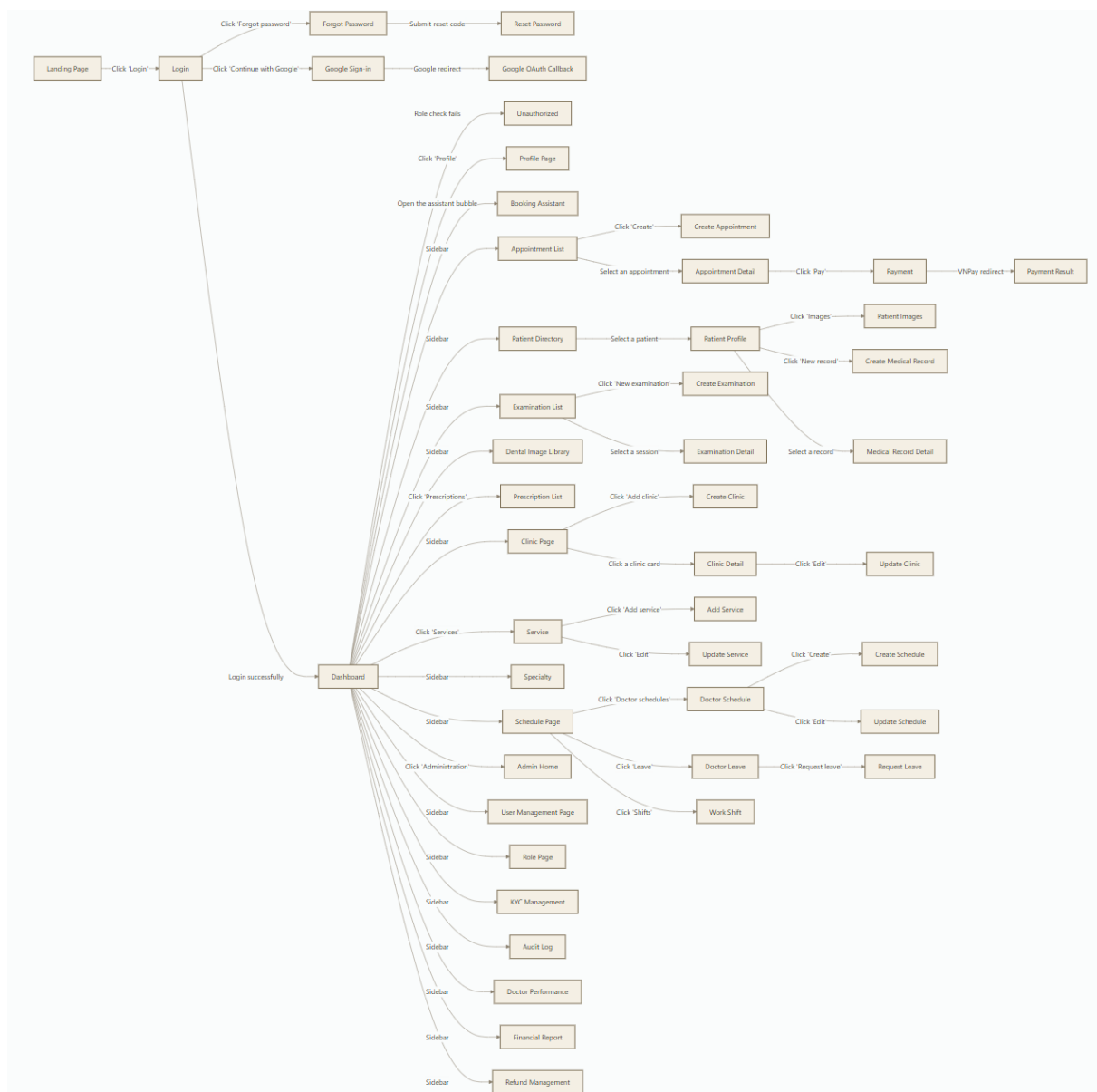


Figure 10. Screen flow of Admin

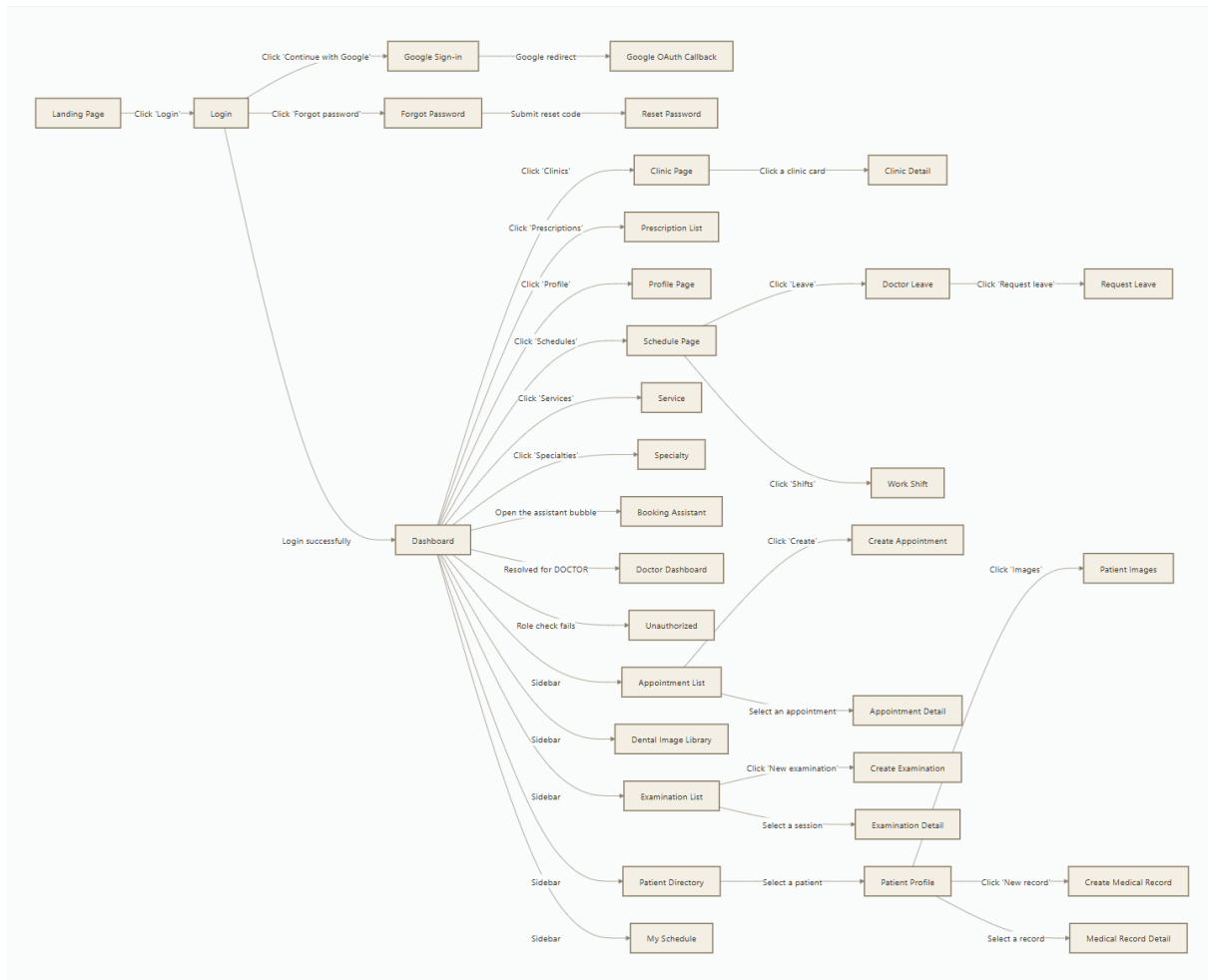


Figure 11. Screen Flow of Doctor

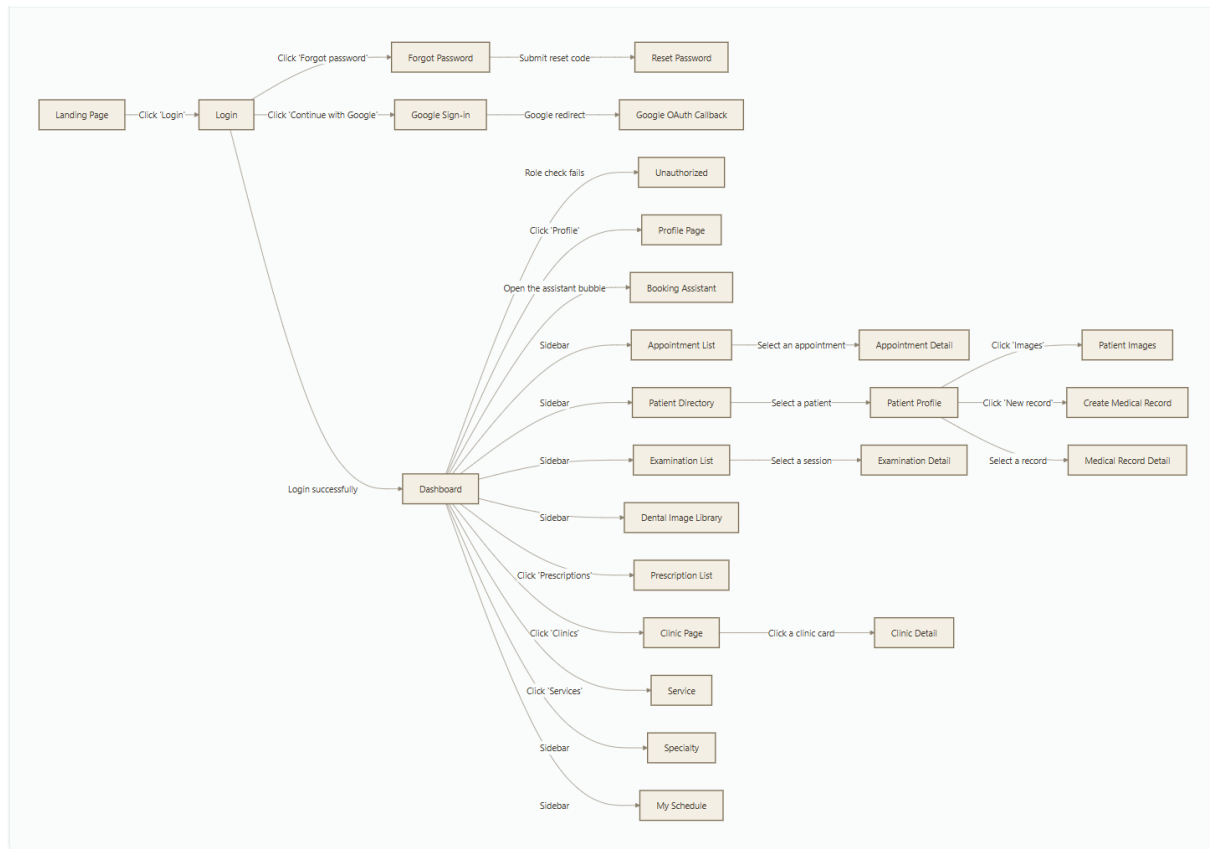


Figure 12. Screen Flow of Nurse

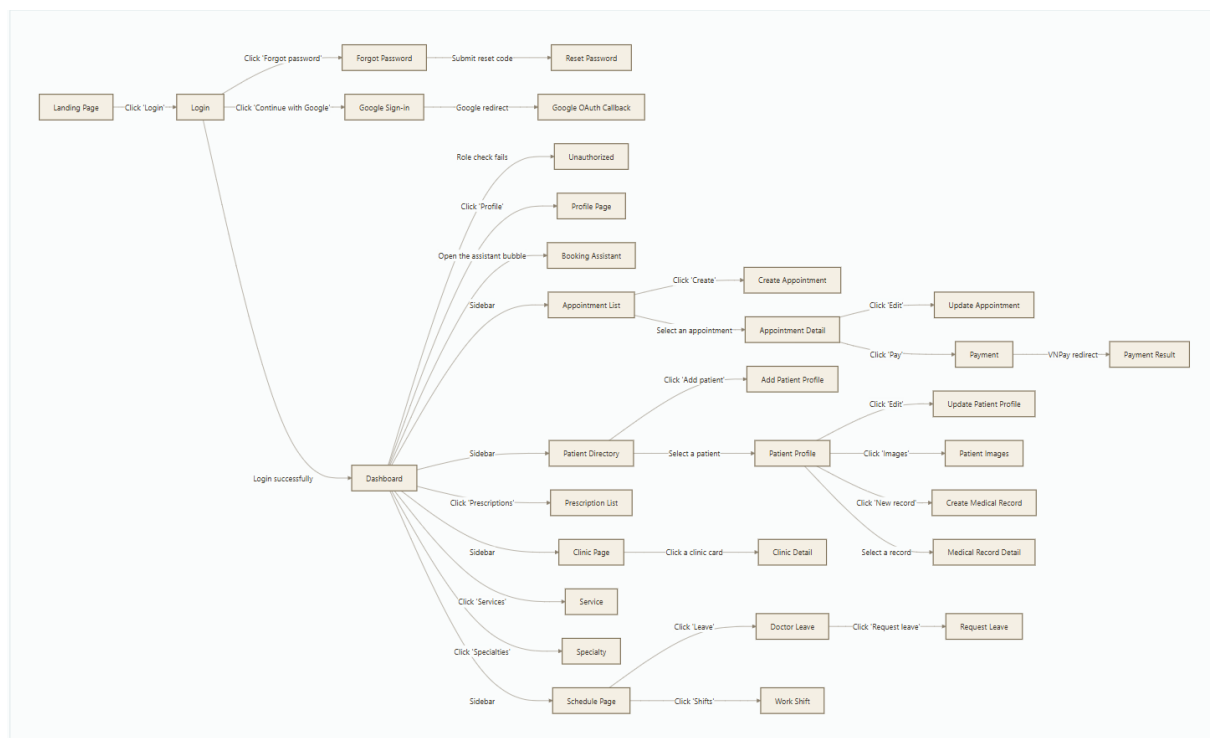


Figure 13. Screen Flow of Receptionist

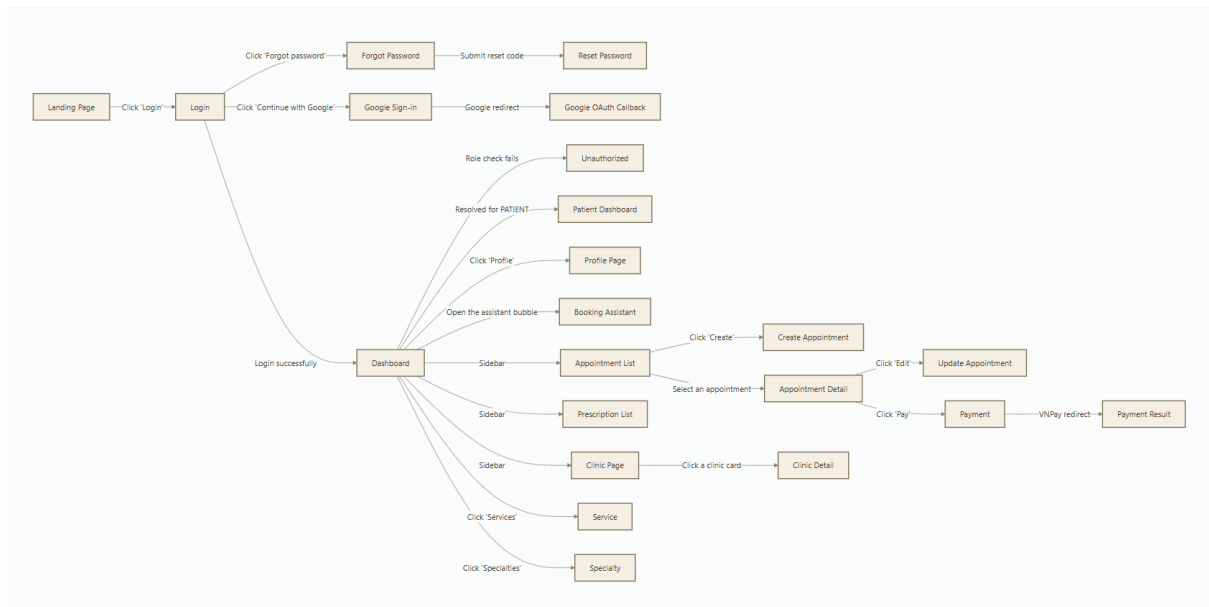


Figure 14. Screen Flow of Patient

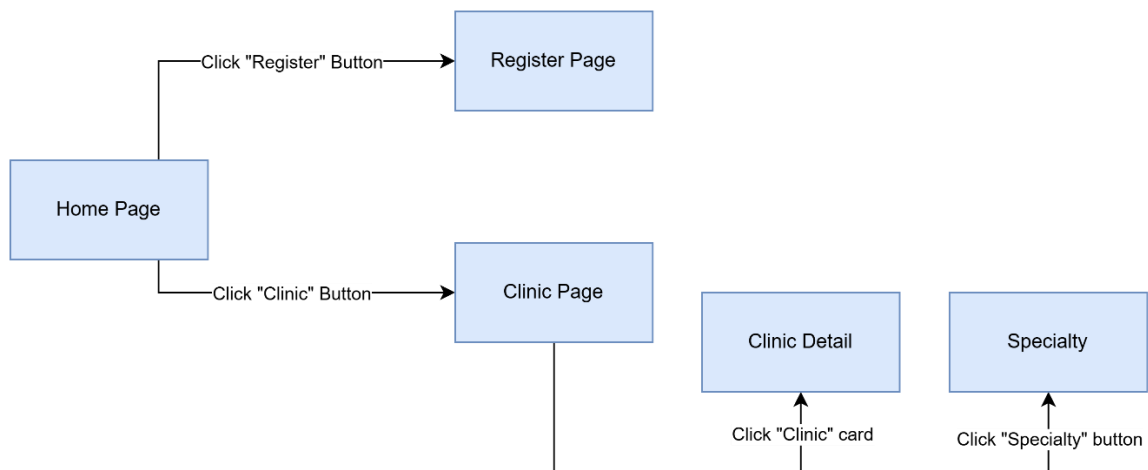


Figure 15. Screen Flow of Guest

b. Screen Details

#	Feature	Screen	Description
1	Authentication Management	Landing Page	Public home page introducing the clinic network and entry points to sign in or register.
2	Authentication Management	Login	Authenticate by email/phone and password and grant role-based access to the system.
3	Authentication Management	Register	Create a patient account with email/phone, password and OTP verification.
4	Authentication Management	Forgot Password	Request a password-reset code for the account email or phone.
5	Authentication Management	Reset Password	Verify the reset code and set a new password.

6	Authentication Management	Google Sign-in	Complete Google OAuth 2.0 sign-in and link the external identity to an account.
7	Authentication Management	Google OAuth Callback	Consume the authorization code returned by Google and establish the session.
8	Authentication Management	Unauthorized	Inform the signed-in user that their role may not access the requested screen.
9	Performance Management	Dashboard	Role-resolved landing workspace showing the operational summary for the signed-in user.
10	Performance Management	Doctor Dashboard	Clinical workspace: today's appointments, assigned patients and open examinations.
11	Performance Management	Patient Dashboard	Patient workspace: upcoming appointments, prescriptions and payment status.
12	User Management	Profile Page	View and update personal profile information, avatar and password.
13	Appointment Management	Booking Assistant	AI chat assistant that books and looks up appointments; available as a floating bubble across the app.
14	Appointment Management	Appointment List	Browse and filter appointments between patients and clinics.
15	Appointment Management	Create Appointment	Create an appointment by clinic, specialty or doctor and reserve the slot.
16	Appointment Management	Appointment Detail	Show one appointment with its status history, payment state and clinical links.
17	Appointment Management	Update Appointment	Modify appointment details, or request/confirm cancellation, and notify related parties.
18	Appointment Management	Payment	Process the appointment payment through the VNPay integration.
19	Appointment Management	Payment Result	Reconcile the VNPay callback and display the settled payment outcome.
20	Patient Management	Patient Directory	Browse and search the patient directory.
21	Patient Management	Add Patient Profile	Register a new patient profile with demographics and contact details.
22	Patient Management	Patient Profile	Show one patient with medical history, records, treatment history and representatives.
23	Patient Management	Update Patient Profile	Edit demographic and contact information of an existing patient.
24	Patient Management	Patient Images	Browse and attach dental images belonging to one patient.
25	Patient Management	Create Medical Record	Create a visit-level medical record linked to patient, clinic, doctor and appointment.
26	Patient Management	Medical Record Detail	Open one medical record with its versions, amendments, attachments and export action.
27	Clinic Examination	Examination List	Browse clinical examination sessions by clinic, doctor and status.
28	Clinic Examination	Create Examination	Open a new examination session for a patient from an appointment.
29	Clinic Examination	Examination Detail	Record symptoms, diagnoses, dental chart, treatment plan, prescription and clinical orders; finalize the session.
30	Dental Imaging	Dental Image Library	Browse, filter, upload and annotate dental images (X-ray, CBCT, endodontic) across patients.
31	Clinic Examination	Prescription List	Browse issued prescriptions for the signed-in patient or treating doctor.
32	Clinic Management	Clinic Page	Display the clinic network with overview information and operational details.
33	Clinic Management	Create Clinic	Register a new clinic branch.

34	Clinic Management	Clinic Detail	Show clinic profile, contact, specialties, treatment rooms and offered services.
35	Clinic Management	Update Clinic	Update clinic information, treatment rooms and configuration data.
36	Service Catalog Management	Service	Manage the dental service catalog and per-clinic availability and pricing.
37	Service Catalog Management	Add Service	Add a service to the catalog with category, duration and base price.
38	Service Catalog Management	Update Service	Update service definition, duration and base price.
39	Service Catalog Management	Specialty	Manage the clinical specialty catalog and its clinic assignments.
40	Schedule Management	Schedule Page	Scheduling hub for doctor shifts, leave and work-shift definitions.
41	Schedule Management	Doctor Schedule	View and manage published doctor schedules by date, clinic and room.
42	Schedule Management	Create Schedule	Create a doctor schedule entry, unique per doctor, date and shift.
43	Schedule Management	Update Schedule	Update or delete a schedule entry while it is still mutable, recording the change.
44	Schedule Management	Doctor Leave	Review and approve doctor leave that blocks availability.
45	Schedule Management	Request Leave	Submit a leave request for a date range with a reason.
46	Schedule Management	Work Shift	Maintain the named work-shift definitions used when building schedules.
47	Schedule Management	My Schedule	Show the signed-in doctor's or nurse's own shifts and request schedule changes.
48	User Management	Admin Home	Entry point to the administration area.
49	User Management	User Management Page	Administer user accounts, including ban and unban of policy-violating accounts.
50	User Management	Role Page	Manage role definitions and the permissions assigned to each role.
51	User Management	KYC Management	Review submitted identity documents and approve or reject verification.
52	User Management	Audit Log	Search the tamper-evident log of security and clinical data events.
53	Performance Management	Doctor Performance	Track doctor productivity and performance indicators and generate reports.
54	Performance Management	Financial Report	Report clinic revenue and payment settlement over a period.
55	Performance Management	Refund Management	Review, approve or reject refund requests raised against payments.

c. Non-Screen Functions

#	Feature	System Function	Description
1	Account management	POST Auth RegisterUser(params)	Create a new account, validate unique email/phone/username, return JWT tokens.

2	Account management	POST Auth Login(params)	Authenticate by email/phone plus password and issue access and refresh tokens.
3	Account management	POST Auth RefreshToken(params)	Rotate the access token using a stored refresh token for session continuity.
4	Account management	POST Auth SendOtp(params)	Issue a one-time passcode for phone/email verification or password reset.
5	User administration	GET Users GetAll(params)	Return the paginated user list with filters for administrative operations.
6	KYC verification	POST KYC SubmitKyc(params)	Submit identity documents; files are stored AES-encrypted for the verification workflow.
7	KYC verification	PUT KYC ApproveKyc(params)	Record the reviewer decision on a pending KYC request and update verification status.
8	AI KYC OCR	POST KycOcr ExtractIdentityFields(params)	Extract CCCD fields from uploaded identity images (YOLOv11 + VietOCR) and return a KYC quality checklist.
9	Appointment management	POST Appointment CreateAtClinic(params)	Create a clinic appointment, enforce the no-double-booking exclusion constraints, and set status SCHEDULED.
10	Appointment management	PUT Appointment Cancel(params)	Cancel an appointment with a reason and audit cancelled_by/cancelled_at.
11	Availability service	GET Availability GetDoctorSlots(params)	Return the available time slots for a doctor on a given day.
12	Booking idempotency	POST Appointment IdempotentSubmit(params)	Deduplicate repeated booking submissions through stored idempotency keys.
13	AI booking orchestration	POST AIBooking CreateFromChat(params)	Convert an assistant conversation into a structured appointment booking request.
14	Payment integration	POST VNPay CreatePaymentUrl(params)	Build the signed VNPay checkout URL for an appointment payment.
15	Payment integration	GET VNPay Callback(params)	Verify the VNPay callback signature and reconcile payment status idempotently.
16	Payment integration	POST Payment Refund(params)	Raise a refund request and record its approval workflow against the payment.
17	Reminder service (background)	INTERVAL Reminder DispatchPendingReminders()	Periodically scan upcoming appointments and dispatch reminders, deduplicated per

			appointment, channel and start time.
18	Notification delivery	POST Notification Send(params)	Deliver a notification through the APP, EMAIL, SMS or PUSH gateway and log the attempt.
19	Patient profile	POST Patient CreatePatient(params)	Create the patient master profile with demographics, allergies and chronic conditions.
20	Medical record	POST MedicalRecord CreateRecord(params)	Create a visit record linked to patient, clinic, doctor and appointment.
21	Medical record	POST MedicalRecord Finalize(params)	Finalize a record, write an immutable snapshot to medical_record_versions, and emit an audit log entry.
22	Record export	POST RecordExport Create(params)	Generate an export file for a medical record and store it with an expiry.
23	Imaging service	POST Imaging UploadDentalImage(params)	Upload a dental image with metadata: category, tooth numbers and PACS identifier.
24	PACS integration worker	ASYNC PACS SyncImageJob(params)	Synchronize image records to the PACS server in the background and write pacs_sync_logs.
25	Schedule service	POST Schedule CreateDoctorSchedule(params)	Create a doctor shift schedule, unique per doctor, date and shift.
26	Audit logging	ASYNC Audit RecordEvent(params)	Append actor, action, timestamp and outcome for security and clinical data events.
27	Health monitoring	GET Health Readiness(params)	Service readiness checks (database and connectivity) for orchestration and deployment probes.
28	User administration	PATCH Users Ban(userId)	Lock a user account that violates system policy, recording who banned it and why.
29	User administration	PATCH Users Unban(userId)	Restore access to a previously banned account.
30	User administration	PUT UserRoles Assign(userId)	Assign or revoke the roles held by a user.
31	Role administration	POST/PUT/DELETE Roles Manage(params)	Create a role, update the permissions attached to it, or delete one no longer required.
32	Profile	POST Accounts ConfirmAvatar(params)	Confirm an uploaded avatar against its signature and attach it to the account.
33	Appointment management	PUT Appointment Confirm(id)	Confirm a scheduled appointment and lock the reserved slot.

34	Payment integration	PUT Payment ReviewRefund(id)	Approve or reject a pending refund request against a payment.
35	Clinic examination	POST Symptom/Diagnosis Manage(params)	Add, update or remove the symptoms and diagnoses recorded on an examination session.
36	Clinic examination	POST TreatmentPlan Manage(params)	Create, revise, delete or export a treatment plan while it is still editable.
37	Clinic examination	POST ClinicalOrder Create(params)	Raise an imaging, laboratory or clinical test order from an examination session.
38	Dental imaging	POST DentalImage Annotate(imageId)	Attach markup, measurements or notes to a stored dental image.
39	Service catalog	POST/PUT/DELETE Specialty Manage(params)	Maintain the specialty catalog and its per-clinic assignments.
40	Clinic management	POST/PUT/DELETE TreatmentRoom Manage(params)	Maintain treatment rooms, their type, equipment inventory and availability.
41	Schedule management	POST ScheduleChange Request(params)	Submit a change against a published schedule and track its approval.

e. Entity Relationship Diagram

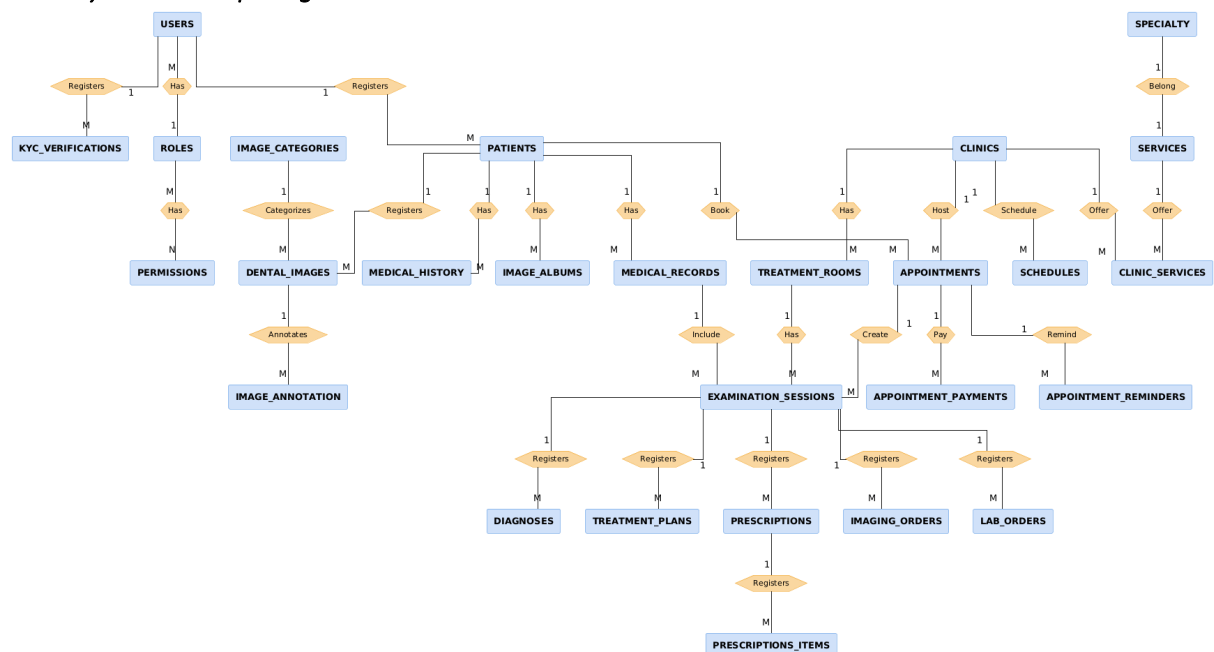


Figure 16. Entity Diagram

Account entity

Attribute	Data type	Primary Key	Foreign Key	Description
account_id	UUID	x		Identifier of account
username	VARCHAR(50)			Username
email	VARCHAR(255)			Email address
phone	VARCHAR(20)			Phone number
password_hash	VARCHAR(60)			Bcrypt output is always 60 chars; NULL for OAuth-only accounts (Google, etc.)
full_name	VARCHAR(255)			Matches users.full_name
gender	SMALLINT			ISO 5218 code: 0 unknown, 1 male, 2 female (chk_accounts_gender)
role	VARCHAR(12)			ADMIN, DOCTOR, RECEPTIONIST, PATIENT, NURSE, MANAGER (chk_accounts_role)
status	VARCHAR(11)			ACTIVE, LOCKED, SUSPENDED, DEACTIVATED
failed_login_attempts	INTEGER			Failed login attempts
locked_at	TIMESTAMP			Locked timestamp
locked_reason	TEXT			Locked reason
locked_by	UUID		x	Admin who locked
email_verified	BOOLEAN			Email verified flag
phone_verified	BOOLEAN			Phone verified flag
last_login_at	TIMESTAMP			Last login timestamp
created_at	TIMESTAMP			Record creation timestamp
updated_at	TIMESTAMP			Last update timestamp
created_by	UUID			User who created the record

updated_by	UUID			User who last updated the record
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User entity

Attribute	Data type	Primary Key	Foreign Key	Description
user_id	UUID	x		Links to auth_service_db.accounts.account_id
full_name	VARCHAR(255)			Full name
email	VARCHAR(255)			Denormalized from Auth
phone	VARCHAR(20)			Denormalized from Auth
date_of_birth	DATE			Date of birth
gender	SMALLINT			ISO 5218 code: 0 unknown, 1 male, 2 female (chk_users_gender)
avatar_url	TEXT			Profile avatar URL
created_at	TIMESTAMP			Record creation timestamp
updated_at	TIMESTAMP			Last update timestamp
created_by	UUID			User who created the record
updated_by	UUID			User who last updated the record
is_banned	BOOLEAN			Banned flag
banned_at	TIMESTAMPTZ			Banned timestamp
ban_reason	TEXT			Ban reason

RefreshToken entity

Attribute	Data type	Primary Key	Foreign Key	Description
token_id	UUID	x		Identifier of token
account_id	UUID		x	FK to accounts (account)
token_hash	CHAR(64)			Sha256 hex
expires_at	TIMESTAMP			Expiry timestamp
revoked_at	TIMESTAMP			Revoked timestamp
device_info	TEXT			Device info

ip_address	VARCHAR(45)			Client IP address
created_at	TIMESTAMP			Record creation timestamp

OtpToken entity

Attribute	Data type	Primary Key	Foreign Key	Description
otp_id	UUID	x		Identifier of otp
account_id	UUID		x	FK to accounts (account)
otp_code	CHAR(6)			6-digit numeric code
otp_type	VARCHAR(15)			Login, password_reset, identity_verify
expires_at	TIMESTAMP			Expiry timestamp
used_at	TIMESTAMP			Used timestamp
created_at	TIMESTAMP			Record creation timestamp
updated_at	TIMESTAMP			Last update timestamp
created_by	UUID			User who created the record
updated_by	UUID			User who last updated the record

PhoneVerification entity

Attribute	Data type	Primary Key	Foreign Key	Description
verification_id	UUID	x		Identifier of verification
user_id	UUID		x	FK to users (user)
phone	VARCHAR(20)			Phone number
verified_at	TIMESTAMP			Verification timestamp
is_verified	BOOLEAN			Verified flag
created_at	TIMESTAMP			Record creation timestamp

OAuthConnection entity

Attribute	Data type	Primary Key	Foreign Key	Description
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connection_id	UUID	x		Identifier of connection
account_id	UUID		x	FK to accounts (account)
provider	VARCHAR(8)			Google, facebook, apple
provider_user_id	VARCHAR(255)			Identifier of provider user
access_token	TEXT			Access token
refresh_token	TEXT			Refresh token
token_expires_at	TIMESTAMP			Token expires timestamp
is_active	BOOLEAN			Active flag
created_at	TIMESTAMP			Record creation timestamp
updated_at	TIMESTAMP			Last update timestamp
created_by	UUID			User who created the record
updated_by	UUID			User who last updated the record

Role entity

Attribute	Data type	Primary Key	Foreign Key	Description
role_id	UUID	x		Identifier of role
role_name	VARCHAR(12)			ADMIN, DOCTOR, RECEPTIONIST, PATIENT, NURSE, MANAGER (matches backend RoleEnum)
description	TEXT			Free-text description
created_at	TIMESTAMP			Record creation timestamp
updated_at	TIMESTAMP			Last update timestamp
created_by	UUID			User who created the record
updated_by	UUID			User who last updated the record

Permission entity

Attribute	Data type	Primary Key	Foreign Key	Description
permission_id	UUID	x		Identifier of permission

permission_name	VARCHAR(100)			Permission name
resource	VARCHAR(50)			Appointment, medical_record, etc.
action	VARCHAR(20)			Create, read, update, delete
description	TEXT			Free-text description
created_at	TIMESTAMP			Record creation timestamp
updated_at	TIMESTAMP			Last update timestamp
created_by	UUID			User who created the record
updated_by	UUID			User who last updated the record

UserRole entity

Attribute	Data type	Primary Key	Foreign Key	Description
id	UUID	x		ID
user_id	UUID		x	FK to users (user)
role_id	UUID		x	FK to roles (role)
assigned_at	TIMESTAMP			Assigned timestamp
assigned_by	UUID			User who assigned the record

RolePermission entity

Attribute	Data type	Primary Key	Foreign Key	Description
id	UUID	x		ID
role_id	UUID		x	FK to roles (role)
permission_id	UUID		x	FK to permissions (permission)
assigned_at	TIMESTAMP			Assigned timestamp
assigned_by	UUID			User who assigned the record

KycVerification entity

Attribute	Data type	Primary Key	Foreign Key	Description
kyc_id	UUID	x		Identifier of kyc
user_id	UUID		x	FK to users (user)
id_type	VARCHAR(50)			Passport, national_id, driver_license
id_number	VARCHAR(100)			ID number
full_name	VARCHAR(255)			As printed on the ID document
date_of_birth	DATE			Date of birth
id_front_image	TEXT			ID front image
id_back_image	TEXT			ID back image
selfie_image	TEXT			Selfie image
verification_status	VARCHAR(14)			NOT_SUBMITTED, PENDING_REVIEW, VERIFIED, REJECTED.
ocr_status	VARCHAR(10)			OCR pipeline (EnhanceKycVerifications)
ocr_confidence	INTEGER			OCR confidence
ocr_payload	JSONB			OCR payload
ocr_attempts	INTEGER			OCR attempts
ocr_last_error	TEXT			OCR last error
ocr_processed_at	TIMESTAMP			OCR processed timestamp
verified_at	TIMESTAMP			Verification timestamp
verified_by	UUID		x	Reviewer who verified
decision_source	VARCHAR(60)			Review decision (AddKycDecisionMetadata)
decision_reason	TEXT			Decision reason

rejection_reason	TEXT			Rejection reason
submitted_at	TIMESTAMP			Submitted timestamp
document_hash	CHAR(64)			Sha256 hex of the ID scan
notes	TEXT			Free-text notes
admin_notes	TEXT			Admin notes
consent_version	VARCHAR(50)			Consent & retention (HardenKycPrivacy)
consent_accepted_at	TIMESTAMP			Consent accepted timestamp
document_storage_consent_accepted_at	TIMESTAMP			Document storage consent accepted timestamp
ocr_processing_consent_accepted_at	TIMESTAMP			OCR processing consent accepted timestamp
no_marketing_consent_accepted_at	TIMESTAMP			No marketing consent accepted timestamp
processing_purpose	VARCHAR(100)			Processing purpose
retention_policy_version	VARCHAR(50)			Retention policy version
retention_expires_at	TIMESTAMP			Retention expires timestamp
deleted_at	TIMESTAMP			Soft delete for privacy erasure
created_at	TIMESTAMP			Record creation timestamp
updated_at	TIMESTAMP			Last update timestamp
created_by	UUID			User who created the record
updated_by	UUID			User who last updated the record

AuditLog entity

Attribute	Data type	Primary Key	Foreign Key	Description
log_id	UUID	x		Identifier of log
user_id	UUID		x	FK to users (user)
action	VARCHAR(100)			Action
resource	VARCHAR(100)			Resource
resource_id	UUID			Identifier of resource
ip_address	VARCHAR(45)			Client IP address
user_agent	TEXT			Client user-agent string
details	JSONB			Details
created_at	TIMESTAMP			Record creation timestamp

Notification entity

Attribute	Data type	Primary Key	Foreign Key	Description
notification_id	UUID	x		Identifier of notification
recipient_id	UUID		x	Same-DB FK -> users
template_id	UUID		x	FK to notification_templates (notification template)
notification_type	VARCHAR(100)			Appointment_reminder, otp, payment_confirmation.
channel	VARCHAR(5)			SMS, EMAIL, PUSH, APP (chk_notifications_channel)
subject	TEXT			Subject
message	TEXT			Message
related_entity_id	UUID			Schedule_id, appointment_id, payment_id
related_entity_type	VARCHAR(100)			Appointment, payment, worker_schedule.
scheduled_at	TIMESTAMP			Scheduled timestamp
sent_at	TIMESTAMP			Sent timestamp
read_at	TIMESTAMP			Read timestamp

status	VARCHAR(9)			Pending, sent, failed, read, cancelled.
retry_count	INT			Reliability logic
max_retries	INT			Max retries
next_retry_at	TIMESTAMP			Next retry timestamp
error_message	TEXT			Error detail on failure
created_at	TIMESTAMP			Record creation timestamp
updated_at	TIMESTAMP			Last update timestamp

NotificationTemplate entity

Attribute	Data type	Primary Key	Foreign Key	Description
template_id	UUID	x		Identifier of template
template_code	VARCHAR(100)			APPOINTMENT_REMINDER, OTP_VERIFY
name	VARCHAR(255)			Name
description	TEXT			Free-text description
subject_template	TEXT			Handlebars-style template for email subject
body_template	TEXT			Handlebars-style template for message body
channel	VARCHAR(5)			SMS, EMAIL, PUSH, APP (chk_notification_templates_channel)
is_active	BOOLEAN			Active flag
created_at	TIMESTAMP			Record creation timestamp
updated_at	TIMESTAMP			Last update timestamp

NotificationPreference entity

Attribute	Data type	Primary Key	Foreign Key	Description
preference_id	UUID	x		Identifier of preference
user_id	UUID		x	Same-DB FK -> users

notification_type	VARCHAR(100)			PROMO, APPOINTMENT, SYSTEM.
channel	VARCHAR(5)			SMS, EMAIL, PUSH, APP (chk_notification_preferences_channel)
is_enabled	BOOLEAN			Enabled flag
created_at	TIMESTAMP			Record creation timestamp
updated_at	TIMESTAMP			Last update timestamp

NotificationDeliveryLog entity

Attribute	Data type	Primary Key	Foreign Key	Description
log_id	UUID	x		Identifier of log
notification_id	UUID		x	FK to notifications (notification)
gateway_name	VARCHAR(50)			Twilio, SendGrid, Firebase.
gateway_response_id	VARCHAR(255)			Identifier of gateway response
status	VARCHAR(20)			Success, failed
error_payload	JSONB			Error payload
created_at	TIMESTAMP			Record creation timestamp

PushSubscription entity

Attribute	Data type	Primary Key	Foreign Key	Description
subscription_id	UUID	x		Primary key
user_id	UUID		x	FK to users (user)
endpoint	TEXT			Unique Web Push endpoint URL supplied by the browser
p256dh	VARCHAR(255)			Client public key for payload encryption
auth	VARCHAR(255)			Client auth secret for payload encryption

user_agent	VARCHAR(255)			Registering browser/device user-agent
created_at	TIMESTAMP			Record creation timestamp
updated_at	TIMESTAMP			Last update timestamp

Patient entity

Attribute	Data type	Primary Key	Foreign Key	Description
patient_id	UUID	x		Identifier of patient
user_id	UUID			References iam-service users.user_id (cross-service, no FK)
patient_code	VARCHAR(50)			Patient code
full_name	VARCHAR(255)			Full name
date_of_birth	DATE			Date of birth
gender	SMALLINT			ISO 5218 code: 0 unknown, 1 male, 2 female (chk_patients_gender)
phone	VARCHAR(20)			Phone number
email	VARCHAR(255)			Email address
address	TEXT			Street address
ward	VARCHAR(100)			Ward
district	VARCHAR(100)			District
city	VARCHAR(100)			City
emergency_contact	VARCHAR(255)			Emergency contact
emergency_phone	VARCHAR(20)			Emergency phone
allergies	TEXT[]			Allergies
chronic_diseases	TEXT[]			Chronic diseases
insurance_number	VARCHAR(100)			Insurance number
insurance_provider	VARCHAR(255)			Insurance provider
created_at	TIMESTAMP			Record creation timestamp
updated_at	TIMESTAMP			Last update timestamp

PatientRepresentative entity

Attribute	Data type	Primary Key	Foreign Key	Description
representative_id	UUID	x		Identifier of representative
patient_id	UUID		x	FK to patients (patient)
full_name	VARCHAR(255)			Full name
relationship	VARCHAR(100)			Relationship
phone	VARCHAR(20)			Phone number
email	VARCHAR(255)			Email address
legal_document_type	VARCHAR(50)			Legal document type
legal_document_number	VARCHAR(100)			Legal document number
is_primary	BOOLEAN			Primary flag
is_active	BOOLEAN			Active flag
authorized_for_treatment	BOOLEAN			Authorized for treatment flag
authorized_for_payment	BOOLEAN			Authorized for payment flag
authorized_for_records	BOOLEAN			Authorized for records flag
verified_at	TIMESTAMP			Verification timestamp
verified_by	UUID			User who verified the record
created_at	TIMESTAMP			Record creation timestamp
updated_at	TIMESTAMP			Last update timestamp

MedicalHistory entity

Attribute	Data type	Primary Key	Foreign Key	Description
history_id	UUID	x		Identifier of history
patient_id	UUID		x	FK to patients (patient)

condition_name	VARCHAR(255)			Condition name
condition_type	VARCHAR(50)			Condition type
diagnosed_date	DATE			Diagnosed date
treatment	TEXT			Treatment
notes	TEXT			Free-text notes
created_at	TIMESTAMP			Record creation timestamp
updated_at	TIMESTAMP			Last update timestamp

MedicalRecord entity

Attribute	Data type	Primary Key	Foreign Key	Description
record_id	UUID	x		Identifier of record
patient_id	UUID		x	FK to patients (patient)
appointment_id	UUID			References clinic-service appointments.appointment_id (cross-service, no FK)
clinic_id	UUID			References clinic-service clinics.clinic_id (cross-service, no FK)
doctor_id	UUID			References iam-service users.user_id (cross-service, no FK)
visit_date	DATE			Visit date
chief_complaint	TEXT			Patient-reported chief complaint
diagnosis	TEXT			Diagnosis
treatment_plan	TEXT			Treatment plan
notes	TEXT			Free-text notes
record_status	VARCHAR(9)			Record status
record_hash	VARCHAR(255)			Record hash
finalized_at	TIMESTAMP			Finalized timestamp
finalized_by	UUID			User who finalized the record
created_at	TIMESTAMP			Record creation timestamp

updated_at	TIMESTAMP			Last update timestamp
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MedicalRecordVersion entity

Attribute	Data type	Primary Key	Foreign Key	Description
version_id	UUID	x		Identifier of version
record_id	UUID		x	FK to medical_records (medical record)
version_number	INTEGER			Version number
snapshot	JSONB			Snapshot
changed_by	UUID			User who changed the record
change_reason	TEXT			Change reason
created_at	TIMESTAMP			Record creation timestamp

RecordExport entity

Attribute	Data type	Primary Key	Foreign Key	Description
export_id	UUID	x		Identifier of export
patient_id	UUID		x	FK to patients (patient)
record_id	UUID		x	FK to medical_records (medical record)
export_type	VARCHAR(50)			Export type
export_format	VARCHAR(20)			Export format
file_url	TEXT			URL of file
exported_by	UUID			User who exported the record
expires_at	TIMESTAMP			Expiry timestamp
created_at	TIMESTAMP			Record creation timestamp

ExaminationSession entity

Attribute	Data type	Primary Key	Foreign Key	Description
session_id	UUID	x		Identifier of session

appointment_id	UUID			References clinic-service appointments.appointment_id (cross-service, no FK)
record_id	UUID		x	FK to medical_records (medical record)
patient_id	UUID		x	FK to patients (patient)
doctor_id	UUID			References iam-service users.user_id (cross-service, no FK)
clinic_id	UUID			References clinic-service clinics.clinic_id (cross-service, no FK)
session_date	TIMESTAMP			Session date
chief_complaint	TEXT			Patient-reported chief complaint
present_illness	TEXT			Present illness
physical_examination	TEXT			Physical examination
vital_signs	JSONB			Vital signs
status	VARCHAR(20)			Current status value
started_at	TIMESTAMP			Started timestamp
completed_at	TIMESTAMP			Completed timestamp
signed_at	TIMESTAMP			Signed timestamp
signed_by	UUID			User who signed the record
created_at	TIMESTAMP			Record creation timestamp

ExaminationSessionAmendment entity

Attribute	Data type	Primary Key	Foreign Key	Description
amendment_id	UUID	x		Identifier of amendment
session_id	UUID		x	FK to examination_sessions (examination session)
record_id	UUID		x	FK to medical_records (medical record)

patient_id	UUID		x	FK to patients (patient)
doctor_id	UUID			Identifier of doctor
amendment_reason	TEXT			Amendment reason
amendment_text	TEXT			Amendment text
amended_by	UUID			User who amended the record
created_at	TIMESTAMP			Record creation timestamp

Symptom entity

Attribute	Data type	Primary Key	Foreign Key	Description
symptom_id	UUID	x		Identifier of symptom
session_id	UUID		x	FK to examination_sessions (examination session)
patient_id	UUID		x	FK to patients (patient)
symptom_name	VARCHAR(255)			Symptom name
body_location	VARCHAR(100)			Body location
severity	VARCHAR(8)			Severity
onset_date	DATE			Onset date
duration	VARCHAR(100)			Duration
description	TEXT			Free-text description
recorded_by	UUID			User who recorded the record
created_at	TIMESTAMP			Record creation timestamp
updated_at	TIMESTAMP			Last update timestamp

Diagnosis entity

Attribute	Data type	Primary Key	Foreign Key	Description
diagnosis_id	UUID	x		Identifier of diagnosis
session_id	UUID		x	FK to examination_sessions (examination session)

icd_code	VARCHAR(20)			ICD code
diagnosis_name	VARCHAR(255)			Diagnosis name
diagnosis_type	VARCHAR(50)			Diagnosis type
severity	VARCHAR(8)			Severity
notes	TEXT			Free-text notes
created_at	TIMESTAMP			Record creation timestamp

DentalChart entity

Attribute	Data type	Primary Key	Foreign Key	Description
chart_id	UUID	x		Identifier of chart
patient_id	UUID		x	FK to patients (patient)
record_id	UUID		x	FK to medical_records (medical record)
tooth_number	INTEGER			Tooth number
tooth_status	VARCHAR(50)			Tooth status
surfaces	JSONB			Surfaces
notes	TEXT			Free-text notes
created_at	TIMESTAMP			Record creation timestamp
updated_at	TIMESTAMP			Last update timestamp

TreatmentPlan entity

Attribute	Data type	Primary Key	Foreign Key	Description
plan_id	UUID	x		Identifier of plan
session_id	UUID		x	FK to examination_sessions (examination session)
patient_id	UUID		x	FK to patients (patient)
record_id	UUID		x	FK to medical_records (medical record)

plan_name	VARCHAR(255)			Plan name
objectives	TEXT			Objectives
duration_weeks	INTEGER			Duration weeks
status	VARCHAR(18)			Longest: partially_accepted
estimated_cost	NUMERIC(12,2)			Estimated cost
quote_currency	CHAR(3)			Quote currency
sent_at	TIMESTAMP			Sent timestamp
sent_to	UUID			Sent to
sent_via	VARCHAR(20)			Sent via
confirmed_at	TIMESTAMP			Confirmed timestamp
proposed_at	TIMESTAMP			Proposed timestamp
accepted_at	TIMESTAMP			Accepted timestamp
accepted_by	UUID			User who accepted the record
declined_at	TIMESTAMP			Declined timestamp
declined_by	UUID			User who declined the record
decline_reason	TEXT			Decline reason
created_by	UUID			User who created the record
created_at	TIMESTAMP			Record creation timestamp
updated_at	TIMESTAMP			Last update timestamp
quote_version	VARCHAR(100)			Quote version
risk_disclosure	TEXT			Risk disclosure
alternative_options	TEXT			Alternative options

acceptance_scope	VARCHAR(7)			Acceptance scope
accepted_scope_note	TEXT			Accepted scope note
accepted_representative_id	UUID		x	FK to patient_representatives (patient representative)
accepted_representative_name	VARCHAR(255)			Accepted representative name
accepted_representative_relationship	VARCHAR(100)			Accepted representative relationship
accepted_representative_phone	VARCHAR(20)			Accepted representative phone

TreatmentHistory entity

Attribute	Data type	Primary Key	Foreign Key	Description
treatment_id	UUID	x		Identifier of treatment
record_id	UUID		x	FK to medical_records (medical record)
patient_id	UUID		x	FK to patients (patient)
treatment_date	DATE			Treatment date
tooth_numbers	INTEGER[]			Tooth numbers
procedure_code	VARCHAR(50)			Procedure code
procedure_name	VARCHAR(255)			Procedure name
description	TEXT			Free-text description
cost	NUMERIC(10,2)			Cost
status	VARCHAR(20)			Current status value
performed_by	UUID			User who performed the record
created_at	TIMESTAMP			Record creation timestamp
updated_at	TIMESTAMP			Last update timestamp

Prescription entity

Attribute	Data type	Primary Key	Foreign Key	Description
prescription_id	UUID	x		Identifier of prescription
session_id	UUID		x	FK to examination_sessions (examination session)
record_id	UUID		x	FK to medical_records (medical record)
patient_id	UUID		x	FK to patients (patient)
doctor_id	UUID			Identifier of doctor
prescription_date	DATE			Prescription date
status	VARCHAR(9)			Current status value
notes	TEXT			Free-text notes
issued_at	TIMESTAMP			Issued timestamp
issued_by	UUID			User who issued the record
cancelled_at	TIMESTAMP			Cancelled timestamp
cancellation_reason	TEXT			Cancellation reason
created_at	TIMESTAMP			Record creation timestamp
updated_at	TIMESTAMP			Last update timestamp
minor_patient_at_issue	BOOLEAN			Minor patient at issue flag
patient_age_years_at_issue	INTEGER			Patient age years at issue
patient_age_months_at_issue	INTEGER			Patient age months at issue
representative_name_snapshot	VARCHAR(255)			Representative name snapshot

representative_phone_snapshot	VARCHAR(20)			Representative phone snapshot
representative_id_snapshot	UUID			Representative ID snapshot
representative_relationship_snapshot	VARCHAR(100)			Representative relationship snapshot

PrescriptionItem entity

Attribute	Data type	Primary Key	Foreign Key	Description
item_id	UUID	x		Identifier of item
prescription_id	UUID		x	FK to prescriptions (prescription)
medication_name	VARCHAR(255)			Medication name
medication_code	VARCHAR(50)			Medication code
dosage	VARCHAR(100)			Dosage
route	VARCHAR(50)			Route
frequency	VARCHAR(100)			Frequency
duration_days	INTEGER			Duration days
quantity	INTEGER			Quantity
instructions	TEXT			Instructions
created_at	TIMESTAMP			Record creation timestamp

ClinicalOrder entity

Attribute	Data type	Primary Key	Foreign Key	Description
order_id	UUID	x		Identifier of order
session_id	UUID		x	FK to examination_sessions (examination session)
record_id	UUID		x	FK to medical_records (medical record)
patient_id	UUID		x	FK to patients (patient)

ordered_by	UUID			User who ordered the record
order_type	VARCHAR(13)			Order type
test_type	VARCHAR(100)			Test type
clinical_indication	TEXT			Clinical indication
teeth_numbers	INTEGER[]			Teeth numbers
urgency	VARCHAR(7)			Urgency
status	VARCHAR(11)			Current status value
ordered_date	TIMESTAMP			Ordered date
scheduled_date	TIMESTAMP			Scheduled date
completed_date	TIMESTAMP			Completed date
result_url	TEXT			URL of result
report	TEXT			Report
created_at	TIMESTAMP			Record creation timestamp
updated_at	TIMESTAMP			Last update timestamp

DiagnosticOrder entity

Attribute	Data type	Primary Key	Foreign Key	Description
order_id	UUID	x		Identifier of order
appointment_id	UUID		x	FK to appointments (appointment)
patient_id	UUID			References medical-service patients.patient_id (cross-database, no FK — see DropAppointmentPatientForeignKey1730000000006)
doctor_id	UUID			References iam-service users.user_id (cross-service, no FK)
order_code	VARCHAR(50)			Order code
order_type	VARCHAR(13)			Order type
description	TEXT			Free-text description

priority	VARCHAR(7)			Priority
tooth_number	VARCHAR(10)			Tooth number
area	VARCHAR(100)			Area
status	VARCHAR(11)			Current status value
result_summary	TEXT			Result summary
result_attachment_url	TEXT			URL of result attachment
notes	TEXT			Free-text notes
ordered_at	TIMESTAMP			Ordered timestamp
completed_at	TIMESTAMP			Completed timestamp
created_at	TIMESTAMP			Record creation timestamp
updated_at	TIMESTAMP			Last update timestamp

LabTestResult entity

Attribute	Data type	Primary Key	Foreign Key	Description
result_id	UUID	x		Identifier of result
order_id	UUID		x	FK to clinical_orders (clinical order)
test_name	VARCHAR(255)			Test name
result_value	TEXT			Result value
result_unit	VARCHAR(50)			Result unit
reference_range	VARCHAR(100)			Reference range
is_abnormal	BOOLEAN			Abnormal flag
notes	TEXT			Free-text notes
created_at	TIMESTAMP			Record creation timestamp

DentallImage entity

Attribute	Data type	Primary Key	Foreign Key	Description
image_id	UUID	x		Identifier of image
patient_id	UUID		x	FK to patients (patient)
record_id	UUID		x	FK to medical_records (medical record)
category_id	UUID		x	FK to image_categories (image category)
image_type	VARCHAR(50)			Image type
image_url	TEXT			URL of image
thumbnail_url	TEXT			URL of thumbnail
file_size_kb	INTEGER			File size kb
file_format	VARCHAR(10)			File format
tooth_numbers	INTEGER[]			Tooth numbers
view_angle	VARCHAR(50)			View angle
description	TEXT			Free-text description
tags	TEXT[]			Tags
metadata	JSONB			Additional structured metadata
pacs_id	VARCHAR(255)			Identifier of pacs
taken_date	DATE			Taken date
taken_by	UUID			User who taken the record
uploaded_by	UUID			User who uploaded the record
is_archived	BOOLEAN			Archived flag
created_at	TIMESTAMP			Record creation timestamp
updated_at	TIMESTAMP			Last update timestamp

ImageCategory entity

Attribute	Data type	Primary Key	Foreign Key	Description
category_id	UUID	x		Identifier of category

category_name	VARCHAR(100)			Category name
description	TEXT			Free-text description
created_at	TIMESTAMP			Record creation timestamp
updated_at	TIMESTAMP			Last update timestamp

ImageAnnotation entity

Attribute	Data type	Primary Key	Foreign Key	Description
annotation_id	UUID	x		Identifier of annotation
image_id	UUID		x	FK to dental_images (dental image)
annotated_by	UUID			User who annotated the record
annotation_type	VARCHAR(50)			Annotation type
annotation_data	JSONB			Annotation data
note	TEXT			Note
created_at	TIMESTAMP			Record creation timestamp
updated_at	TIMESTAMP			Last update timestamp

PacsSyncLog entity

Attribute	Data type	Primary Key	Foreign Key	Description
sync_id	UUID	x		Identifier of sync
image_id	UUID		x	FK to dental_images (dental image)
sync_type	VARCHAR(50)			Sync type
pacs_server	VARCHAR(255)			PACS server
status	VARCHAR(20)			Current status value
error_message	TEXT			Error detail on failure
synced_at	TIMESTAMP			Synced timestamp
created_at	TIMESTAMP			Record creation timestamp
updated_at	TIMESTAMP			Last update timestamp

Appointment entity

Attribute	Data type	Primary Key	Foreign Key	Description
appointment_id	UUID	x		Identifier of appointment
appointment_code	VARCHAR (50)			Appointment code
patient_id	UUID			References medical-service patients.patient_id (cross-database, no FK — see DropAppointmentPatientForeignKey1730000000006)
doctor_id	UUID			References iam-service users.user_id (cross-service, no FK)
clinic_id	UUID		x	FK to clinics (clinic)
room_id	UUID		x	FK to treatment_rooms (treatment room)
service_id	UUID		x	FK to services (service)
appointment_date	DATE			Appointment date
appointment_time	TIME			Appointment time
duration_minutes	INTEGER			Duration minutes
appointment_type	VARCHAR (12)			Appointment type
status	VARCHAR (11)			Current status value
chief_complaint	TEXT			Patient-reported chief complaint
notes	TEXT			Free-text notes
cancellation_reason	TEXT			Cancellation reason
cancellation_requested	BOOLEAN			Cancellation requested flag
cancelled_by	UUID			User who cancelled the record
cancelled_at	TIMESTAMP			Cancelled timestamp

is_outside_hours	BOOLEAN			Outside hours flag
outside_hours_reason	TEXT			Outside hours reason
approved_by	UUID			User who approved the record
payment_id	UUID			Identifier of payment
payment_status	VARCHAR(14)			Payment status
created_by	UUID			User who created the record
created_at	TIMESTAMP			Record creation timestamp
updated_at	TIMESTAMP			Last update timestamp
occupied_during	TSRANGE			Generated booking window used by the overlap guards below
session_id	UUID			Identifier of session
treatment_plan_id	UUID			Identifier of treatment plan

AppointmentStatusHistory entity

Attribute	Data type	Primary Key	Foreign Key	Description
history_id	UUID	x		Identifier of history
appointment_id	UUID		x	FK to appointments (appointment)
old_status	VARCHAR(11)			Old status
new_status	VARCHAR(11)			New status
changed_by	UUID			User who changed the record
reason	TEXT			Reason text
created_at	TIMESTAMP			Record creation timestamp

AppointmentReminderPreference entity

Attribute	Data type	Primary Key	Foreign Key	Description
preference_id	UUID	x		Identifier of preference

patient_id	UUID			Identifier of patient
channel	VARCHAR(5)			Delivery channel (APP/EMAIL/SMS)
enabled	BOOLEAN			Enabled flag
reminder_minutes_before	INTEGER			Reminder minutes before
created_at	TIMESTAMP			Record creation timestamp
updated_at	TIMESTAMP			Last update timestamp

AppointmentNotificationLog entity

Attribute	Data type	Primary Key	Foreign Key	Description
log_id	UUID	x		Identifier of log
appointment_id	UUID		x	FK to appointments (appointment)
notification_type	VARCHAR(50)			Notification type
channel	VARCHAR(5)			Delivery channel (APP/EMAIL/SMS)
status	VARCHAR(20)			Current status value
attempt_count	INTEGER			Number of delivery attempts
notification_id	VARCHAR(100)			Identifier of notification
preference_enabled	BOOLEAN			Preference enabled flag
reminder_minutes_before	INTEGER			Reminder minutes before
last_attempt_at	TIMESTAMP			Last attempt timestamp
next_retry_at	TIMESTAMP			Next retry timestamp
error_message	TEXT			Error detail on failure
read_at	TIMESTAMP			Read timestamp
responded_at	TIMESTAMP			Responded timestamp

created_at	TIMESTAMP			Record creation timestamp
updated_at	TIMESTAMP			Last update timestamp
scheduled_for	TIMESTAMP			Reminder window used to dedupe automatic sends; NULL for manual sends

IdempotencyKey entity

Attribute	Data type	Primary Key	Foreign Key	Description
idempotency_key	VARCHAR(255)	x		Idempotency key
method	VARCHAR(7)			Method
path	VARCHAR(512)			Path
status	VARCHAR(11)			Current status value
response_status	INTEGER			Response status
response_body	JSONB			Response body
created_at	TIMESTAMP			Record creation timestamp
expires_at	TIMESTAMP			Expiry timestamp

DoctorSchedule entity

Attribute	Data type	Primary Key	Foreign Key	Description
schedule_id	UUID	x		Identifier of schedule
doctor_id	UUID			References iam-service users.user_id (cross-service, no FK)
clinic_id	UUID		x	FK to clinics (clinic)
shift_id	UUID		x	FK to work_shifts (work shifts)
work_date	DATE			Work date
room_id	UUID		x	FK to treatment_rooms (treatment room)

max_patients	INTEGER			Max patients
status	VARCHAR(9)			Current status value
notes	TEXT			Free-text notes
created_at	TIMESTAMP			Record creation timestamp
updated_at	TIMESTAMP			Last update timestamp

ScheduleChange entity

Attribute	Data type	Primary Key	Foreign Key	Description
change_id	UUID	x		Identifier of change
schedule_id	UUID		x	FK to doctor_schedules (doctor schedule)
changed_by	UUID			User who changed the record
change_type	VARCHAR(14)			Change type
old_values	JSONB			Old values
new_values	JSONB			New values
reason	TEXT			Reason text
approved_by	UUID			User who approved the record
approval_status	VARCHAR(8)			Approval status
created_at	TIMESTAMP			Record creation timestamp

DoctorLeave entity

Attribute	Data type	Primary Key	Foreign Key	Description
leave_id	UUID	x		Identifier of leave
doctor_id	UUID			References iam-service users.user_id (cross-service, no FK)
leave_type	VARCHAR(9)			Leave type
start_date	DATE			Start date
end_date	DATE			End date
reason	TEXT			Reason text

status	VARCHAR(8)			Current status value
approved_by	UUID			User who approved the record
created_at	TIMESTAMP			Record creation timestamp
updated_at	TIMESTAMP			Last update timestamp

WorkShift entity

Attribute	Data type	Primary Key	Foreign Key	Description
shift_id	UUID	x		Identifier of shift
shift_name	VARCHAR(100)			Shift name
start_time	TIME			Start time
end_time	TIME			End time
description	TEXT			Free-text description
created_at	TIMESTAMP			Record creation timestamp

Clinic entity

Attribute	Data type	Primary Key	Foreign Key	Description
clinic_id	UUID	x		Identifier of clinic
clinic_name	VARCHAR(255)			Clinic name
clinic_code	VARCHAR(50)			Clinic code
address	TEXT			Street address
ward	VARCHAR(100)			Ward
district	VARCHAR(100)			District
city	VARCHAR(100)			City
phone	VARCHAR(20)			Phone number
email	VARCHAR(255)			Email address
website	VARCHAR(255)			Website
logo_url	TEXT			URL of logo
operating_hours	JSONB			Operating hours
status	VARCHAR(11)			Current status value

license_number	VARCHAR(100)			License number
license_expiry	DATE			License expiry
created_at	TIMESTAMP			Record creation timestamp
updated_at	TIMESTAMP			Last update timestamp

TreatmentRoom entity

Attribute	Data type	Primary Key	Foreign Key	Description
room_id	UUID	x		Identifier of room
clinic_id	UUID		x	FK to clinics (clinic)
room_name	VARCHAR(100)			Room name
room_code	VARCHAR(50)			Room code
room_type	CLINIC_ROOM_TYPE			Room type
floor_number	INTEGER			Floor number
equipment_list	JSONB			Equipment list
status	VARCHAR(11)			Current status value
created_at	TIMESTAMP			Record creation timestamp
updated_at	TIMESTAMP			Last update timestamp

Specialty entity

Attribute	Data type	Primary Key	Foreign Key	Description
specialty_id	UUID	x		Identifier of specialty
specialty_name	VARCHAR(255)			Specialty name
specialty_code	VARCHAR(50)			Specialty code
description	TEXT			Free-text description
icon_url	TEXT			URL of icon
is_active	BOOLEAN			Active flag
display_order	INTEGER			Sort order for display
created_at	TIMESTAMP			Record creation timestamp
updated_at	TIMESTAMP			Last update timestamp

ClinicSpecialty entity

Attribute	Data type	Primary Key	Foreign Key	Description
clinic_id	UUID	x	x	FK to clinics (clinic)
specialty_id	UUID	x	x	FK to specialties (specialty)
created_at	TIMESTAMP			Record creation timestamp

DoctorSpecialty entity

Attribute	Data type	Primary Key	Foreign Key	Description
doctor_id	UUID	x		References iam-service users.user_id (cross-service, no FK)
specialty_id	UUID	x	x	FK to specialties (specialty)
certification_number	VARCHAR(100)			Certification number
certified_date	DATE			Certified date
is_primary	BOOLEAN			Primary flag
created_at	TIMESTAMP			Record creation timestamp

ServiceCategory entity

Attribute	Data type	Primary Key	Foreign Key	Description
category_id	UUID	x		Identifier of category
category_name	VARCHAR(255)			Category name
description	TEXT			Free-text description
parent_category_id	UUID		x	FK to service_categories (service category)
is_active	BOOLEAN			Active flag
display_order	INTEGER			Sort order for display
created_at	TIMESTAMP			Record creation timestamp

Service entity

Attribute	Data type	Primary Key	Foreign Key	Description
service_id	UUID	x		Identifier of service
service_code	VARCHAR(50)			Service code
service_name	VARCHAR(255)			Service name
category_id	UUID		x	FK to service_categories (service category)
specialty_id	UUID		x	FK to specialties (specialty)
description	TEXT			Free-text description
duration_minutes	INTEGER			Duration minutes
base_price	NUMERIC(10,2)			Base price
currency	CHAR(3)			Currency
is_active	BOOLEAN			Active flag
requires_appointment	BOOLEAN			Requires appointment flag
preparation_instructions	TEXT			Preparation instructions
created_at	TIMESTAMP			Record creation timestamp
updated_at	TIMESTAMP			Last update timestamp
required_room_type	CLINIC_ROOM_TYPE			Required room type

ClinicService entity

Attribute	Data type	Primary Key	Foreign Key	Description
clinic_service_id	UUID	x		Identifier of clinic service
clinic_id	UUID		x	FK to clinics (clinic)
service_id	UUID		x	FK to services (service)

custom_price	NUMERIC(10,2)			Custom price
is_available	BOOLEAN			Available flag
created_at	TIMESTAMP			Record creation timestamp
updated_at	TIMESTAMP			Last update timestamp

Payment entity

Attribute	Data type	Primary Key	Foreign Key	Description
payment_id	UUID	x		Identifier of payment
appointment_id	UUID			References clinic-service appointments.appointment_id (cross-service, no FK)
amount	NUMERIC(12,2)			Amount
currency	CHAR(3)			Currency
status	VARCHAR(8)			Pending, paid, failed, refunded
provider	VARCHAR(30)			Provider
provider_txn_ref	VARCHAR(100)			Provider txn ref
order_info	TEXT			Order info
refund_amount	NUMERIC(12,2)			Refund tracking
refunded_at	TIMESTAMP			Refunded timestamp
refund_status	VARCHAR(12)			Refund approval workflow (null = no refund activity)
refund_reason	TEXT			Refund reason
refund_requested_by	UUID			References iam-service users.user_id (cross-service, no FK)
refund_requested_at	TIMESTAMP			Refund requested timestamp
refund_reviewed_by	UUID			References iam-service users.user_id (cross-service, no FK)
refund_reviewed_at	TIMESTAMP			Refund reviewed timestamp

created_at	TIMESTAMP			Record creation timestamp
updated_at	TIMESTAMP			Last update timestamp

Entities List

#	Entity	Description
1	Account	Login credentials, primary role and account status
2	User	User profile: name, contact, date of birth, gender, avatar
3	RefreshToken	Issued refresh tokens for session renewal and revocation
4	OtpToken	One-time passcodes for phone/email verification and password reset
5	PhoneVerification	Phone-number verification state per user
6	OAuthConnection	Linked external identity provider accounts (Google, etc.)
7	Role	RBAC role definition
8	Permission	RBAC action/resource permission
9	UserRole	Assignment of roles to users
10	RolePermission	Assignment of permissions to roles
11	KycVerification	Identity verification workflow with AES-encrypted documents
12	AuditLog	Tamper-evident log of security and clinical data events
13	Notification	In-app/push notification instance delivered to a user
14	NotificationTemplate	Reusable message templates per notification type and channel
15	NotificationPreference	Per-user opt-in/opt-out settings by channel
16	NotificationDeliveryLog	Delivery attempts, status and provider responses
17	PushSubscription	Web push endpoints registered per device
18	Patient	Core patient demographic profile
19	PatientRepresentative	Guardian or legal representative authorized to act for a patient

20	MedicalHistory	Historical conditions, allergies and medications by patient
21	MedicalRecord	Visit-level clinical record; immutable once finalized
22	MedicalRecordVersion	Version snapshots of a medical record for amendment tracking
23	RecordExport	Generated record export files with expiry for patient/compliance download
24	ExaminationSession	Clinical examination encounter
25	ExaminationSessionAmendment	Post-finalization corrections to an examination session
26	Symptom	Reported symptoms with body location, severity and onset
27	Diagnosis	Diagnoses attached to an examination session
28	DentalChart	Per-tooth status and surface findings for a patient record
29	TreatmentPlan	Therapy plan, steps, quote and patient consent
30	TreatmentHistory	Performed procedures by tooth, with code, cost and status
31	Prescription	Prescribed medications issued from a session
32	PrescriptionItem	Medication lines in a prescription
33	ClinicalOrder	Lab/imaging order raised from an examination session
34	DiagnosticOrder	Requested diagnostic study with priority, tooth/area and result summary
35	LabTestResult	Result values, units, reference ranges and abnormal flags for an order
36	DentalImage	Uploaded diagnostic images
37	ImageCategory	Image classification taxonomy
38	ImageAnnotation	Markup and measurement comments on images
39	PacsSyncLog	Synchronization attempts between stored images and the PACS server
40	Appointment	Booking and attendance object
41	AppointmentStatusHistory	Status transition trail for an appointment
42	AppointmentReminderPreference	Reminder channels and lead time chosen per appointment

43	AppointmentNotificationLog	Reminder dispatch attempts, channel and delivery state
44	IdempotencyKey	Deduplication keys preventing duplicate booking requests
45	DoctorSchedule	Doctor shift schedule by date, clinic and room
46	ScheduleChange	Audit trail of edits to a published schedule
47	DoctorLeave	Approved or requested leave blocking availability
48	WorkShift	Named shift definitions with start and end times
49	Clinic	Clinic branch metadata
50	TreatmentRoom	Rooms in clinic branches, including equipment inventory
51	Specialty	Clinical specialty taxonomy
52	ClinicSpecialty	Specialties offered at each clinic branch
53	DoctorSpecialty	Specialties held by each doctor
54	ServiceCategory	Grouping of dental services
55	Service	Dental services catalog
56	ClinicService	Per-clinic service availability and pricing
57	Payment	Payment transaction with status, currency and gateway reference

3.2 Authentication

a. Signup

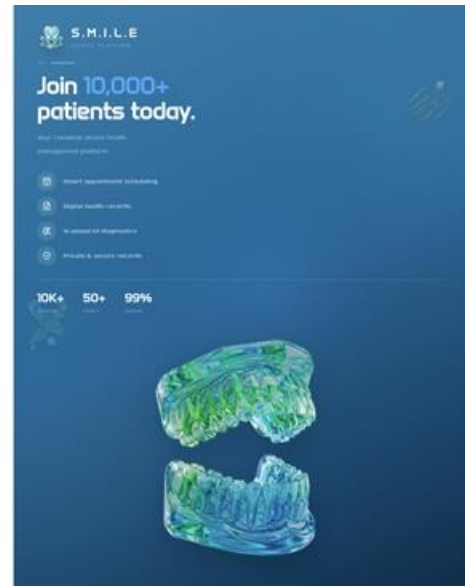
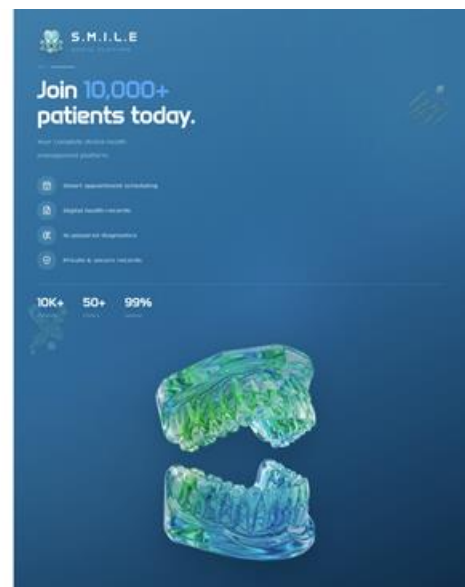
[A function can be a screen or a non-screen function (listed in the part 5.1 above). In this part, you need to provide the details on the related function, focus on mentioning below information

- *Function trigger: how this function is triggered (navigation path, a timing frequency, etc.*
- *Function description: actors/roles, purpose, interface, data processing, etc.*
- *Screen layout: mockup prototype of the screen, sample below is for Manage Products screen*
- *Function Details: provide explanation for the data, validation, functionalities (for both normal cases and abnormal cases), etc. of the function so that the reader can image how it work.*

]

- **Function trigger:** A Guest selects registration and submits the signup form.
- **Function description:** The function creates a patient account from email, password, username, phone, and profile details through the account service.

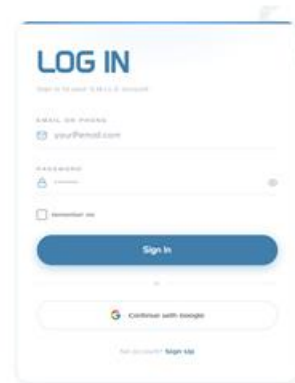
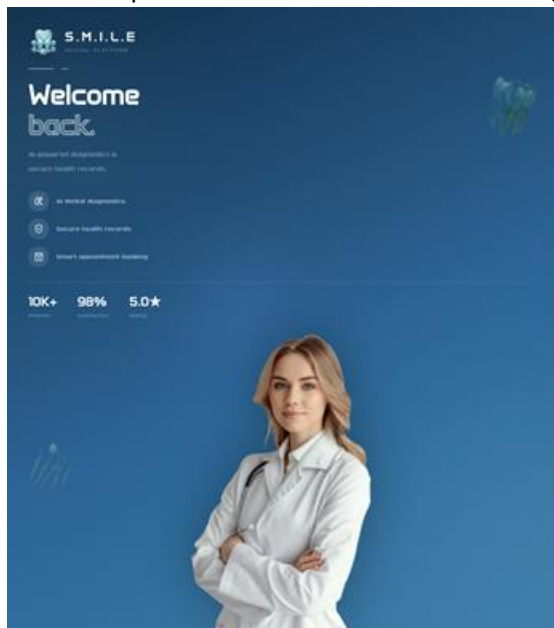
- **Screen layout:** The Signup screen contains account and profile fields, password confirmation, a create account action, a Google signup action, validation, and a result message.
- **Function Details:** Email, phone, and username must be unique. The password must contain at least eight characters with letters and numbers, and its confirmation must match. Valid data creates the user, stores a secure password hash, and returns access and refresh tokens. Duplicate or invalid data remains unsaved and shows the documented field error.

b. Login

- **Function trigger:** An Admin, Nurse, Doctor, Receptionist, or Patient opens Login and submits email or phone with a password.
- **Function description:** The function authenticates the user and grants access according to assigned roles.
- **Screen layout:** The Login screen contains an email or phone field, a password field, a sign in action, a Google login action, a forgot password link, and an inline result area.
- **Function Details:** The account service verifies the credentials and account status. Valid credentials issue access and refresh tokens and open the authorized system area. Incorrect

credentials show an inline error. Repeated failures temporarily lock the account, and an expired session returns the user to Login.

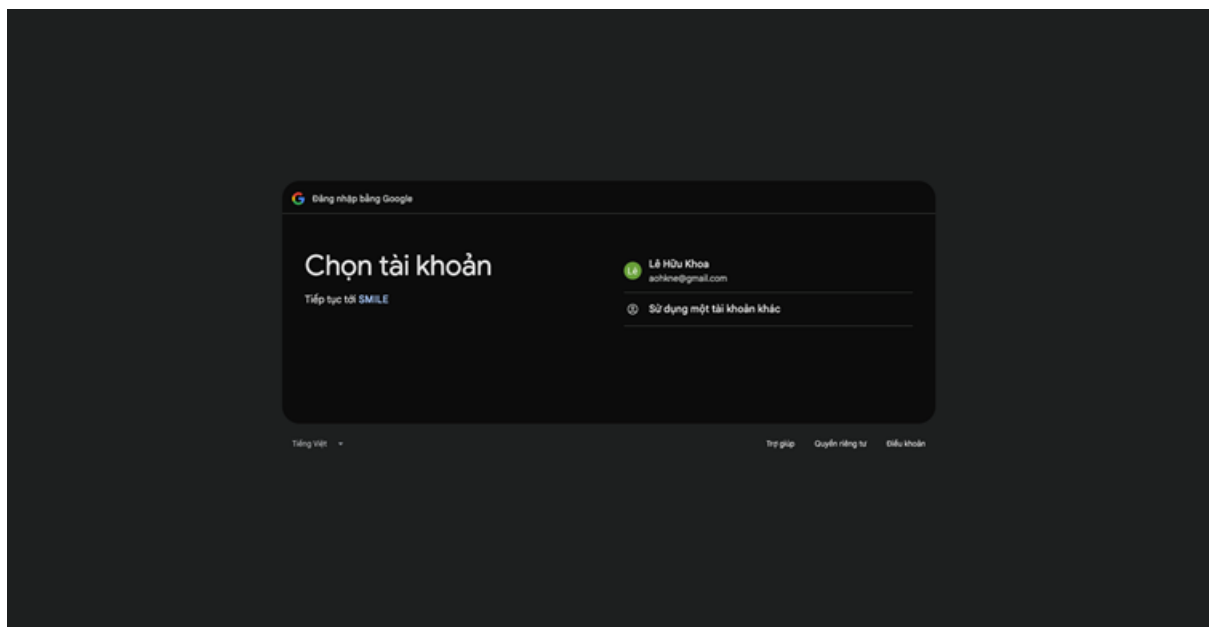
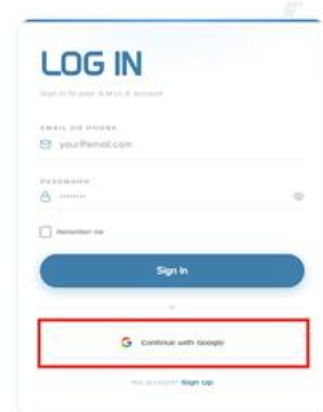
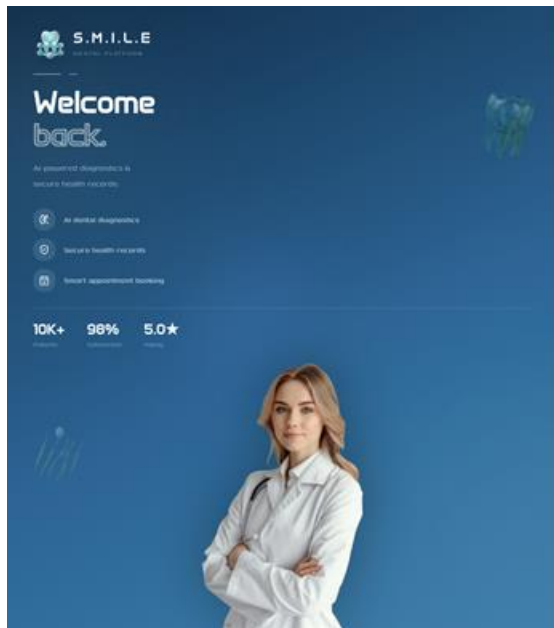


c. Logout

- **Function trigger:** A signed in Admin, Nurse, Doctor, Receptionist, or Patient selects Logout.
- **Function description:** The function ends the current session and prevents reuse of its access token.
- **Screen layout:** There is no separate screen. The result appears as a return to the Login screen.
- **Function Details:** The system adds the token identifier to the Redis blacklist for the rest of the token lifetime. A valid request ends access and returns the user to Login. A later request with the same token is rejected.

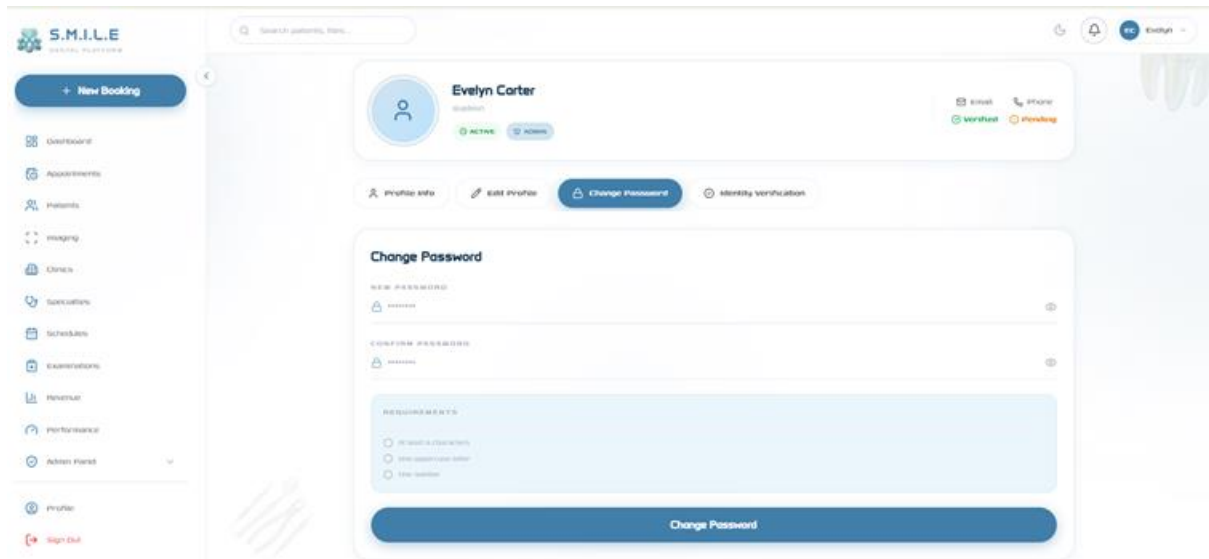
d. Login with Google

- **Function trigger:** An Admin, Nurse, Doctor, Receptionist, or Patient selects the Google login action on Login and completes Google authentication.
- **Function description:** The function uses Google OAuth 2.0 to authenticate the user, then links or creates the local account.
- **Screen layout:** The Login screen contains the Google login action. Google provides the authentication interface, and the local result appears on Login or in the authorized system area.
- **Function Details:** The system verifies the Google token before issuing its own access and refresh tokens. A valid token links or creates the local account and completes login. A failed Google authentication leaves the user on Login and shows a failure message.



e. Reset Password

- **Function trigger:** An Admin, Nurse, Doctor, Receptionist, or Patient opens Reset Password with a valid reset token and submits a new password.
- **Function description:** The function replaces the stored password hash after reset token verification.
- **Screen layout:** The Reset Password screen contains the reset code or token input, new password and confirmation fields, a submit action, inline validation, and a result message.
- **Function Details:** The token must be valid and the password must contain at least eight characters with letters and numbers. The confirmation must match. Valid input stores the new secure hash and shows successful completion. An invalid or expired code or invalid password prevents the update and shows the relevant error.



f. Forgot Password

- **Function trigger:** An Admin, Nurse, Doctor, Receptionist, or Patient selects Forgot Password from Login and submits a registered email.
- **Function description:** The function starts password recovery by generating a limited duration reset token for the registered account.
- **Screen layout:** There is no separate screen listed. The email field, send action, validation, and result appear in the recovery flow opened from Login.
- **Function Details:** The email must be valid and registered. The account service generates a limited duration token and sends it through the configured email provider. A valid request proceeds to reset verification. Invalid input or a service failure keeps the user on the interface and shows an error.

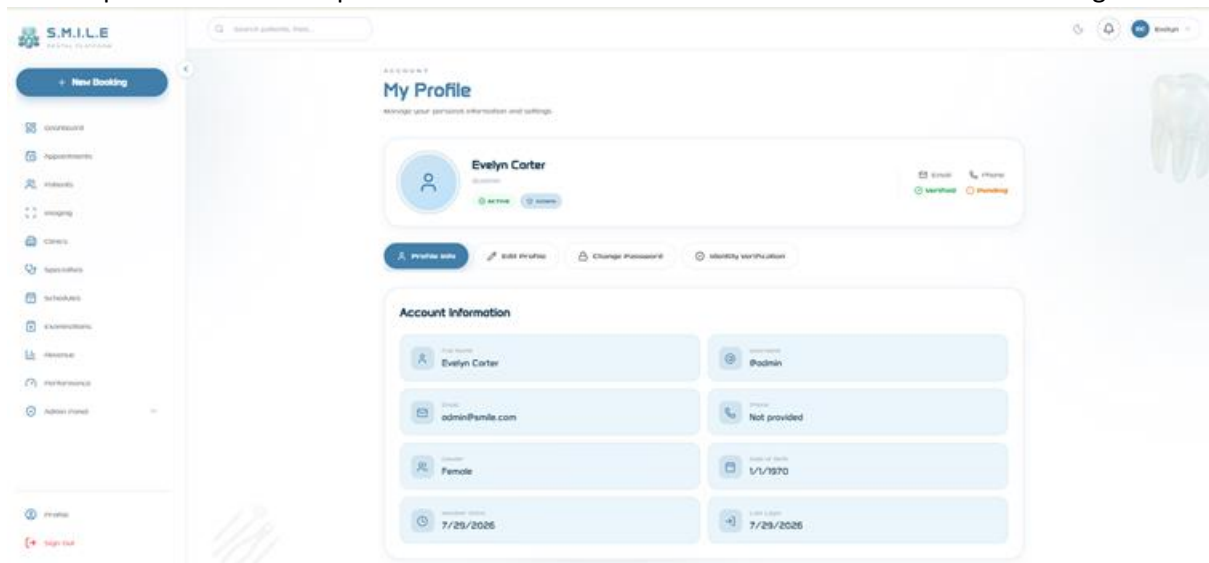
g. Change Password

- **Function trigger:** A signed in Admin, Nurse, Doctor, Receptionist, or Patient selects the password update option in account settings.
- **Function description:** The function changes the current account password after verifying the existing password.
- **Screen layout:** The password update interface contains current password, new password, confirmation, a save action, inline validation, and a result message.
- **Function Details:** The current password must match. The new password must contain at least eight characters with letters and numbers, and confirmation must match. Valid input stores a new secure hash. A wrong current password or invalid new password is rejected and leaves the existing password unchanged.

a. View Profile

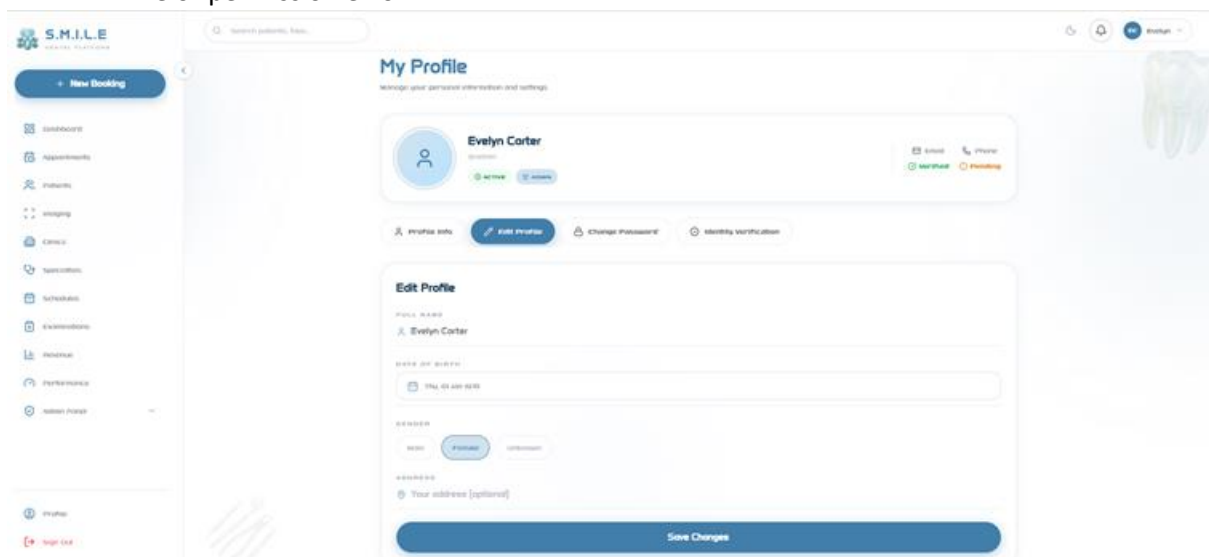
- **Function trigger:** A signed in Admin, Nurse, Doctor, Receptionist, or Patient opens Profile Page.
- **Function description:** The function returns the account and profile record owned by the authenticated user.
- **Screen layout:** Profile Page displays username, email, phone, full name, date of birth, gender, avatar, verification state, and account status.

- **Function Details:** The system identifies the user from the active session and restricts access to owned data unless a role grants broader access. A valid session displays the profile. Missing permission or an expired session denies access and returns the documented message.



b. Update Profile

- **Function trigger:** A signed in Admin, Nurse, Doctor, Receptionist, or Patient selects Update Profile from Profile Page and saves changes.
- **Function description:** The function updates name, date of birth, gender, avatar, or contact details for the authenticated user.
- **Screen layout:** Update Profile contains editable profile fields, avatar control, save and cancel actions, inline validation, and a result message.
- **Function Details:** The system checks ownership and validates required values, email format, phone format, and field length before saving. Valid changes update the user record and show success. Invalid data or insufficient permission leaves the record unchanged and shows an inline or permission error.



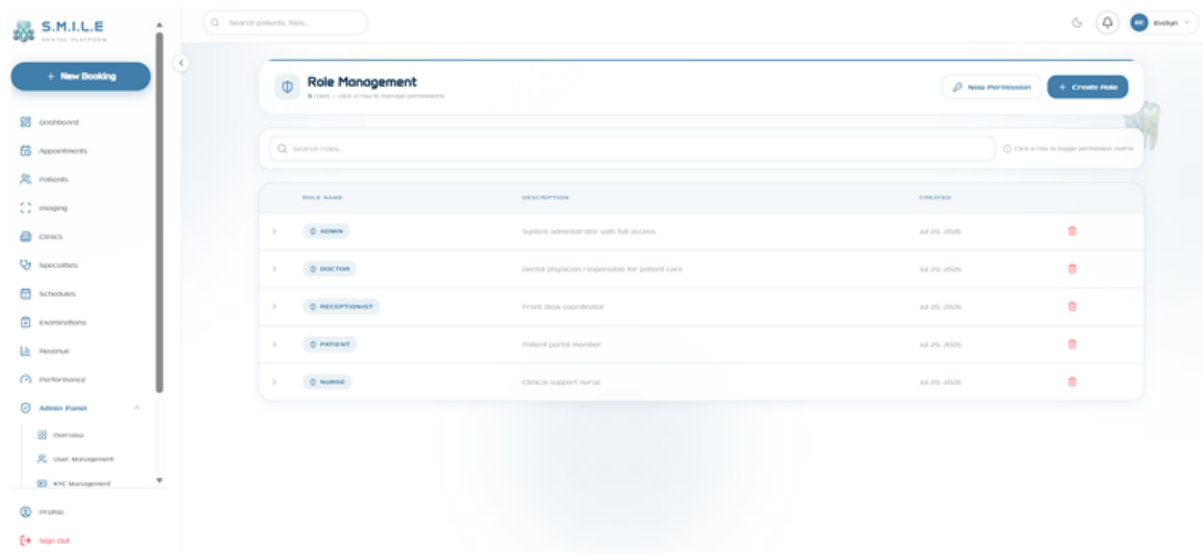
c. Send OTP

- **Function trigger:** An Admin, Doctor, Receptionist, or Patient requests phone verification from the relevant verification interface.

- **Function description:** The function generates and dispatches a single use verification code for the phone number of the user.
- **Screen layout:** There is no separate screen. The send action and result appear in the current phone verification interface, followed by the code entry area.
- **Function Details:** The phone number must use a valid format. The system stores the code in Redis with an expiry and dispatches it. Success shows that the code was sent. A wrong or expired code is rejected, and a connection or service failure shows the documented error.

d. View Role

- **Function trigger:** An Admin opens Role Page.
- **Function description:** The function lists defined roles and their descriptions for access control administration.
- **Screen layout:** Role Page displays role name, description, and available management actions.
- **Function Details:** The system returns the stored role records. A successful request shows all available roles. No records show the documented empty state, while insufficient permission or a server failure shows the relevant error.



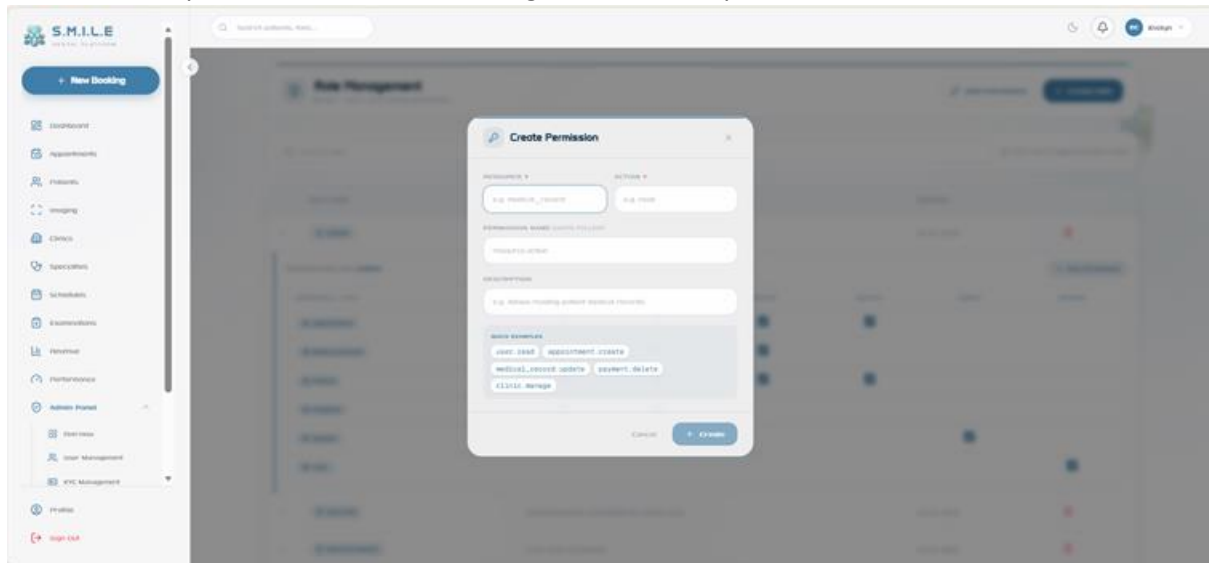
d. Update Role

- **Function trigger:** An Admin selects a role on Role Page, edits its name or description, and saves.
- **Function description:** The function updates a role definition and records the administrative change.
- **Screen layout:** The role editor contains role name and description fields, save and cancel actions, inline validation, and a result message.
- **Function Details:** The role name must be unique. Valid changes update the role, record the updating user and time, write an audit entry, and show success. A duplicate name, missing permission, or invalid input prevents the update and shows an error.

e. Create Permission

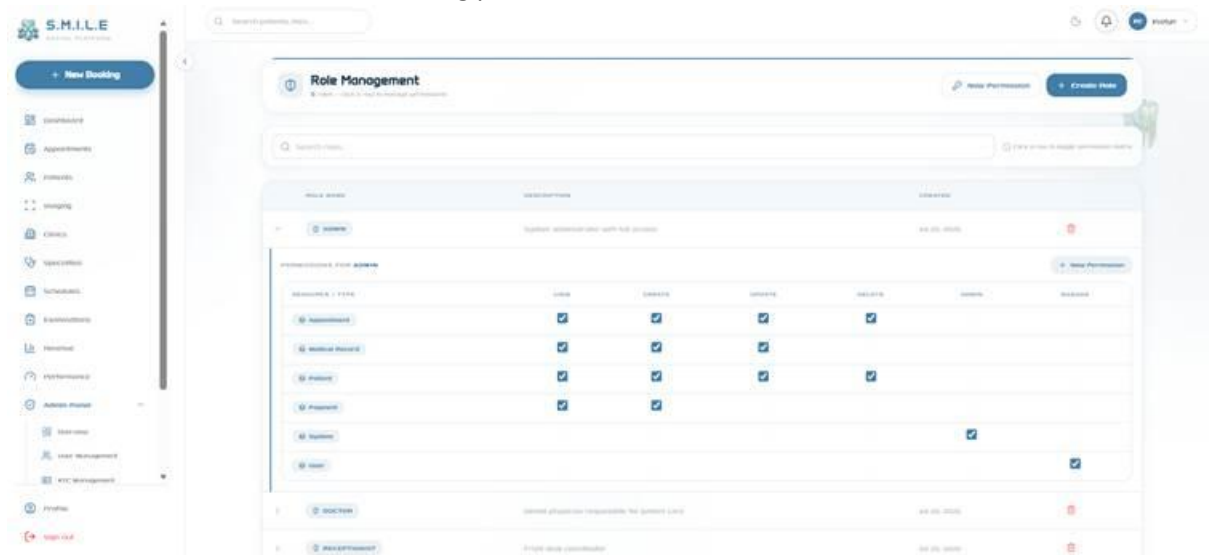
- **Function trigger:** An Admin selects the permission creation action from Role Page and submits the form.
- **Function description:** The function defines a permission from a target resource, an action, a name, and an optional description.
- **Screen layout:** The permission dialog on Role Page contains name, resource, action, description, role selections, save and cancel controls, validation, and a result message.

- **Function Details:** The permission name must be unique and required values must be present. Valid data creates the permission with creator and timestamp data. Duplicate or missing values prevent creation, and missing administrative permission denies the action.



f. View Permissions by Role

- **Function trigger:** An Admin selects a role and opens its permission scope on Role Page.
- **Function description:** The function returns permission records filtered by the selected role.
- **Screen layout:** The role permission area displays the selected role and matching permission names, resources, actions, and descriptions.
- **Function Details:** The system uses the selected role to return its permission mappings. A valid selection displays the matching records. A role with no permissions shows an empty state, while an invalid role or missing permission shows an error.



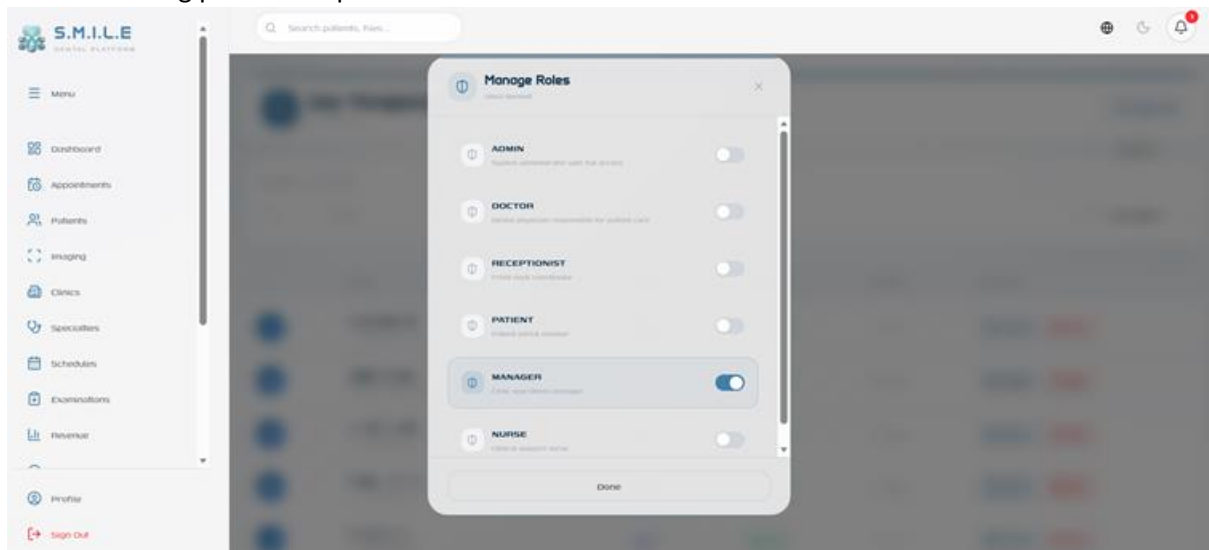
g. Update Role Permission

- **Function trigger:** An Admin selects a role on Role Page, changes its permission selections, and saves.
- **Function description:** The function attaches or detaches permissions on a role to maintain its access scope.

- **Screen layout:** Update Role Permission displays the role, available permissions, current selections, save and cancel actions, and a result message.
- **Function Details:** The system validates the role and permission records before updating their mappings. A valid change saves the new scope, records the actor and time in the audit log, and shows success. Invalid records or missing authorization leave the existing mappings unchanged.

h. Assign Role

- **Function trigger:** An Admin opens User Detail, selects an available role, and confirms assignment.
- **Function description:** The function grants the selected role to a user and records the sensitive action.
- **Screen layout:** There is no separate screen. The role selector and result appear on User Detail.
- **Function Details:** Only an authorized Admin may control account roles. A valid user and role create the user role link, write an audit entry, and show successful assignment. Invalid data or missing permission prevents the link and shows an error.



i. Revoke Role

- **Function trigger:** An Admin opens User Detail, selects an assigned role, and confirms revocation.
- **Function description:** The function removes a role from a user and records the sensitive action.
- **Screen layout:** There is no separate screen. Assigned roles, the revoke action, and its result appear on User Detail.
- **Function Details:** Only an authorized Admin may revoke roles. A valid user role link is removed and an audit entry records the actor, action, target, and time. An invalid link or missing permission leaves assignments unchanged and shows an error.

j. Access Audit Log

- **Function trigger:** An Admin requests recorded security and administrative activity from user administration.
- **Function description:** The function returns audit entries so the Admin can review sensitive actions.
- **Screen layout:** The Access Audit Log screen contains activity filters and a table with actor, action, target, and timestamp data.

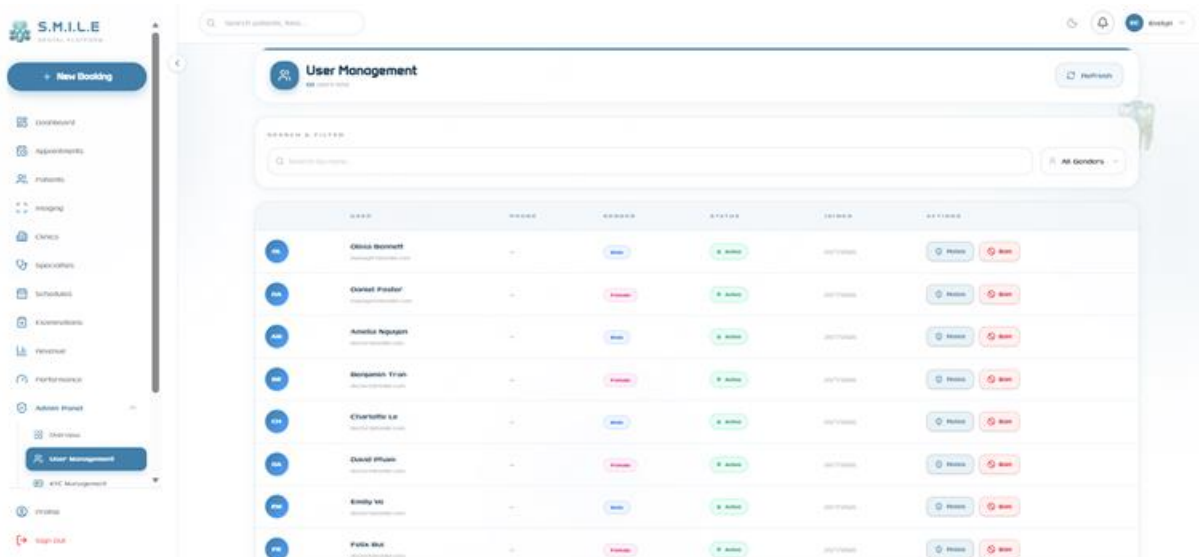
- [illegible]

k. Verify Identity KYC (Phone Number)

- **Function trigger:** An Admin, Nurse, Doctor, or Receptionist submits identity documents, accepts required consent, and verifies a phone code.
- **Function description:** The function stores identity evidence securely, may extract document fields through OCR, and advances the verification status for review.
- **Screen layout:** There is no separate screen listed. Identity type and number, document images, selfie, notes, consent, phone code, status, and results appear in the identity verification flow.
- **Function Details:** The phone code must be valid and unexpired. Consent is required, and each image must use an allowed format and size. Valid files are stored with AES encryption and the request moves to pending review. A duplicate pending request, invalid consent, invalid file, or wrong code is rejected. Approval changes status to verified, while rejection displays the reviewer reason.

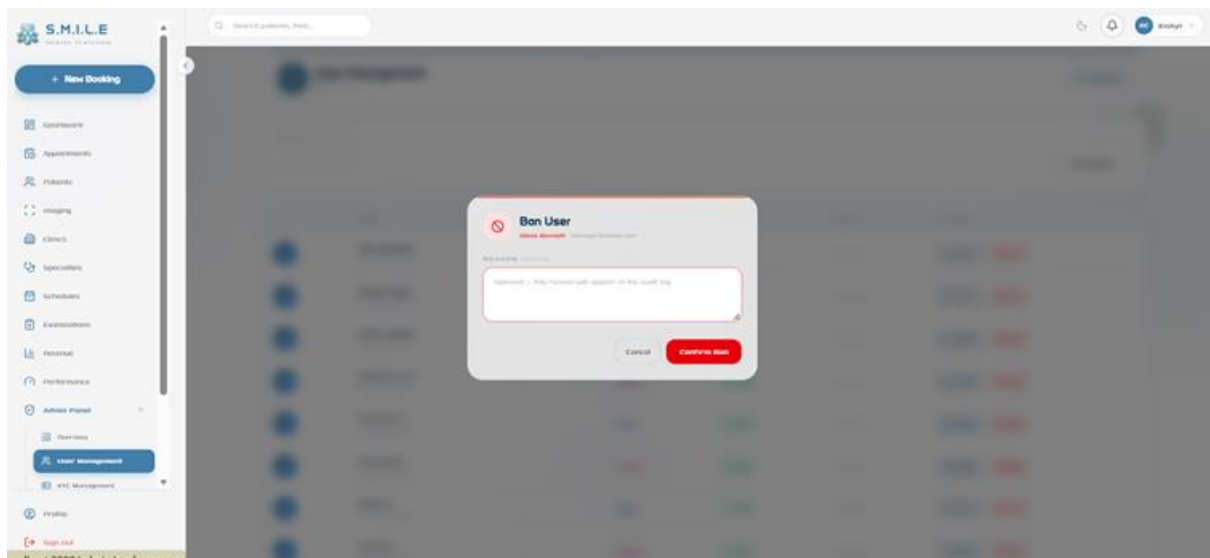
1. View User List

- **Function trigger:** An Admin opens User Management Page and applies optional filters or pagination.
- **Function description:** The function returns matching registered user profiles for administration.
- **Screen layout:** User Management Page displays a searchable user list with profile, role, verification, and account status information, filters, pagination, and user actions.
- **Function Details:** The service validates administrative access and returns the requested page of users. Matching records appear in the list, and no matches show an empty result. Missing permission or a failed request shows the documented error.



m. Lock/Ban User Account

- **Function trigger:** An Admin selects a user on User Management Page or User Detail and confirms the lock action.
- **Function description:** The function locks an account that violates policy, ends active sessions, and records the action.
- **Screen layout:** The user administration interface shows account status, a lock action, confirmation, and a result message.
- **Function Details:** The system validates administrative permission and the target account. A valid action changes status to locked, forces logout of active sessions, writes an audit entry, and shows success. Invalid targets or missing permission leave the account unchanged and show an error.



n. Unlock/Unban User Account

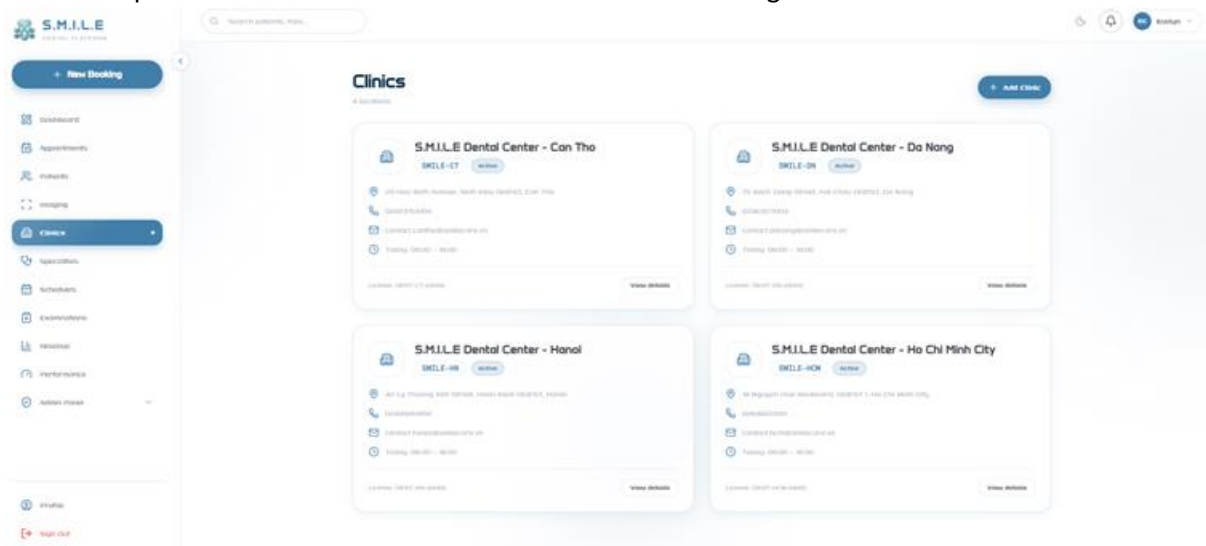
- **Function trigger:** An Admin selects a locked user on User Management Page or User Detail and confirms the unlock action.
- **Function description:** The function restores access to a locked account and records the administrative action.
- **Screen layout:** The user administration interface shows the locked status, an unlock action, confirmation, and a result message.

- **Function Details:** The system validates administrative permission and the target account. A valid action changes the account to active, writes an audit entry, and shows success. An invalid target or missing permission leaves the status unchanged and shows an error.

3.4 Clinic Management

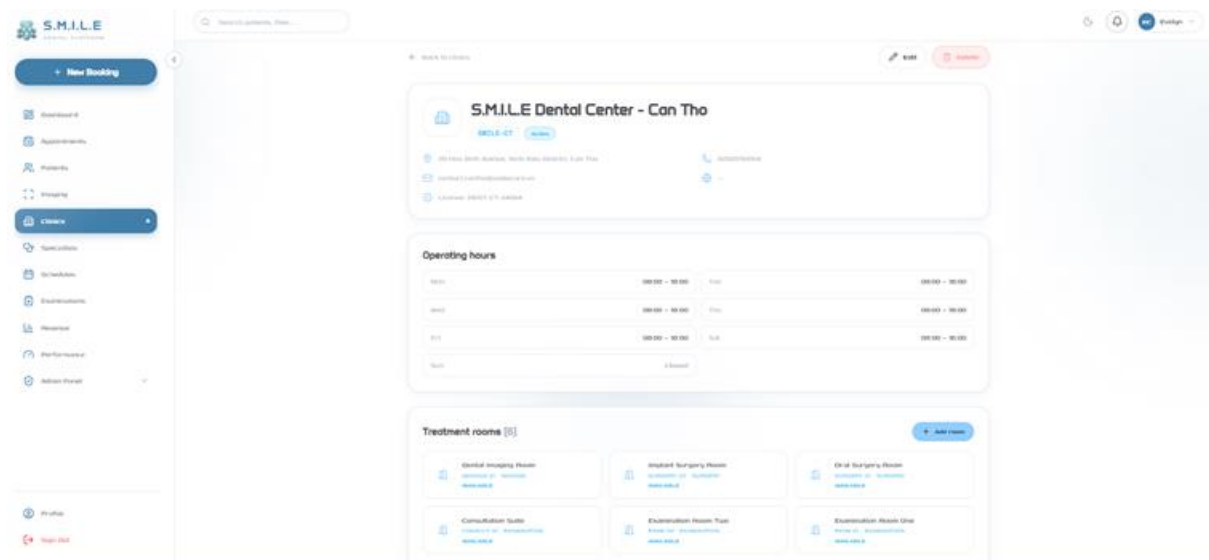
a. View Clinic Information

- **Function trigger:** An Admin, Nurse, Doctor, Receptionist, Patient, or Guest opens Clinic Page.
- **Function description:** The function provides the publicly readable list of clinics in the network.
- **Screen layout:** Clinic Page displays clinic name, address, contact data, operating hours, status, logo, and an action to open details.
- **Function Details:** Authentication is not required. The system returns available clinic records. Matching clinics appear on the page, while no records show an empty state and a failed request shows the standard connection or server message.



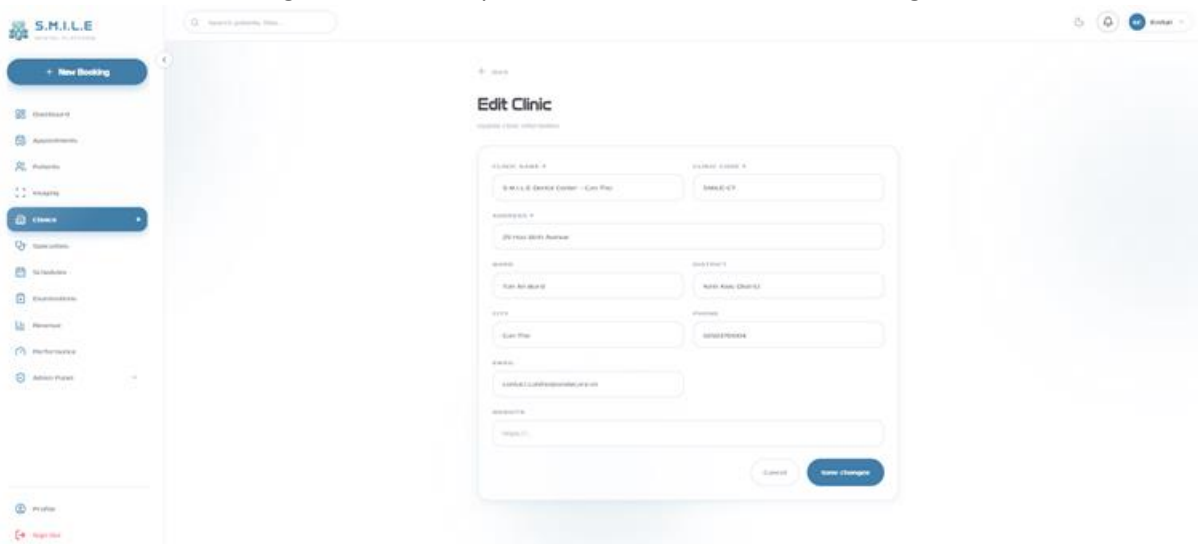
b. View Clinic Details

- **Function trigger:** An Admin, Nurse, Doctor, Receptionist, Patient, or Guest selects a clinic from Clinic Page.
- **Function description:** The function returns the selected clinic address, contact details, opening hours, and offered services.
- **Screen layout:** Clinic Detail displays the clinic profile, address, phone, email, website, operating hours, status, license data, logo, and service information.
- **Function Details:** The system validates the selected clinic identifier and returns its stored details. A valid clinic opens its profile. A missing clinic or failed request shows an empty or standard error result.



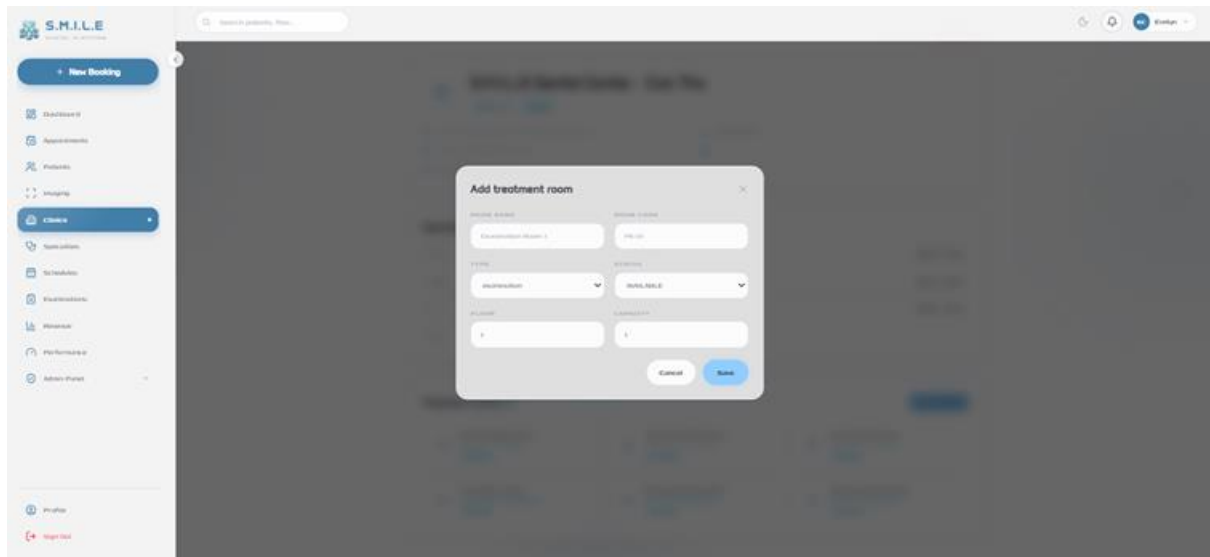
c. Update Clinic Information

- **Function trigger:** An Admin opens Update Clinic from Clinic Detail, edits information, and saves.
- **Function description:** The function maintains clinic profile and configuration data.
- **Screen layout:** Update Clinic contains fields for name, code, address, contact data, website, logo, operating hours, status, license number, and license expiry, with save and cancel actions.
- **Function Details:** The system validates required values, contact formats, field lengths, and the unique clinic code before saving. Valid data updates the clinic and shows success. Invalid values or missing administrative permission leave the record unchanged and show an error.



d. Add Treatment Room

- **Function trigger:** An Admin selects the room creation action for a clinic and submits the room form.
- **Function description:** The function registers a treatment room within the selected clinic.
- **Screen layout:** The Add Treatment Room dialog contains clinic, room name, room code, room type, floor, capacity, status, save and cancel controls, and a result message.
- **Function Details:** Room code must be unique within the clinic, and required values must be valid. Successful input creates the room and shows success on Treatment Room. A duplicate code, invalid data, or missing permission prevents creation and shows the relevant error.

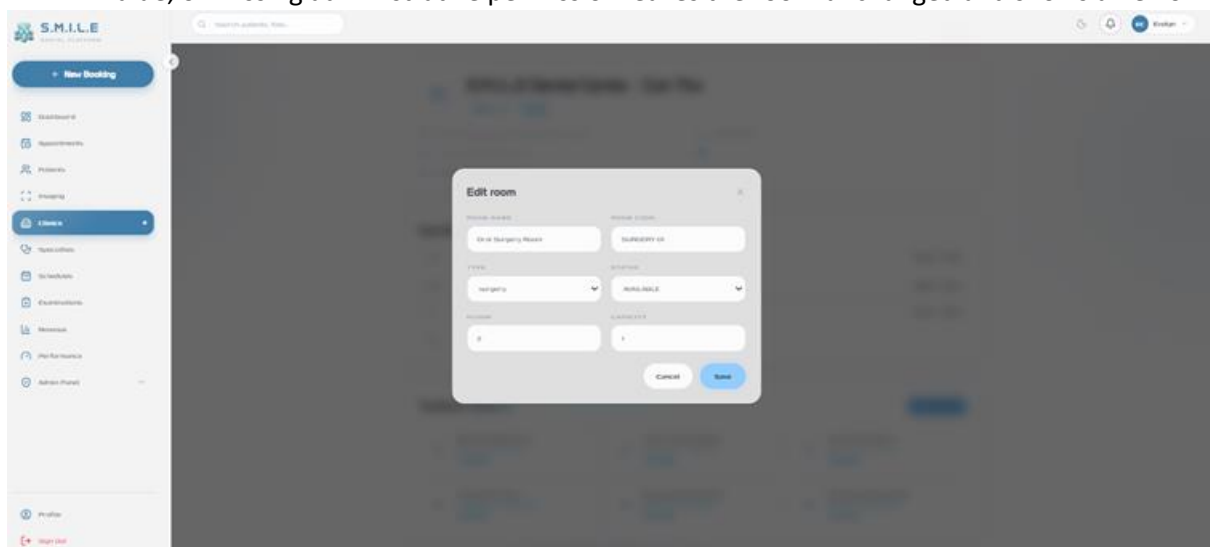


e. View Treatment Room

- **Function trigger:** An Admin, Doctor, Nurse, or Receptionist opens Treatment Room for a selected clinic.
- **Function description:** The function lists clinic rooms with their type and current status.
- **Screen layout:** Treatment Room displays room name, code, type, floor, capacity, equipment summary, status, and permitted management actions.
- **Function Details:** The system validates clinic access and returns its room records. Matching rooms appear in the inventory, and no rooms show an empty state. Missing permission or a request failure shows the documented error.

f. Update Treatment Room

- **Function trigger:** An Admin selects a room from Treatment Room, edits its data, and saves.
- **Function description:** The function changes room name, type, availability, and other stored room details.
- **Screen layout:** Update Treatment Room contains the existing room fields, save and cancel actions, inline validation, and a result message.
- **Function Details:** The system validates required values and preserves room code uniqueness within the clinic. Valid changes update the room and show success. A duplicate code, invalid value, or missing administrative permission leaves the room unchanged and shows an error.



g. Delete Treatment Room

- **Function trigger:** An Admin selects Delete for a room and confirms the action.
- **Function description:** The function removes a room that is no longer used.
- **Screen layout:** Delete Treatment Room appears as a confirmation dialog over Treatment Room, followed by a result message in the room inventory.
- **Function Details:** The system checks whether schedules or appointments still reference the room. An unreferenced room is deleted and success appears in the inventory. Active dependencies prevent deletion, retain the room, and show the documented error.

3.5 Schedule Management

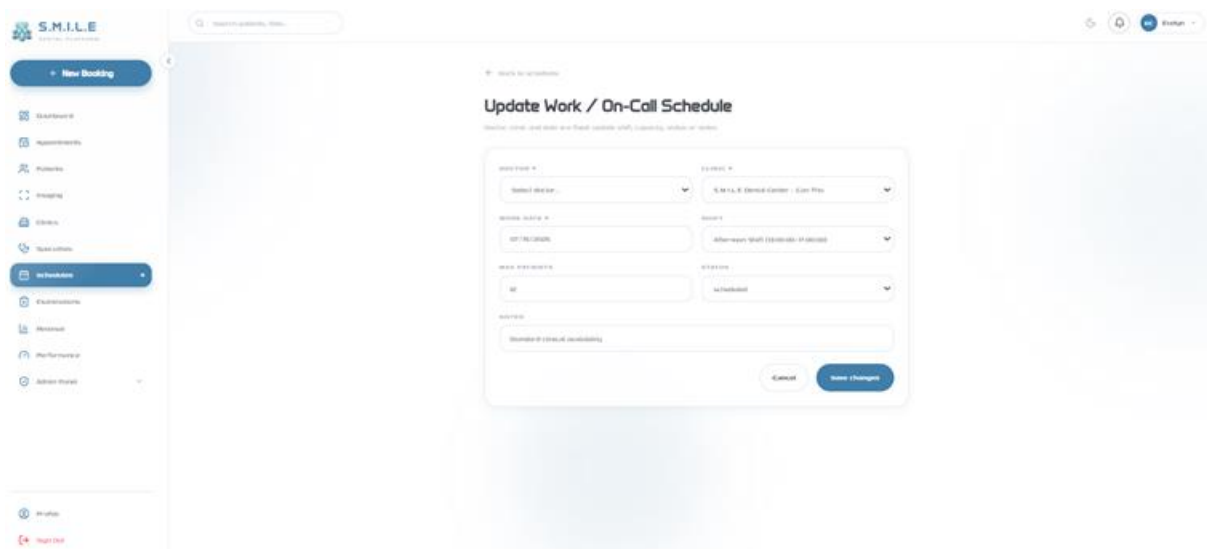
a. Create Work Schedule - On-Call Schedule

- **Function trigger:** An Admin selects Create Schedule from Schedule Page and submits a doctor shift assignment.
- **Function description:** The function creates a work schedule for a doctor, clinic, date, shift, and treatment room.
- **Screen layout:** Create Schedule contains doctor, clinic, work date, shift, room, maximum patients, status, notes, save and cancel actions.
- **Function Details:** A room is required. The doctor, date, and shift combination must be unique, and assignments for the same doctor must not overlap. Valid input creates the schedule and shows success on Schedule Page. Missing room, duplicate shift, overlap, or invalid data prevents creation and shows the relevant error.

The screenshot displays the S.M.I.L.E. system interface. On the left is a sidebar with navigation links: Dashboard, Appointments, Patients, Imaging, Clinics, Specialties, Schedules (highlighted), Examinations, Review, Performance, and Admin Panel. The main content area is titled 'Create Work / On-Call Schedule' with a subtitle 'Assign a doctor to a clinic shift on a given date'. The form contains several fields: 'Select doctor' (dropdown), 'Select clinic' (dropdown), 'Work date' (calendar icon), 'Shift' (dropdown with 'No shift' selected), 'Max patients' (input field with '20'), 'Status' (dropdown with 'Scheduled' selected), and 'Notes' (text area with 'Optional room'). At the bottom right are 'Cancel' and 'Create schedule' buttons. The top of the interface shows a search bar, user profile, and 'Logout' button.

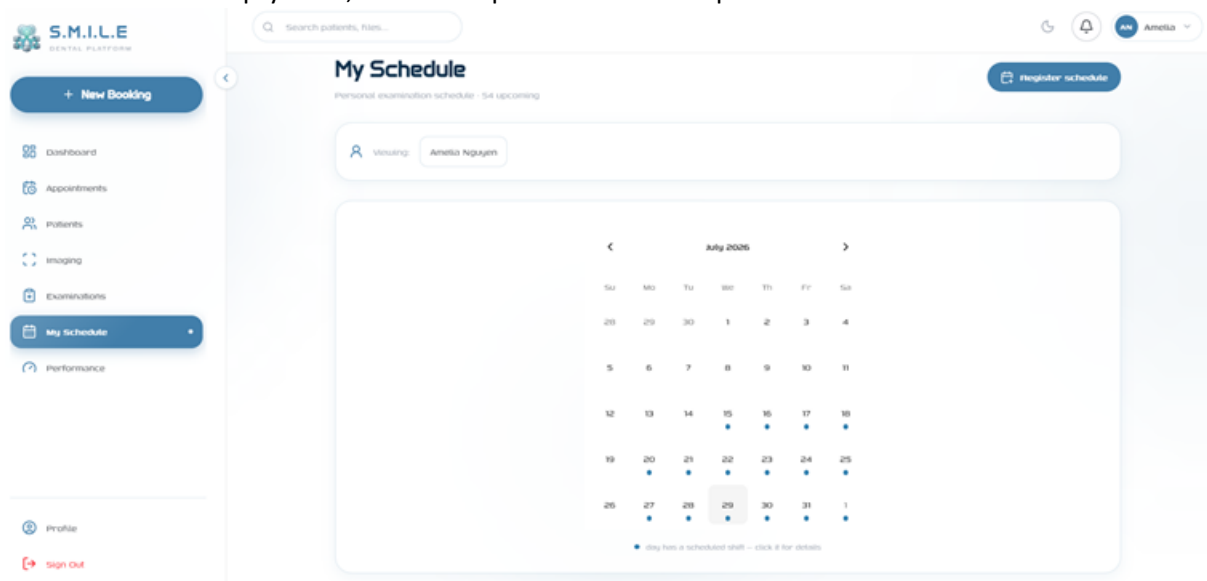
b. Update Work Schedule - On-Call Schedule

- **Function trigger:** An Admin selects an existing entry on Schedule Page, edits it, and saves.
- **Function description:** The function changes shift, room, capacity, status, or notes for an existing schedule of a doctor.
- **Screen layout:** Update Schedule contains the current doctor, clinic, date, shift, room, maximum patients, status, notes, save and cancel actions.
- **Function Details:** The schedule must still allow changes, the user making the change must be recorded, a room is required, and shifts of the doctor must not overlap. Valid changes are saved with old and new values in the schedule change log and show success. Invalid state, missing room, or overlap leaves the schedule unchanged and shows an error.



c. View Personal Schedule – Examination

- **Function trigger:** A Doctor opens Schedule Page and selects a date period.
- **Function description:** The function returns the assigned examination shifts of the authenticated Doctor.
- **Screen layout:** Schedule Page displays work date, clinic, shift, treatment room, capacity, status, notes, and period controls.
- **Function Details:** The system identifies the Doctor from the session and restricts results to that Doctor over the selected period. Assigned shifts appear on the page. No assignments show an empty state, while an expired session or request failure shows the documented error.



d. Register Personal Schedule – Examination

- **Function trigger:** A Doctor selects Request Schedule Change and submits a leave or change request.
- **Function description:** The function records a personal schedule request with a date range and reason for approval.
- **Screen layout:** Request Schedule Change contains start date, end date, reason, submit and cancel actions, request status, and result messages.

- Function Details:** The end date must follow the start date, and the request must not overlap another pending or approved request. Valid input is stored with pending status and shows that approval is awaited. An invalid date range or overlap prevents submission and shows an inline error.

e. Update Personal Schedule – Examination

- Function trigger:** A Doctor selects an owned personal schedule entry and submits changes.
- Function description:** The function amends the personal examination schedule of the Doctor and records the change.
- Screen layout:** There is no separate personal update screen listed. The owned entry, editable schedule data, save controls, and the result appear on Schedule Page.
- Function Details:** The system validates ownership, schedule values, and overlap rules. Changes that affect existing bookings require approval. A valid change is applied and recorded in the schedule change log. Invalid data, overlap, or a required approval state prevents direct completion and shows the relevant result.

f. Notify Schedule Change

- Function trigger:** The system triggers the function after a schedule change affects staff.
- Function description:** The function creates schedule change alerts through the notification service for affected staff, including Doctors.
- Screen layout:** There is no separate screen. The result appears as a notification for affected staff and as a success message on the current schedule interface.
- Function Details:** The service receives the change and recipient data and sends the schedule alert. Successful delivery notifies affected staff. A connection or service failure returns the standard error and does not report successful delivery.

g. Notify Shift Change

- Function trigger:** A Doctor hands a shift to a colleague and confirms the transfer.
- Function description:** The function reassigns the shift and informs the receiving Doctor.
- Screen layout:** There is no separate screen. The transfer action appears on the current schedule interface, and the result appears as a notification to the receiving Doctor and a success message to the initiating Doctor.
- Function Details:** The system validates the shift, receiving Doctor, and assignment rules before reassignment. A valid transfer updates the schedule and sends the alert. An invalid target,

conflicting shift, or notification service failure shows the relevant error and does not report successful notification.

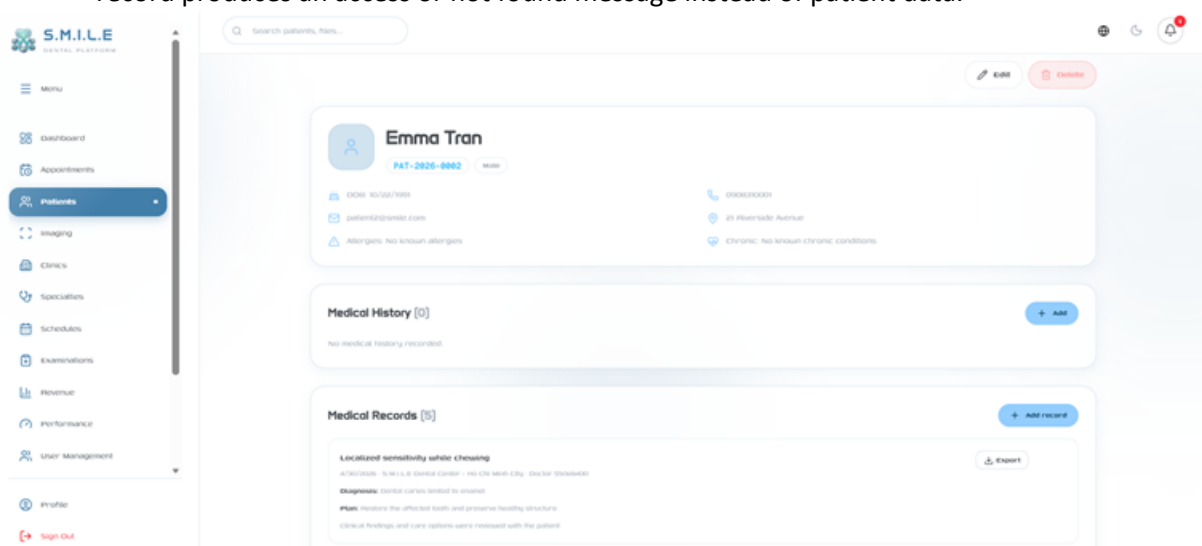
3.6 Patient Management

a. Add Patient Profile

- **Function trigger:** A doctor or receptionist selects Add Patient Profile to register a patient.
- **Function description:** The function creates the patient master record with demographic, contact, allergy, chronic disease, emergency contact, and insurance information.
- **Screen layout:** The Add Patient Profile screen presents input fields for identity, date of birth, gender, phone, email, address, emergency contact, blood type, allergies, chronic diseases, and insurance, with Save and Cancel controls.
- **Function Details:** The patient creation service validates the entered data and requires a unique patient code. A valid submission saves the profile and shows a success message on the Patient Profile screen. A duplicate patient code or invalid data is rejected and no patient record is created.

b. View Patient Profile

- **Function trigger:** A doctor, receptionist, nurse, or patient opens a patient profile.
- **Function description:** The function displays the patient master record for care and contact management while limiting patients to their own record and records they represent.
- **Screen layout:** The Patient Profile screen shows the patient code, demographic details, contact details, address, emergency contact, blood type, allergies, chronic diseases, and insurance information.
- **Function Details:** The system identifies the requested patient and checks access ownership. Authorized users receive the stored profile. An unauthorized patient or an unknown patient record produces an access or not found message instead of patient data.



c. Update Patient Profile

- **Function trigger:** A receptionist or patient selects Edit on an accessible patient profile.
- **Function description:** The function updates patient demographic or contact information while enforcing ownership.

- **Screen layout:** The Update Patient Profile screen loads the current demographic, contact, address, emergency contact, and insurance values in editable fields with Save and Cancel controls.
- **Function Details:** The system checks that the actor may edit the patient, validates changed values, and updates the record and updated time. Valid changes appear on the Patient Profile screen. Invalid data or failed ownership checks leave the record unchanged and show an error message.

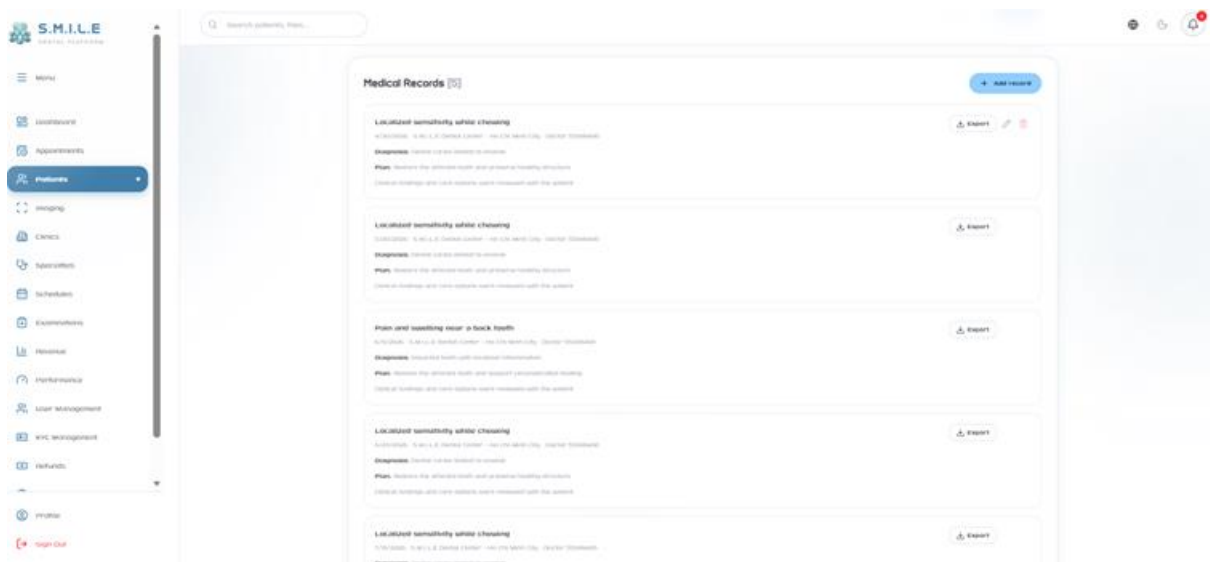
c. Add Medical History

- **Function trigger:** A doctor adds a medical history entry from a patient profile.
- **Function description:** The function records a patient condition and its clinical history for later care decisions.
- **Screen layout:** The medical history entry area contains condition name, condition type, diagnosed date, treatment summary, notes, Save, and Cancel controls.
- **Function Details:** The system links the entry to the selected patient and stores the condition as chronic, acute, or past with its date, treatment, and notes. A valid entry appears in the medical history area of the Patient Profile screen. Invalid data or an unknown patient prevents saving and shows an error message.

The screenshot displays the S.M.I.L.E. Patient Profile interface. On the left is a sidebar menu with options: Menu, Dashboard, Appointments, Patients (selected), Imaging, Clinics, Specialists, Schedules, Examinations, Revenue, Performance, User Management, XMC Management, Refunds, Profile, and Sign Out. The main content area has a search bar at the top. Below it, the 'Medical History' section shows 'No medical history recorded' and an '+ Add' button. A modal form titled 'Add medical history' is open, containing fields for 'CONDITION NAME' (with 'hypertension' entered), 'CONDITION TYPE' (with 'CHRONIC / ACUTE' selected), 'DIAGNOSED DATE' (with 'MM/DD/YYYY' placeholder), 'TREATMENT', and 'NOTES'. At the bottom of the modal are 'Cancel' and 'Save' buttons. Below the modal, the 'Medical Records' section shows '0' records and an '+ ADD RECORD' button.

c. View Medical Record

- **Function trigger:** An admin, doctor, receptionist, or patient opens a medical record.
- **Function description:** The function presents the selected visit record and its version history while limiting patients to their own records.
- **Screen layout:** The Medical Record screen shows patient, appointment, clinic, doctor, visit date, chief complaint, diagnosis, treatment plan, notes, record status, finalization data, and version history.
- **Function Details:** The system checks record access and returns the record and available versions. Authorized actors see the current clinical content and status. An unauthorized patient or missing record receives an access or not found message and no clinical content.

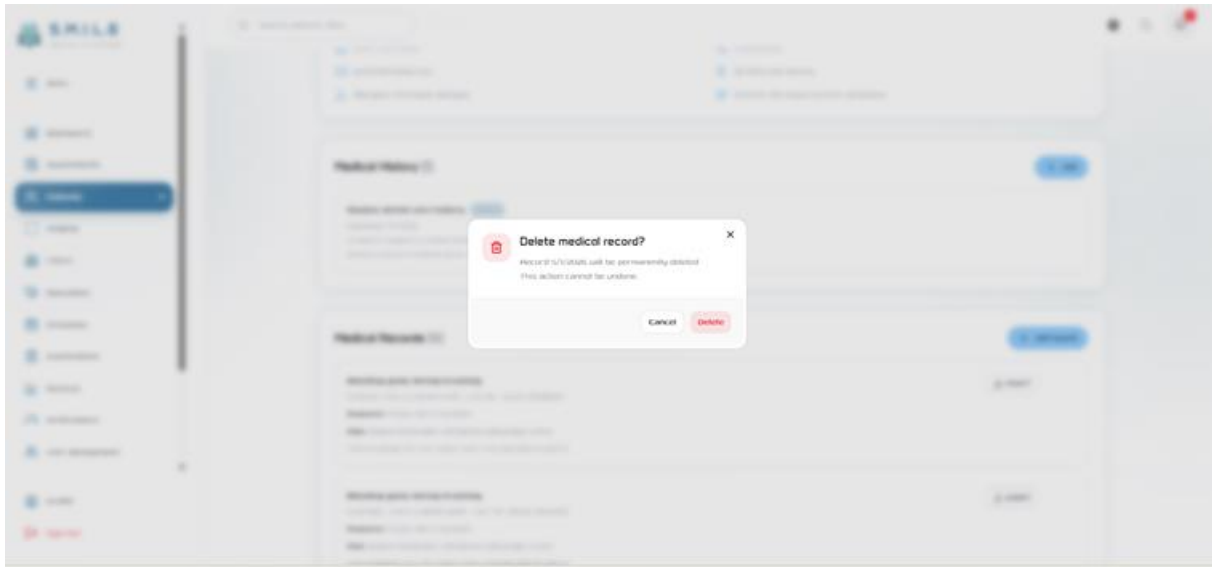


c. Update Medical Record

- **Function trigger:** A doctor selects Edit on a draft medical record.
- **Function description:** The function updates complaint, diagnosis, treatment plan, or notes and preserves a version snapshot.
- **Screen layout:** The Update Medical Record screen provides editable clinical fields, an update reason field, current status, Save, and Cancel controls.
- **Function Details:** The system verifies that the record is still draft, validates the clinical changes, actor, and reason, then stores a new version snapshot. Valid changes appear on the Medical Record screen and in version history. A finalized record cannot be edited and requires an amendment instead. Invalid data leaves the existing version unchanged.

d. Delete Medical Record

- **Function trigger:** A doctor chooses Delete for a medical record created in error and confirms the action.
- **Function description:** The function removes the erroneous medical record and captures the deletion in the audit trail.
- **Screen layout:** The Medical Record screen provides a Delete control and a confirmation dialog that identifies the selected record.
- **Function Details:** The system verifies the record and doctor authorization before deletion. A successful deletion removes the record and returns the doctor to the patient medical record list with confirmation. A missing record, failed authorization, or failed deletion keeps the record and shows an error message.



e. Add Treatment Profile

- **Function trigger:** A doctor selects Add Treatment Record for a patient medical record.
- **Function description:** The function records treatment performed on a tooth or region and links it to the patient and medical record.
- **Screen layout:** The Add Treatment Record screen contains patient and medical record context, tooth or region selection, treatment details, related visit or procedure information, Save, and Cancel controls.
- **Function Details:** The system validates the patient, medical record, tooth or region, and entered treatment details before saving. A valid entry appears in the Treatment Record area for the patient. Invalid data or a missing related record prevents creation and shows an error message.

f. View Treatment Profile

- **Function trigger:** A doctor or patient opens treatment history for a patient, medical record, or tooth number.
- **Function description:** The function returns matching treatment entries so the actor can review performed care.
- **Screen layout:** The Treatment Record screen provides patient, medical record, and tooth filters and displays the matching treatment history and related details.
- **Function Details:** The system applies the selected filters and checks patient access before returning treatment entries. Matching records appear in the Treatment Record screen. Unauthorized access is rejected, while no matches produce an empty result message.



g. Update Treatment Profile

- **Function trigger:** A doctor selects Edit on an existing treatment entry.
- **Function description:** The function corrects or completes treatment details as care evolves.
- **Screen layout:** The Update Treatment Record screen loads the current patient, medical record, tooth or region, and treatment details with Save and Cancel controls.
- **Function Details:** The system verifies the doctor and related records, validates the changed treatment data, and saves the update. Valid changes appear in the Treatment Record screen. Invalid data, a missing record, or failed authorization leaves the entry unchanged and shows an error message.

h. Export Medical Record Profile

- **Function trigger:** A doctor or patient selects Export for an accessible medical record.
- **Function description:** The function creates a document for record transfer or long term storage.
- **Screen layout:** The Medical Record screen provides an Export control and shows export preparation status and the document when ready.
- **Function Details:** There is no separate screen. The system checks record access, creates an export job, and returns the generated document on the Medical Record screen when processing finishes. Unauthorized access or export failure shows an error and no document is released.

3.7 Appointment Management

a. Create Appointment at Facility

- **Function trigger:** A receptionist or patient selects a clinic and starts a new appointment.
- **Function description:** The function books an appointment at a chosen clinic and available time slot.
- **Screen layout:** The Create Appointment screen presents patient, clinic, service, doctor or room when available, date, time, duration, appointment type, complaint, notes, and booking controls.
- **Function Details:** The system validates the patient and clinic, checks doctor, room, and patient slot conflicts, then creates the appointment with scheduled status. A valid booking appears on

the Appointment screen. An unavailable or overlapping slot is rejected and the actor must choose another time.

The image displays two screenshots of the 'Book an Appointment' screen in the S.M.I.L.E. Dental Platform. The top screenshot shows the initial selection screen with four options: 'At Facility' (selected), 'By Specialty', 'By Doctor', and 'Outside Hours'. The bottom screenshot shows the 'At Facility' selection screen with a list of four dental centers: 'S.M.I.L.E. Dental Center - Can Tho', 'S.M.I.L.E. Dental Center - Da Nang', 'S.M.I.L.E. Dental Center - Hanoi', and 'S.M.I.L.E. Dental Center - Ho Chi Minh City'. The left sidebar contains a menu with options: Home, Dashboard, Appointments, Patients, Imaging, Clinics, Specialties, Schedules, Examinations, Revenue, Performance, User Management, KMC Management, Referrals, Profile, and Sign Out.

Book an Appointment

Follow the steps to choose how you'd like to schedule, fill the details, and confirm.

1 Method 2 Clinic 3 Schedule 4 Patient & Review

At Facility
Book a visit with a doctor at a clinic

By Specialty
Get the clinic, choose a specialist

By Doctor
Choose a specific doctor

Outside Hours
Booked on other hours appointment

← Back Continue →

Prefer chat? Try the Booking Assistant
Describe what you need and let the assistant find a slot

Book an Appointment

Follow the steps to choose how you'd like to schedule, fill the details, and confirm.

1 Method 2 Clinic 3 Schedule 4 Patient & Review

A specific doctor will choose get -- reception will assign your doctor, times, and service when you check in at the clinic.

CLINIC 4

Search clinic by name or id...

S.M.I.L.E. Dental Center - Can Tho
20 Tran Binh Khanh Road, Binh Thuan, Can Tho

S.M.I.L.E. Dental Center - Da Nang
20 Tran Binh Khanh Road, Binh Thuan, Da Nang

S.M.I.L.E. Dental Center - Hanoi
20 Tran Binh Khanh Road, Binh Thuan, Hanoi

S.M.I.L.E. Dental Center - Ho Chi Minh City
20 Tran Binh Khanh Road, Binh Thuan, Ho Chi Minh City

← Back Continue →

Prefer chat? Try the Booking Assistant
Describe what you need and let the assistant find a slot

S.M.I.L.E. DENTAL PLATFORM

Menu: Dashboard, **Appointments**, Patients, Imaging, Clinics, Specialists, Schedules, Examinations, Revenue, Performance, User Management, KVC Management, Refunds, Profile, Sign Out

Search patients, files...

Book an Appointment

Follow the steps to choose how you'd like to schedule, fill the details, and confirm.

1. Select ✓ 2. Clinic ✓ 3. **Schedule** 4. Patient & Review

A specific doctor isn't chosen yet – reception will assign your doctor, room, and service when you check in at the clinic.

SERVICE (OPTIONAL)
Panoramic Dental Radiograph

DATE
Sat, 01 Aug 2026

TIME
12:00

← BACK Continue →

Prefer chat? Try the Booking Assistant

Search patients, files...

Book an Appointment

Follow the steps to choose how you'd like to schedule, fill the details, and confirm.

1. Select ✓ 2. Clinic ✓ 3. **Schedule** 4. Patient & Review

A specific doctor isn't chosen yet – reception will assign your doctor, room, and service when you check in at the clinic.

PATIENT
Search patient by name or code...
Additional notes
Select patient...

WORKS

ALL VIEW

Doctor	At Facility	Patient
Clinic	S.M.I.L.E. Dental Center - Can Tho	Service
Date	2026-08-01	Time
		12:00

← BACK Confirm Booking

b. Create Appointment by Specialty

- **Function trigger:** A receptionist or patient selects a dental specialty and requests an appointment.
- **Function description:** The function books by specialty and allocates an available doctor in that specialty.
- **Screen layout:** The Create Appointment screen shows specialty, clinic, patient, date, available time, service, complaint, notes, and booking controls.
- **Function Details:** The system validates the selected specialty and clinic, finds an available doctor in that specialty, and applies doctor, room, and patient conflict checks. Success creates a scheduled appointment visible on the Appointment screen. If no doctor or slot is available, the request is rejected with an availability message.

S.M.I.L.E. Book an Appointment

Follow the steps to choose how you'd like to schedule, fill the details, and confirm.

1. Method | 2. Specialty | 3. Clinic | 4. Patient & Member

At Facility
Book a visit with a doctor at a clinic.

By Doctor
Choose a specific doctor.

By Specialty
Get the latest updates & specialties.

Outside Hours
Request an after-hours appointment.

← BACK | Continue →

Prefer chat? Try the Booking Assistant
Describe what you need and let the assistant find a slot.

S.M.I.L.E. Book an Appointment

Follow the steps to choose how you'd like to schedule, fill the details, and confirm.

1. Method | 2. Specialty | 3. Clinic | 4. Patient & Member

SPECIALTY

Oral Radiology

← BACK | Continue →

Prefer chat? Try the Booking Assistant
Describe what you need and let the assistant find a slot.

S.M.I.L.E. Book an Appointment

Follow the steps to choose how you'd like to schedule, fill the details, and confirm.

1. Method | 2. Specialty | 3. Clinic | 4. Patient & Member

PATIENT

Search patient by name or ID#

Hedy Nguyen (DENT-0006)

CHOOSE COMPLAINT

ORL

APPOINTMENT SLOTS

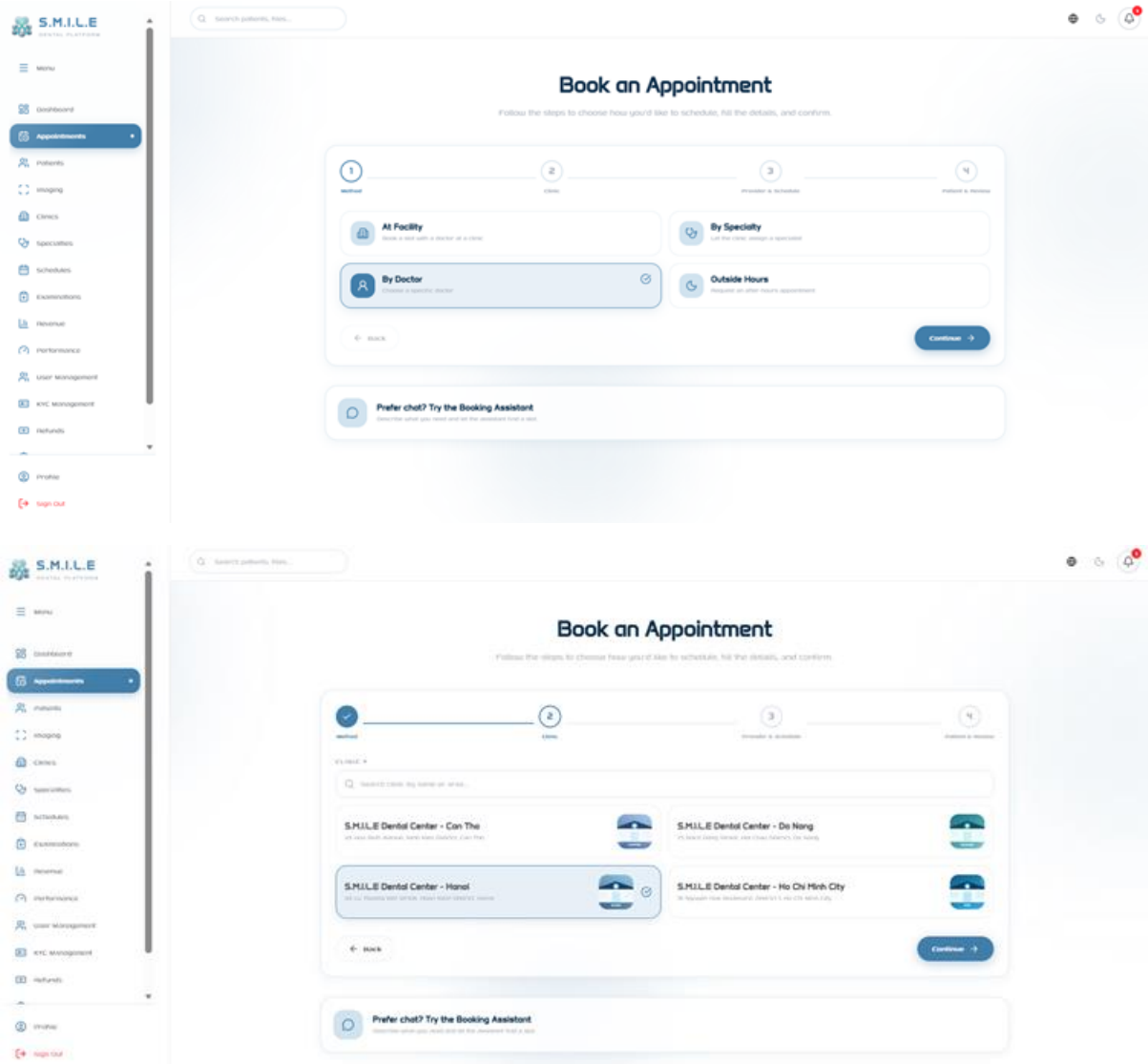
DATE	TIME	STATUS	DOCTOR
01/15/2025	09:00 AM	Available	Hedy Nguyen
01/15/2025	10:00 AM	Booked	Hedy Nguyen
01/15/2025	11:00 AM	Available	Hedy Nguyen
01/15/2025	12:00 PM	Available	Hedy Nguyen
01/15/2025	01:00 PM	Available	Hedy Nguyen
01/15/2025	02:00 PM	Available	Hedy Nguyen
01/15/2025	03:00 PM	Available	Hedy Nguyen
01/15/2025	04:00 PM	Available	Hedy Nguyen
01/15/2025	05:00 PM	Available	Hedy Nguyen

← BACK | Confirm Booking

c. Create Appointment by Doctor

- **Function trigger:** A receptionist or patient selects a specific doctor and starts a booking.
- **Function description:** The function creates an appointment with the chosen doctor after checking the schedule of that doctor.

- **Screen layout:** The Create Appointment screen shows doctor, clinic, patient, available date and time slots, service, duration, appointment type, complaint, notes, and booking controls.
- **Function Details:** The availability service returns slots for the doctor and day. The system validates the selection and checks doctor, room, and patient conflicts before creating a scheduled appointment. Success appears on the Appointment screen. An unavailable doctor or conflicting slot produces an error and no appointment.



d. Create Appointment Outside Working Hours

- **Function trigger:** A receptionist or patient requests a time outside published working hours.
- **Function description:** The function records an exceptional appointment request for approval.
- **Screen layout:** The Create Appointment screen accepts the patient, clinic, doctor or specialty, requested date and time, service, complaint, and notes.
- **Function Details:** The system detects that the requested time is outside working hours, checks required appointment data and conflicts, then flags the request as an exception for approval. The pending exception appears on the Appointment screen. Invalid data or a conflicting slot is rejected with an error message.

The screenshots show the 'Book an Appointment' form in the S.M.I.L.E system, progressing through four steps:

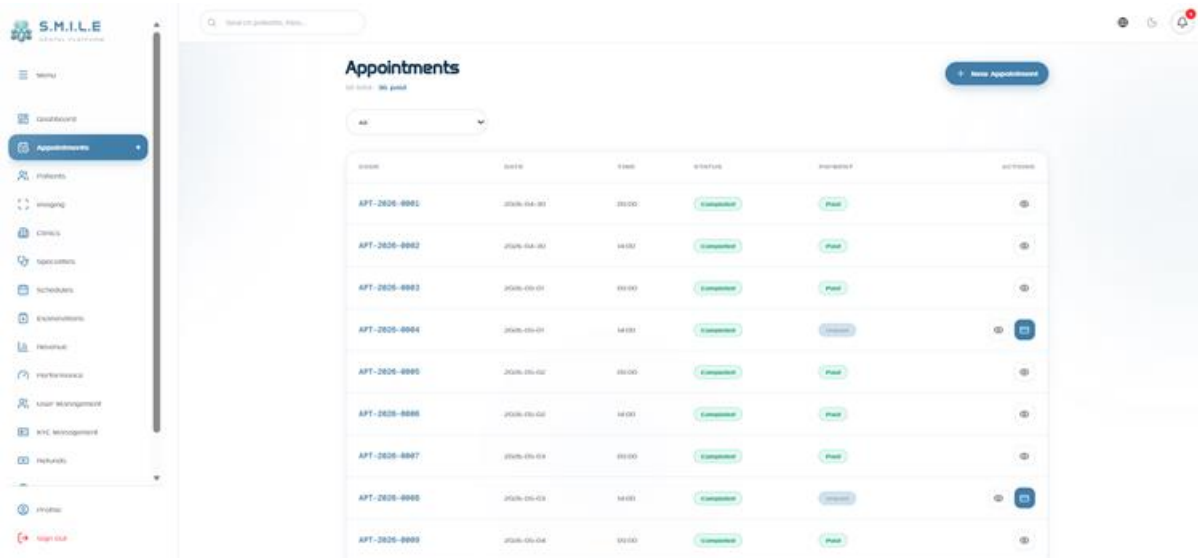
- Step 1: Clinic** - The user selects a clinic from a list. The list includes:
 - S.M.I.L.E Dental Center - Can Tho (40 Tran Anh Khanh Street, New City District, Can Tho)
 - S.M.I.L.E Dental Center - Da Nang (40 Tran Anh Khanh Street, New City District, Da Nang)
 - S.M.I.L.E Dental Center - Hanoi (40 Tran Anh Khanh Street, New City District, Hanoi)
 - S.M.I.L.E Dental Center - Ho Chi Minh City (40 Tran Anh Khanh Street, New City District, Ho Chi Minh City)
- Step 2: Specialty & Schedule** - The user selects a specialty and a service option.
 - Specialty: Pediatric Dentistry
 - Service (Optional): Fluoride Varnish Application
 - Date: Sat, 01 Aug 2020
 - Time: 10:00
- Step 3: Patient & Review** - The user enters patient information and reviews the booking details.
 - Patient Name: JACK LAM (090-123456789)
 - Notes (Reason for After Hours): ABC
 - Review Table:

Outside Hours?	Doctor	Specialty	Service	Date	Time
	JACK LAM	Pediatric Dentistry	Fluoride Varnish Application	2020-08-01	10:00
- Step 4: Confirm Booking** - The final step to confirm the appointment.

e. View Appointment

- Function trigger:** A doctor, receptionist, nurse, or patient opens the Appointment screen or applies appointment filters.

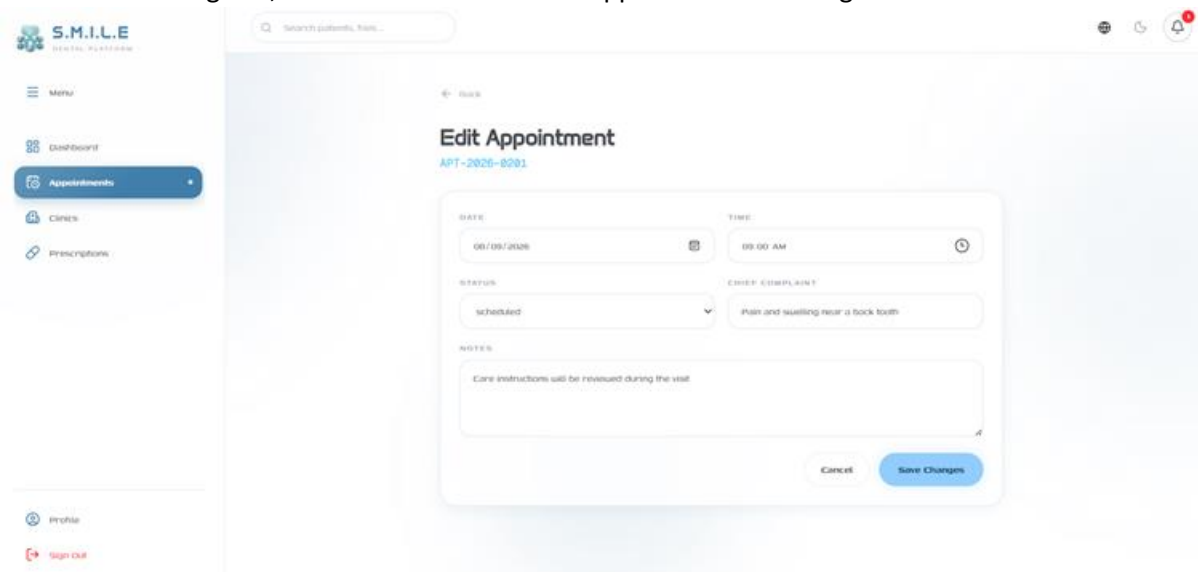
- **Function description:** The function lists appointments and their current state for operational review.
- **Screen layout:** The Appointment screen provides date, doctor, patient, and status filters and displays appointment code, clinic, room, service, date, time, type, complaint, and status.
- **Function Details:** The system applies the requested filters and returns matching appointments with current status. Authorized results appear in the list and can be opened for details. No matches produce an empty result, while invalid or unauthorized requests show an error.



ID	Date	Time	Status	Patient
APT-2020-0001	2020-04-01	09:00	Completed	Patient
APT-2020-0002	2020-04-01	14:00	Completed	Patient
APT-2020-0003	2020-04-01	09:00	Completed	Patient
APT-2020-0004	2020-04-01	14:00	Completed	Patient
APT-2020-0005	2020-04-02	09:00	Completed	Patient
APT-2020-0006	2020-04-02	14:00	Completed	Patient
APT-2020-0007	2020-04-03	09:00	Completed	Patient
APT-2020-0008	2020-04-03	14:00	Completed	Patient
APT-2020-0009	2020-04-04	09:00	Completed	Patient

f. Edit Appointment

- **Function trigger:** A doctor or receptionist selects Edit for an existing appointment.
- **Function description:** The function reschedules or amends appointment details and updates related parties.
- **Screen layout:** The Update Appointment screen loads patient, doctor, clinic, room, service, date, time, duration, type, complaint, notes, and status with Save and Cancel controls.
- **Function Details:** The system first requires a modifiable appointment status, then validates changed data, checks slot availability again, and applies the status rules. A valid update appears on the Appointment screen and related parties are notified. A terminal status, conflicting slot, or invalid data leaves the appointment unchanged and shows an error.



Edit Appointment
APT-2020-0001

DATE: 08/09/2020 TIME: 09:00 AM

STATUS: scheduled COMPLAINT: Pain and swelling near a tooth

NOTES: Care instructions will be reviewed during the visit.

Cancel Save Changes

g. Cancel Appointment

- **Function trigger:** A doctor, receptionist, or patient selects Cancel and enters a reason.
- **Function description:** The function cancels the appointment, releases its reserved slot, and records the cancellation actor and time.
- **Screen layout:** The Appointment screen provides a Cancel control and a confirmation dialog with the appointment summary and required cancellation reason.
- **Function Details:** The system validates actor access, appointment state, cutoff time, and fee policy. It stores the reason, canceller, and cancellation time, changes status to cancelled, and releases the slot. Success appears on the Appointment screen. A terminal state or failed authorization blocks cancellation, while cancellation inside the cutoff window requires fee confirmation.

h. Confirm Appointment

- **Function trigger:** A receptionist selects Confirm for a scheduled appointment.
- **Function description:** The function confirms the booking and locks the selected time slot.
- **Screen layout:** The Confirm Appointment screen or dialog shows the patient, clinic, doctor, date, time, service, and current status with Confirm and Cancel controls.
- **Function Details:** The system verifies receptionist access and the current appointment state before changing status to confirmed and locking the selected slot. Success appears on the Appointment screen. A terminal or otherwise invalid state prevents confirmation and shows an error.

i. Send Appointment Confirmation

- **Function trigger:** A receptionist selects Send Confirmation for a confirmed appointment.
- **Function description:** The function dispatches an appointment confirmation and records the delivery attempt.
- **Screen layout:** There is no separate screen. The action is available from the Appointment screen, where delivery status and any error appear.
- **Function Details:** The system verifies the appointment and recipient details, sends the confirmation notification, and logs the attempt. A successful delivery status appears on the Appointment screen. Missing contact data or delivery failure produces a failed result without changing the appointment.

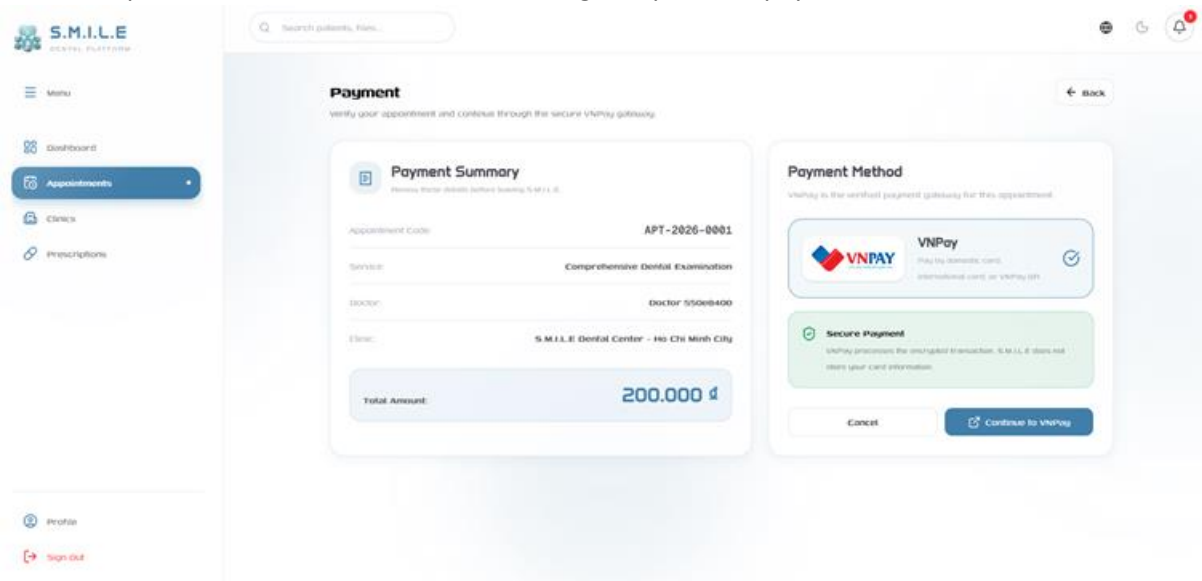
j. Send Appointment Reminder

- **Function trigger:** A receptionist sends a reminder, or the reminder service reaches a scheduled reminder time.
- **Function description:** The function sends an SMS, email, or push reminder before the appointment according to patient preference.
- **Screen layout:** There is no separate screen. Reminder status appears on the Appointment screen with the planned time, channel, sent time, and result.
- **Function Details:** The reminder service scans pending reminders, respects the patient preference, dispatches the selected channel, and records pending, sent, or failed status. Successful delivery appears in the appointment reminder information. A failed attempt is logged and may be retried.

k. Initiate Payment

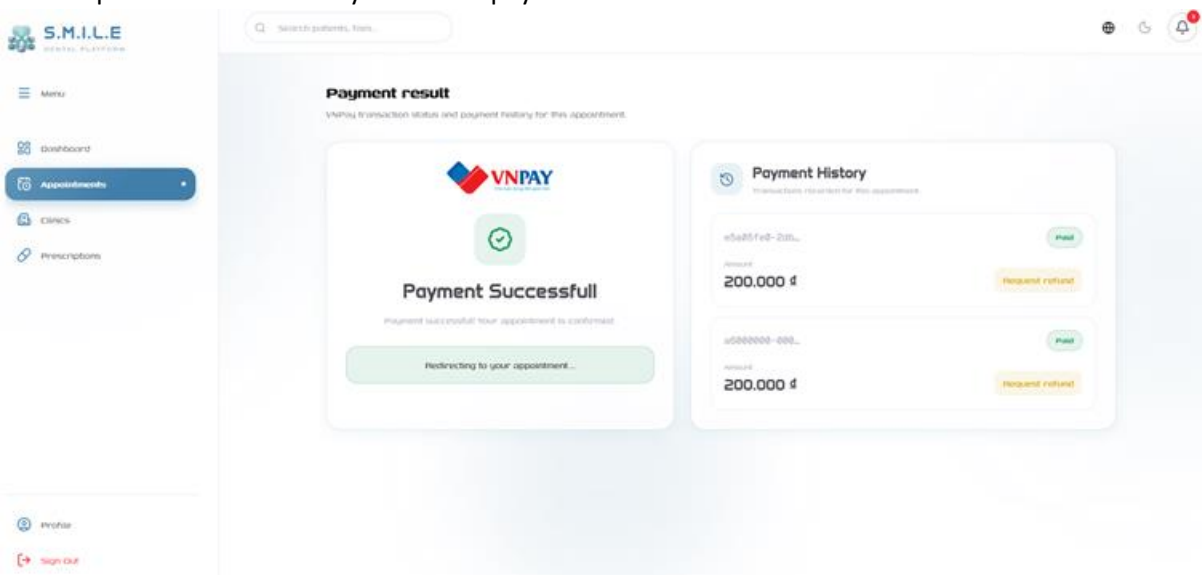
- **Function trigger:** A receptionist or patient selects Pay for an appointment.
- **Function description:** The function starts VNPay payment for the appointment.
- **Screen layout:** The Payment screen shows the appointment, patient, amount, currency, payment method, current status, and a control to continue to VNPay.

- Function Details:** The system validates the appointment and payment data, creates a pending payment with a unique transaction identifier, and builds a signed VNPay checkout URL. Success redirects the actor to VNPay and retains pending status until callback processing. A duplicate transaction identifier or missing URL prevents payment and shows an error.



I. Confirm Payment / View Payment History

- Function trigger:** VNPay returns a callback after checkout, or an admin or patient opens Payment History.
- Function description:** The function reconciles payment status and displays past transactions with amount, provider, and refund state.
- Screen layout:** The Payment History screen provides a status filter and shows appointment, amount, currency, method, transaction, payment status, paid time, and refund information.
- Function Details:** Payment confirmation has no separate screen. The system verifies the VNPay callback signature, stores the gateway response, and reconciles the result shown in Payment History. A valid callback marks payment completed or failed. A callback with an invalid signature cannot update payment status. Admins can view all matching history, while patients can view only their own payments.



m. Refund / Cancel Payment

- **Function trigger:** A patient requests a refund for an eligible paid appointment, then an admin approves or rejects it.
- **Function description:** The function creates and processes a refund request through the payment refund states.
- **Screen layout:** The Payment History screen shows the paid transaction, refund amount, reason, refund state, and request control. Admin review shows approval and rejection controls.
- **Function Details:** The system validates that the appointment is paid and eligible, records the refund request, and lets an admin approve or reject it. Approval creates the refund transaction and updates refund amount, reason, time, and payment history. An ineligible request or rejected review leaves the payment unrefunded and shows the outcome.

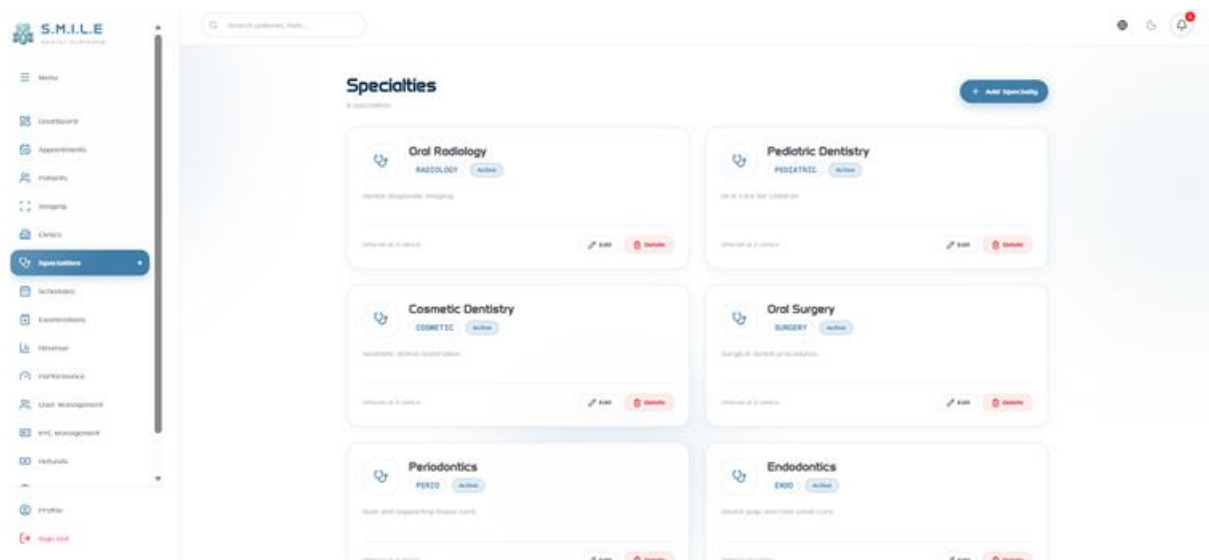
n. Chatbot Support for Booking

- **Function trigger:** A patient describes a booking need in natural language to the chatbot.
- **Function description:** The conversational agent extracts booking intent and details and calls appointment functions for the patient.
- **Screen layout:** There is no separate management screen. The conversation appears in the chatbot interface and any created booking appears on the Appointment screen.
- **Function Details:** The agent identifies clinic, specialty, doctor, date, time, and patient intent from the conversation, then uses the appointment endpoints and normal availability checks. A complete valid request creates or proposes a booking in the conversation and Appointment screen. Missing, unclear, or conflicting details cause a clarification or availability message instead.

3.8 Service Catalog Management

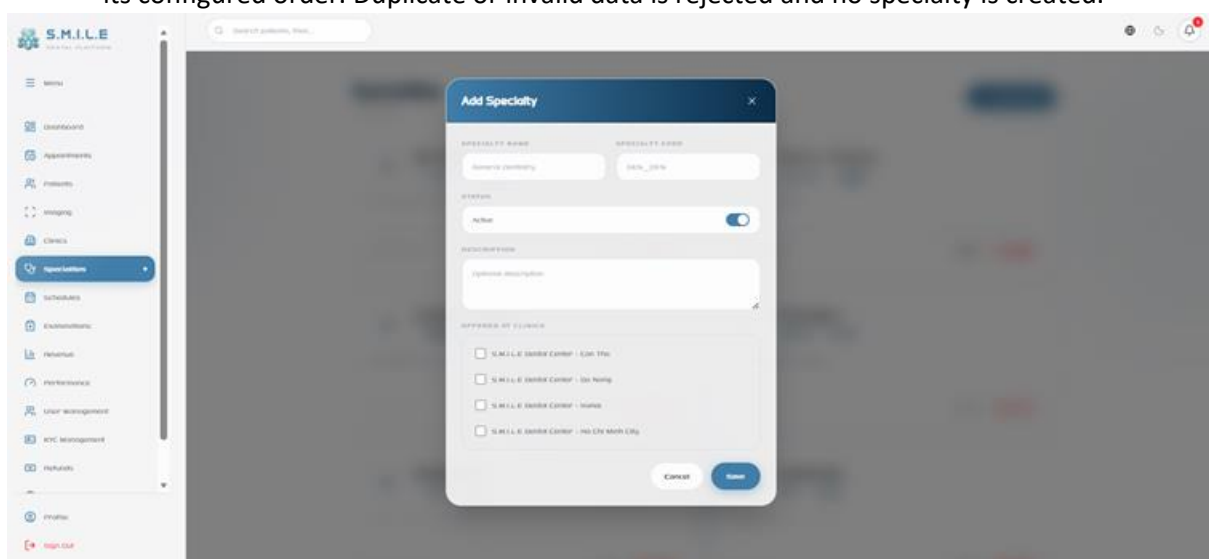
a. View Specialty

- **Function trigger:** An admin, doctor, nurse, receptionist, patient, or guest opens the specialty catalogue.
- **Function description:** The function displays the dental specialties offered by the service catalogue.
- **Screen layout:** The Specialty screen lists specialty name, unique code, description, icon, active state, and display order.
- **Function Details:** The system returns the specialty catalogue for the actor, including public guest access. Available specialties appear in configured display order. If no specialties match, the screen shows an empty result, while a retrieval failure shows an error message.



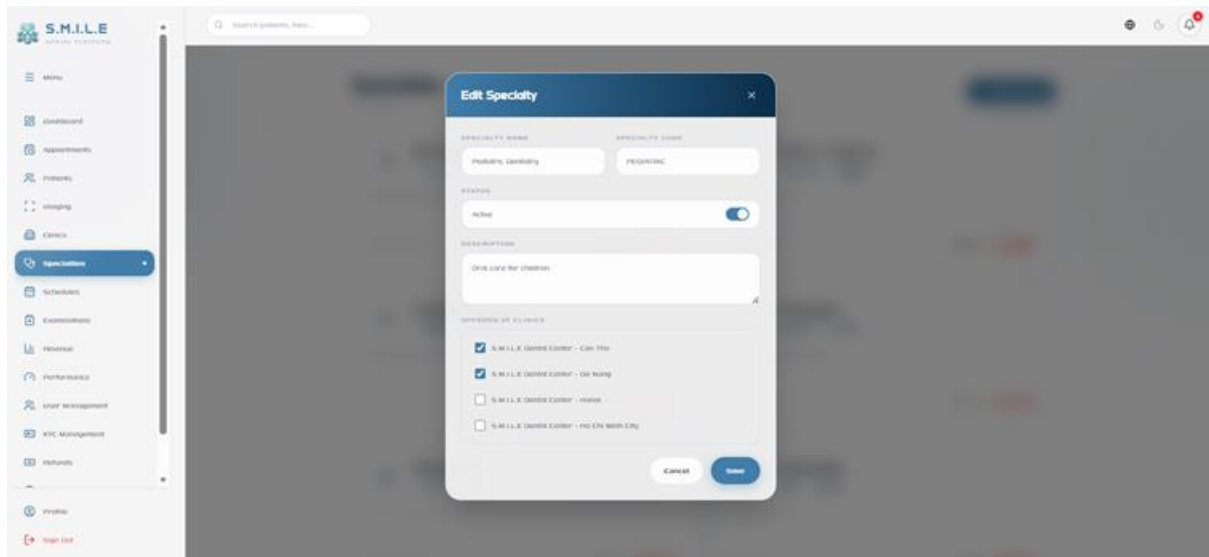
b. Add Specialty

- **Function trigger:** An admin selects Add Specialty from the Specialty screen.
- **Function description:** The function adds a new dental specialty to the service catalogue.
- **Screen layout:** The Add Specialty screen contains specialty name, unique code, description, icon URL, active state, display order, Save, and Cancel controls.
- **Function Details:** The system validates required metadata and confirms that the specialty name and code are unique before saving. A valid specialty appears on the Specialty screen in its configured order. Duplicate or invalid data is rejected and no specialty is created.



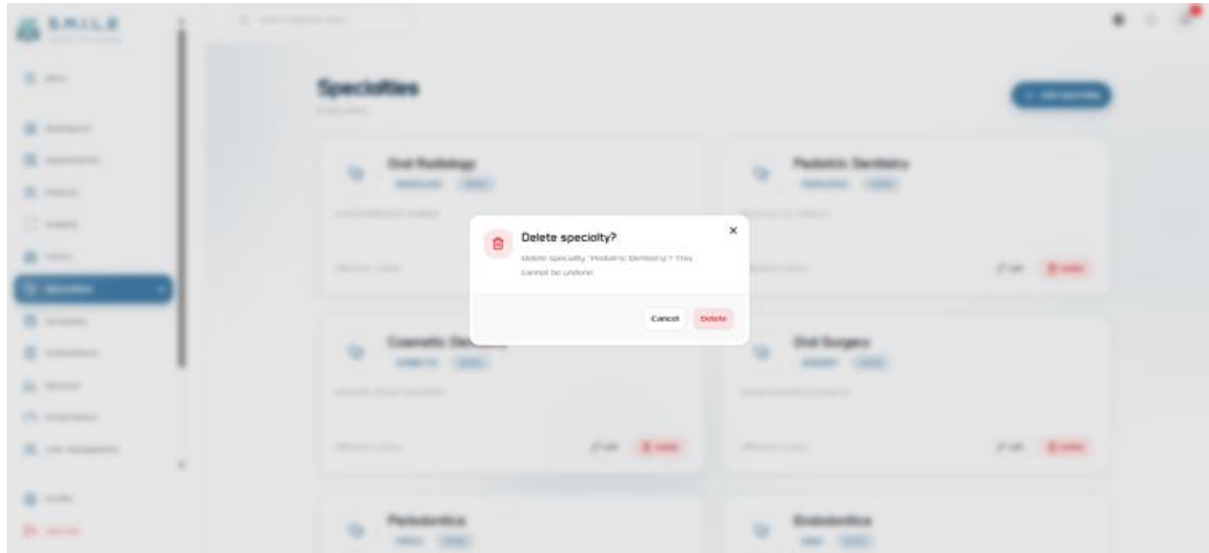
c. Update Specialty

- **Function trigger:** An admin selects Edit for an existing specialty.
- **Function description:** The function updates specialty definitions and catalogue metadata.
- **Screen layout:** The Update Specialty screen loads name, code, description, icon URL, active state, and display order with Save and Cancel controls.
- **Function Details:** The system validates the changed name, code, description, and metadata before saving the updated time. Valid changes appear on the Specialty screen. Duplicate values, invalid data, or a missing specialty leaves the existing record unchanged and shows an error.



d. Delete Specialty

- **Function trigger:** An admin selects Delete for a specialty and confirms the action.
- **Function description:** The function removes a specialty that is no longer offered.
- **Screen layout:** The Specialty screen provides a Delete control and a confirmation dialog showing the selected specialty name and code.
- **Function Details:** The system checks whether doctors or services still reference the specialty. If there are no references, deletion succeeds and the Specialty screen refreshes without the record. If references exist or the specialty is missing, deletion is blocked and an error message explains the result.



3.9 Clinical Examination

Sub-a. Create Session

- **Function trigger:** The doctor opens Clinical Examination for an appointment and starts a new examination session.
- **Function description:** This function creates the clinical context used to record findings and decisions for the patient.

- **Screen layout:** The Clinical Examination screen shows the appointment, patient, doctor, clinic, treatment room, session date, chief complaint, present illness, physical examination, vital signs, and session status. It provides a start action and shows the outcome.
- **Function Details:** The system links the session to the appointment, medical record, patient, doctor, clinic, and treatment room. Required patient identifiers and treatment context must be complete. On success, the session is saved with its start time and an in progress status, and the screen confirms that the examination session started. If required context is missing or storage fails, no session is confirmed and the screen shows an error.

a. Enter Symptoms

- **Function trigger:** The doctor selects Add Symptom during an examination session.
- **Function description:** This function records symptoms reported for the current patient and examination session.
- **Screen layout:** The Add Symptom screen provides symptom entry controls and a save action. It identifies the patient and current session and shows an error or success message.
- **Function Details:** The doctor enters the symptom information and submits it. The system verifies that the patient and examination session are valid, then stores the entry against both records. On success, the symptom appears in the Symptoms view and a success message is shown. If the context or data is invalid, the entry is not stored and the screen shows the problem.

b. View Symptoms

- **Function trigger:** A doctor or patient opens Symptoms for a patient or examination session.
- **Function description:** This function returns recorded symptoms that the actor is permitted to view.
- **Screen layout:** The Symptoms screen lists matching symptom entries and supports selection by session or patient. It shows an empty result when no matching entry exists and an access error when the actor is not permitted.
- **Function Details:** The system reads symptoms for the selected session or patient. A patient is limited to personal records, while the doctor uses patient data needed for treatment. Matching entries are displayed normally. Invalid context, denied access, or retrieval failure produces an error and no protected data is shown.

c. Edit Symptoms

- **Function trigger:** The doctor selects an existing symptom and chooses Edit.
- **Function description:** This function corrects or expands symptom information after updated observation.
- **Screen layout:** The Update Symptom screen displays the selected symptom in editable controls with save and cancel actions. Validation appears near the relevant control and the result appears as a confirmation message.
- **Function Details:** The system loads the symptom linked to the patient and session, then saves valid changes submitted by the doctor. On success, the Symptoms view shows the revised entry and a success message. A finalized examination session is locked, so an edit is rejected with a locked session message. Invalid data or a failed save also leaves the prior entry unchanged.

d. Delete Symptoms

- **Function trigger:** The doctor selects a symptom entered in error and chooses Delete.
- **Function description:** This function removes an incorrect or irrelevant symptom from the examination.

- **Screen layout:** The Delete Symptom screen identifies the selected entry and provides a delete action. The outcome is shown in the Symptoms view through a success or error message.
- **Function Details:** The system verifies the doctor and the linked patient and examination session before deleting the symptom. On success, the entry disappears from the Symptoms view and a success message is shown. If the session is finalized, access is invalid, or deletion fails, the symptom remains and the screen reports the error.

e. Create Treatment Plan

- **Function trigger:** The doctor selects Add Treatment Plan for a patient examination session or medical record.
- **Function description:** This function drafts a treatment plan that records planned care for the clinical case.
- **Screen layout:** The Add Treatment Plan screen provides controls for the plan name, objectives, planned duration, treatment steps, estimated cost, and status. It shows the linked patient, examination session or medical record, a save action, field errors, and the result.
- **Function Details:** The system links the plan to the examination session or medical record and records the creator and timestamps. A new plan must begin as draft. On success, it is stored and appears in Treatment Plan with a draft confirmation. Missing clinical context or invalid data prevents creation and produces an error.

f. View Treatment Plan

- **Function trigger:** A doctor or patient opens a treatment plan.
- **Function description:** This function presents the planned treatment steps, estimated cost, duration, objectives, and current status.
- **Screen layout:** The Treatment Plan screen shows the plan details and its link to the examination session or medical record. Available actions depend on the actor and current plan state.
- **Function Details:** The system retrieves the selected plan and related clinical context. A patient can view personal treatment plans, while a doctor can access plans needed for treatment. Valid data is displayed normally. A missing plan, denied access, or retrieval failure produces an error without exposing protected information.

g. Edit Treatment Plan

- **Function trigger:** The doctor opens an editable treatment plan and chooses Edit.

- **Function description:** This function revises treatment steps or plan details as patient care changes.
- **Screen layout:** The Update Treatment Plan screen displays editable plan details, the current status, save and cancel actions, and validation messages. A renewed consent confirmation appears when an estimated cost increase exceeds the defined threshold.
- **Function Details:** The doctor changes valid plan data and submits it. The system saves changes only while the plan is editable and updates the timestamp. If a cost increase exceeds the threshold, patient consent must be renewed before the change can proceed. On success, the revised plan appears in Treatment Plan. An invalid state, missing consent, invalid data, or failed save leaves the prior plan unchanged and shows an error.

h. Send Treatment Plan

- **Function trigger:** The doctor finishes a draft treatment plan and chooses Send.
- **Function description:** This function proposes the completed plan to the patient for acceptance or decline.
- **Screen layout:** There is no separate screen. The action is available in the Treatment Plan screen, and the proposed status and confirmation message appear there for the doctor and in the patient treatment plan view.
- **Function Details:** The system verifies that the plan is still in draft state, then changes it to proposed so the patient can accept or decline it. On success, both views show the updated status and a sent confirmation. If the plan is not a draft or the update fails, the status does not change and the Treatment Plan screen shows an error.

i. Delete Treatment Plan

- **Function trigger:** The doctor selects a treatment plan that is no longer needed and chooses Delete.
- **Function description:** This function removes an invalid or obsolete treatment plan.
- **Screen layout:** The Delete Treatment Plan screen identifies the selected plan and provides a delete action. The result appears in the Treatment Plan view as a success or error message.
- **Function Details:** The system verifies the doctor and the selected plan before deletion. On success, the plan is deleted and no longer appears in the Treatment Plan view. If access is invalid or storage fails, the plan remains available and the screen reports the error.

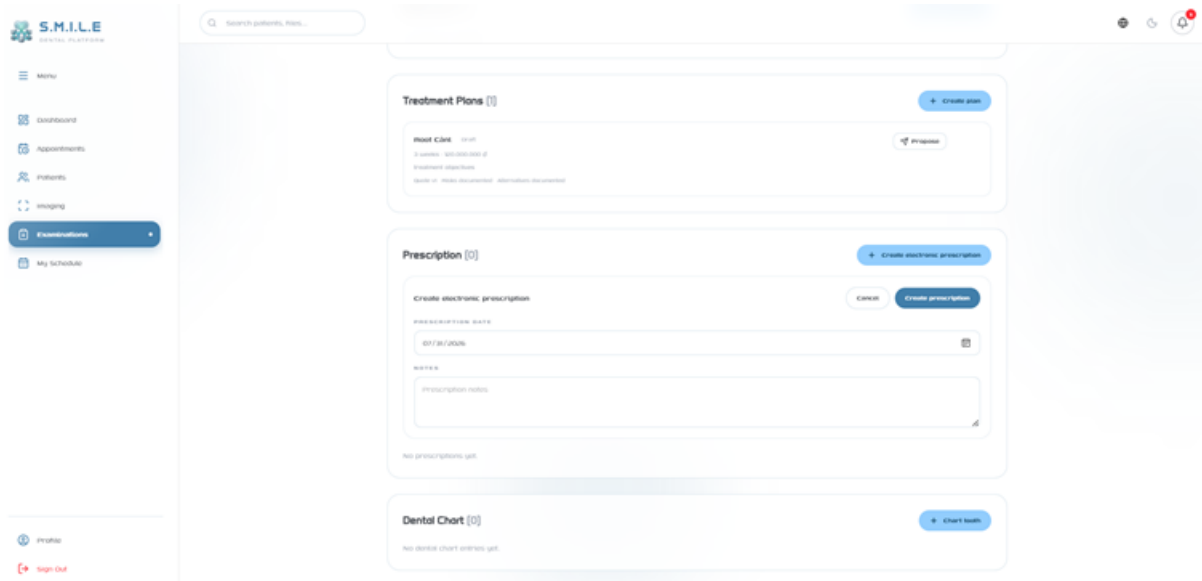
Sub-j. View Electronic Prescription

- **Function trigger:** The doctor opens an electronic prescription linked to an examination session or medical record.
- **Function description:** This function presents the prescription and its medication items for clinical review.
- **Screen layout:** There is no separate screen defined. The result appears within Clinical Examination and shows the patient, prescribing doctor, prescription date, status, notes, digital signature reference, medication, dosage, route, frequency, duration, quantity, and instructions.
- **Function Details:** The system retrieves the prescription linked to the session, medical record, and patient with its medication items. Valid prescription data is displayed with its active, dispensed, or cancelled status. If the prescription is missing, the actor lacks access, or retrieval fails, no prescription data is shown and Clinical Examination displays an error.

j. Create Electronic Prescription

- **Function trigger:** The doctor completes a prescription draft after examination and chooses Issue.

- **Function description:** This function issues an electronic prescription for the patient from a validated draft.
- **Screen layout:** The Issue Digital Prescription screen shows the linked patient and clinical context, prescription notes, and medication rows for name, code, dosage, route, frequency, duration, quantity, and instructions. It provides an issue action, explicit confirmation, field errors, and a result message.
- **Function Details:** The system validates every medication item and requires medication, dosage, frequency, and duration. For paediatric cases, it captures a consent snapshot. A high risk issuance requires explicit confirmation and an audit trail. On success, the prescription is linked to the session and record, its status becomes issued, and a confirmation appears. Missing dosing data, consent, clinical context, or storage failure prevents issuance and shows an error.



k. Order X-ray/CBCT

- **Function trigger:** The doctor selects Order X ray or CBCT from the clinical examination.
- **Function description:** This function creates a diagnostic imaging order for the patient.
- **Screen layout:** The imaging order screen shows the patient and clinical context with controls for imaging type, body part, tooth numbers, clinical indication, urgency, and planned date. It provides a submit action and displays validation and order status.
- **Function Details:** The system links the order to the examination session, medical record, patient, and ordering clinician. It records the order date, creates a tracking code, and starts the order in pending status. On success, the clinical examination shows the new order and a confirmation. Missing context, invalid order data, or storage failure prevents creation and displays an error.

l. Order Laboratory Test Service

- **Function trigger:** The doctor selects Order Laboratory Test Service during an examination.
- **Function description:** This function requests laboratory work to support diagnosis and later records returned results.
- **Screen layout:** The laboratory order screen shows the patient and clinical context with controls for test type, test name, clinical indication, urgency, and order date. It displays order status, sample collection date, result date, and the laboratory integration identifier when available.
- **Function Details:** The system links the order to the session, medical record, patient, and ordering clinician, then tracks ordered, scheduled, completed, or cancelled status. On success, the order appears in Clinical Examination and a confirmation is shown. Returned results are accepted and abnormal values are flagged. Missing context, invalid data, or failed processing leaves the order unconfirmed and displays an error.

m. Order Clinical Test

- **Function trigger:** The doctor selects Order Clinical Test during an examination session.

- **Function description:** This function records an in clinic clinical test order and tracks its progress.
- **Screen layout:** The clinical test order screen shows the patient and current examination session with controls for test details, clinical reason, urgency, and order date. It provides a submit action and displays the current order status and result messages.
- **Function Details:** The system validates the patient and examination session, records the ordering doctor, and tracks the order status. On success, the order appears in Clinical Examination and a confirmation is shown. Missing session context, invalid test data, or a processing failure prevents confirmation and the screen reports the error.

The screenshot displays the S.M.I.L.E. Dental Platform interface. On the left is a sidebar menu with options: Menu, Dashboard, Appointments, Patients, Imaging, **Examinations** (highlighted), and My Schedule. The main content area has a search bar at the top. Below it are three summary cards: 'Dental Chart [0]' with a 'Chart teeth' button, 'Diagnostic Orders — X-ray / CBCT [0]' with an 'Order X-ray / CBCT' button, and 'Clinical / Lab Orders [0]' with 'Order Lab Test' and 'Order Clinical Test' buttons. The 'Clinical / Lab Orders' card is expanded, showing a form titled 'Order Clinical Test'. The form includes fields for 'Order Type' (set to 'Clinical Test'), 'Test Type' (set to 'CBC / Imaging / Sensitivity'), 'Urgency' (set to 'routine'), 'Status' (set to 'ordered'), 'Patient Address' (set to '10, 12, 20'), and a 'Clinical Indication' text area (containing 'Reason for the test'). There are 'Cancel' and 'Create order' buttons at the top right of the form. At the bottom of the card, it says 'No clinical or lab orders get.'.

3.10 Dental Imaging

a. Upload Endodontic Image

- **Function trigger:** The doctor selects Upload Endodontic Image for a patient case.
- **Function description:** This function stores endodontic images and metadata for case documentation.
- **Screen layout:** The upload screen provides file selection and controls for patient, medical record, appointment, clinic, category, tooth numbers, view angle, description, tags, capture date, and uploader. It shows an image preview, validation, and upload result.
- **Function Details:** The system validates the clinical context and allowed file format and size, then stores the image URL, thumbnail, metadata, category, tooth numbers, and uploader. On success, the image appears in the dental image library and a success message is shown. An invalid file, missing context, or failed upload produces an error and no image record is confirmed.

The screenshot shows the 'Dental Imaging' upload interface. On the left is a sidebar with navigation links: Home, Dashboard, Appointments, Patients, Imaging (selected), Examinations, and My Schedule. The main area is titled 'Dental Imaging' with a subtitle 'Image library, examinations and categories'. Below this is the 'Upload Image' form. The form includes a file selection button, a 'Patient Name' field, a 'Record Type' dropdown menu, a 'Record Date' field, an 'Image Type' dropdown menu, a 'Tooth Numbers' dropdown menu, and a 'Description' text area. At the bottom right of the form are 'Cancel' and 'Upload' buttons.

b. Upload X-ray/CBCT Image

- **Function trigger:** The doctor selects Upload X ray or CBCT Image for a patient or treatment record.
- **Function description:** This function stores radiographic images for clinical storage and analysis.
- **Screen layout:** The upload screen provides file selection and controls for patient, record, appointment, clinic, image category, image type, tooth numbers, description, capture date, and uploader. It shows a preview, validation, and the upload outcome.
- **Function Details:** The system validates the allowed file format and size, stores the image and metadata, and links it to the patient and record. Supported imaging data includes X ray, CBCT, panoramic, and DICOM metadata with an optional PACS identifier. On success, the image appears in the library and a confirmation is shown. Invalid files or failed storage are rejected with an error.

c. View Dental Image Library

- **Function trigger:** The doctor opens Clinical Image or Dental Image Library.
- **Function description:** This function returns stored dental images for patient care and case review.
- **Screen layout:** The Clinical Image screen displays image thumbnails and metadata with filters for patient, medical record, category, and uploader. Selecting an entry shows the original image, tooth numbers, view angle, description, tags, dates, annotations, and archive status.
- **Function Details:** The system applies the selected filters and returns matching image records that the doctor may access. Matching images and annotations are displayed normally. If no image matches, the screen shows an empty result. Invalid access or retrieval failure produces an error and no protected image is shown.

d. Attach Image to Treatment Profile

- **Function trigger:** The doctor selects one or more images and chooses Attach to Treatment Profile.
- **Function description:** This function links relevant dental images to a treatment record so they appear with that record.

- **Screen layout:** The attachment interface shows selected images and a target treatment record selector with confirm and cancel actions. The result appears in the treatment record image area and through a confirmation or error message.
- **Function Details:** The system verifies the selected image records, patient context, and target treatment record, then creates the association. On success, the images appear with the treatment record and a confirmation is shown. If an image or record is invalid, access is denied, or saving fails, no association is created and the interface reports the error.

e. Edit Image by Treatment Profile

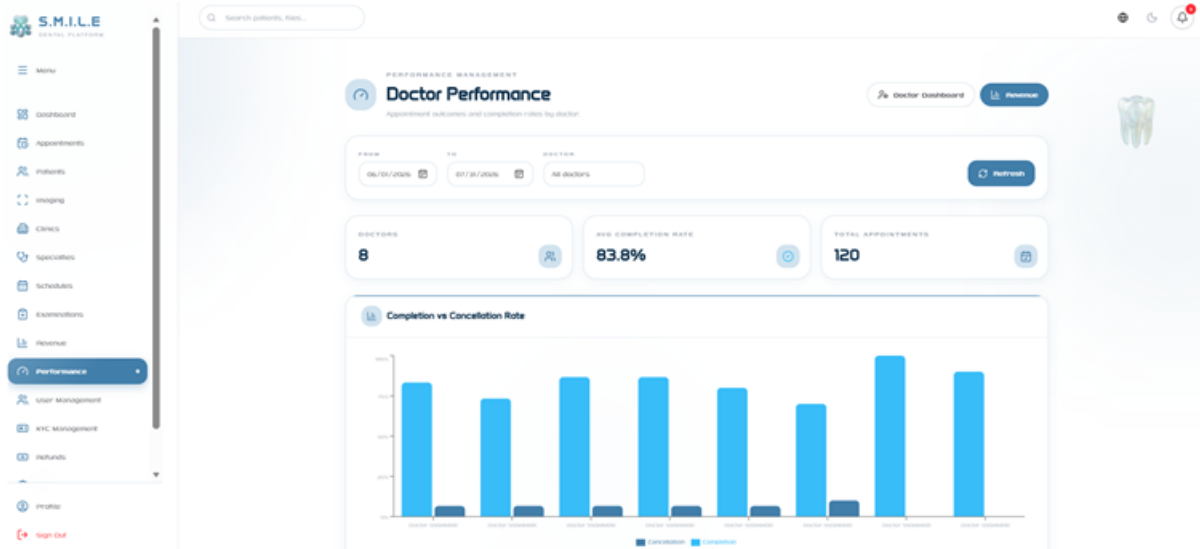
- **Function trigger:** The doctor or nurse opens an image from a treatment profile and chooses Edit.
- **Function description:** This function updates image metadata or annotations and can archive an image.
- **Screen layout:** The image update screen shows the image and editable category, tooth numbers, view angle, description, tags, metadata, and annotations. It provides save and archive actions with validation and result messages.
- **Function Details:** The doctor or nurse edits permitted metadata or annotation data and submits the change. The system validates the image and saves updated values and timestamps. On success, the treatment profile and image library show the revised image and a confirmation. Only a doctor or admin may archive or delete an image. Invalid data, denied access, or failed storage leaves the prior image unchanged and shows an error.

3.11 Performance Management

a. View Doctor Performance Report

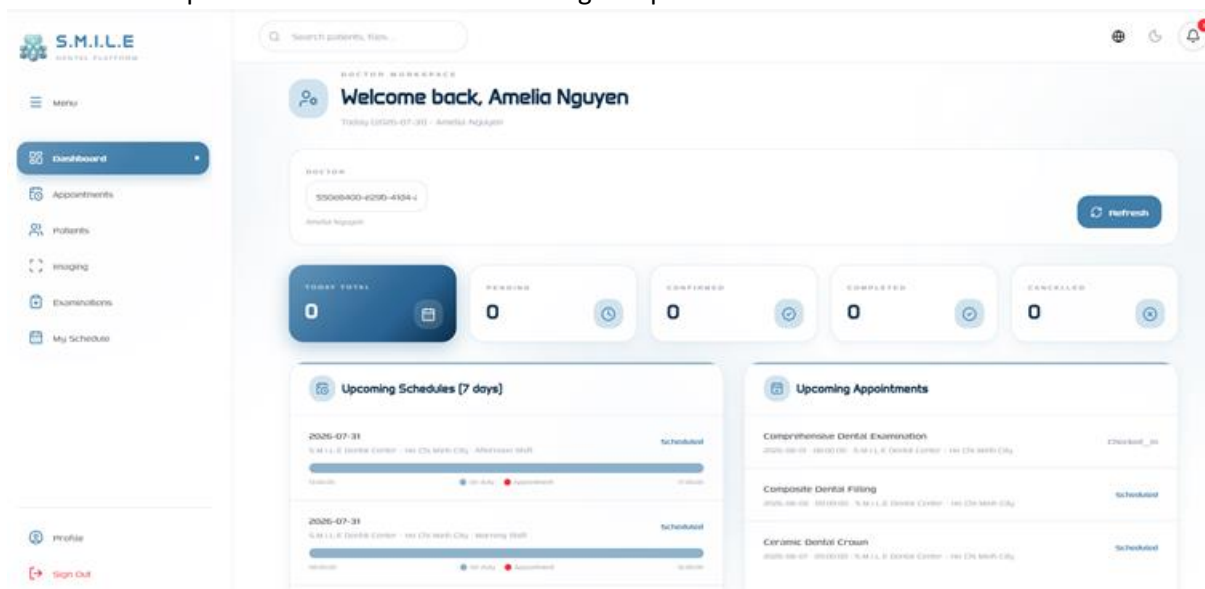
- **Function trigger:** The admin selects Doctor Performance Report and chooses a doctor and reporting period.
- **Function description:** This function summarizes doctor activity and productivity for management review.
- **Screen layout:** The Doctor Report screen provides doctor and period controls and displays aggregated appointment and treatment counts with report status or error messages.

- **Function Details:** The system validates the admin access and reporting period, then aggregates activity for the selected doctor. On success, the report shows the calculated counts for the period. An invalid doctor, invalid period, denied access, or aggregation failure produces an error and no report data is returned.



b. View Dashboard by Doctor

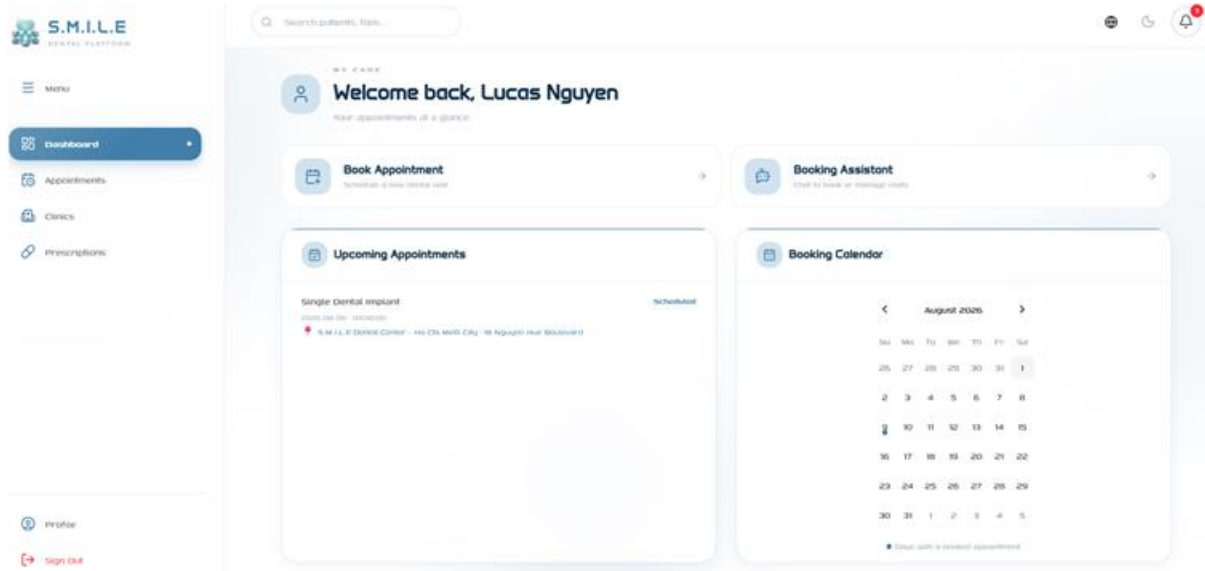
- **Function trigger:** The admin opens the doctor oriented Dashboard.
- **Function description:** This function presents aggregated doctor workload and schedule metrics.
- **Screen layout:** The Dashboard screen displays aggregated doctor workload and schedule indicators. It shows the returned metrics or an error when data cannot be returned.
- **Function Details:** The system validates admin access and aggregates doctor workload and schedule data. On success, the metrics appear on the Dashboard. Denied access or processing failure prevents the metrics from loading and produces an error.



c. View Dashboard by Customer

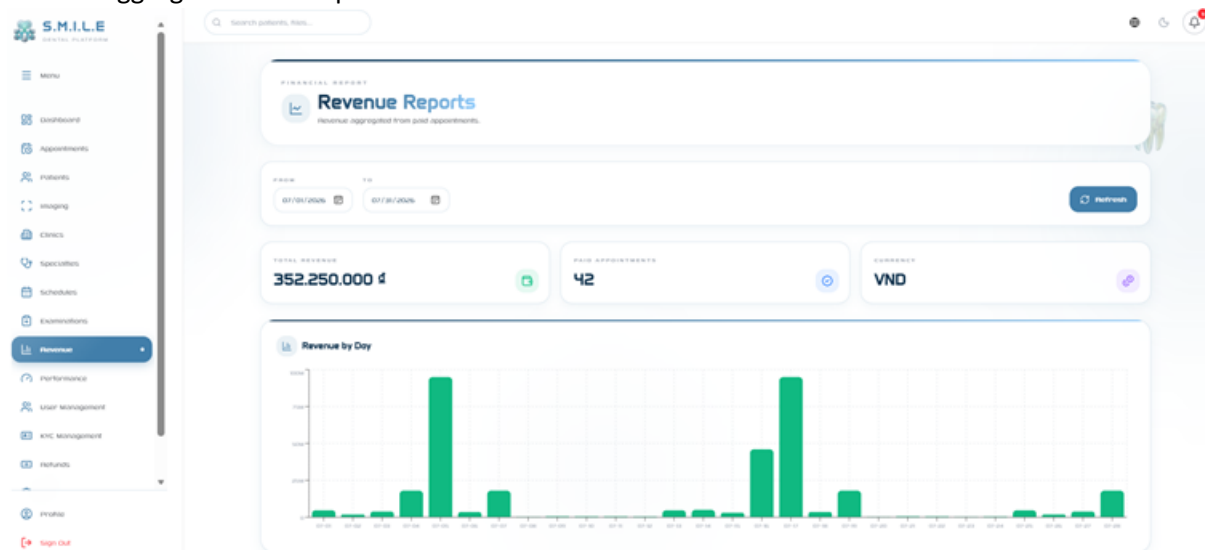
- **Function trigger:** The admin opens the patient oriented Dashboard.
- **Function description:** This function presents aggregated patient volume and visit metrics.

- **Screen layout:** The Dashboard screen displays aggregated patient volume and visit indicators. It shows the returned metrics or an error when data cannot be returned.
- **Function Details:** The system validates admin access and aggregates patient volume and visit data. On success, the metrics appear on the Dashboard. Denied access or processing failure prevents the metrics from loading and produces an error.



d. Revenue / Financial Report

- **Function trigger:** An authorized admin opens Financial Report and selects a reporting period.
- **Function description:** This function reports revenue and financial performance for management review.
- **Screen layout:** The Financial Report screen provides period controls and displays paid amounts, refunds, gross revenue, and net revenue. It shows report status, empty results, and access or processing errors.
- **Function Details:** Access is restricted to authorized management roles. The system validates the period, aggregates paid amounts, subtracts refunds, and returns gross and net revenue. On success, the report displays the calculated values. Invalid dates, denied access, or aggregation failure produces an error and no financial data is shown.



4. Non-Functional Requirements

4.1 External Interfaces

a. User Interfaces

- UI-1: All web and mobile web user interfaces shall conform to WCAG 2.1 Level AA accessibility requirements, including color contrast, focus visibility, form labels, and screen-reader compatibility.
- UI-2: Every major page shall provide contextual help (inline guidance, tooltip, or help link) that explains key actions and required fields.
- UI-3: All critical patient-care workflows (authentication, appointment booking, medical record viewing, treatment plan review, payment confirmation) shall be fully operable by keyboard only.
- UI-4: User interface error handling shall use standardized, non-technical language and must not expose sensitive internal details, stack traces, or protected health information (PHI).
- UI-5: The frontend shall support responsive layouts for common breakpoints (mobile, tablet, desktop) and maintain functional parity for safety-critical workflows.

b. Software Interfaces

- SI-1: Appointment Service and Schedule Service
 - o SI-1.1: The Appointment Service shall validate doctor/clinic slot availability through a programmatic REST interface to the Schedule Service before confirming a booking.

- SI-1.2: Availability queries shall use authenticated service-to-service calls with timeout controls and retry/backoff policies.
- SI-1.3: If scheduling data changes after booking (for example, doctor leave), the system shall emit a notification event and update affected appointment status according to business rules.
- SI-2: Appointment Service and VNPay
 - SI-2.1: The system shall communicate with VNPay through signed requests and verify callback signatures before updating payment status.
 - SI-2.2: Payment callbacks shall be idempotent to prevent duplicate settlement updates.
 - SI-2.3: The system shall support payment reversal/refund operations for eligible cancellation scenarios.
- SI-3: Account Service and Identity/Notification Providers
 - SI-3.1: The Account Service shall integrate with Google OAuth 2.0 for federated authentication.
 - SI-3.2: The Account Service shall integrate with SMTP email provider(s) for OTP, password reset, and transactional notifications.
- SI-4: Dental Image Service and AI Services
 - SI-5.1: Dental Image Service shall submit image payloads/metadata to AI detection endpoints for pathology support and receive structured inference results.
 - SI-5.2: AI integration interfaces shall support request validation, timeout handling, and confidence score reporting.

c. Hardware Interfaces

No hardware interfaces have been identified.

d. Communications Interfaces

- CI-1: The system shall send appointment, payment, and status notifications through email and/or SMS channels based on user communication preferences.
- CI-2: The system shall send problem alerts (payment failure, booking conflict, schedule change, delivery failure of notification) through configured communication channels.
- CI-3: All external and inter-service HTTP communication shall use TLS (minimum TLS 1.2 in production), and bearer token authentication shall be applied according to service role.
- CI-4: API payloads shall use JSON with UTF-8 encoding and standardized error/response envelopes to ensure cross-service compatibility.

4.2 Quality Attributes

The quality attribute requirements are established based on quality models and measurement guidance referenced in ISO/IEC 25022:2016¹, with measurable evaluation criteria aligned to ISO/IEC 25023:2016².

¹ [Systems and software engineering — Systems and software quality requirements and evaluation \(SQuaRE\) — Measurement of quality in use](#)

² [Systems and software engineering — Systems and software Quality Requirements and Evaluation \(SQuaRE\) — Measurement of system and software product quality](#)

a. Usability

- U-1 Training Time: New receptionists shall complete standard appointment-booking tasks after no more than 2 hours of guided training.
- U-2 Task Time: Doctors shall create or update a treatment plan in 3 minutes or less for common workflows under normal network conditions.
- U-3 Internationalization and Accessibility in Use: The platform shall provide bilingual support (Vietnamese and English) for patient-facing and staff-facing critical flows.
- U-4 Interface Consistency: Navigation consistency shall be maintained across major modules (Appointments, Patients, Examination, Imaging, Billing), with persistent global actions for profile, notifications, and help.

b. Reliability

- R-1 Availability: Core production services shall achieve at least 99.9% monthly uptime, excluding approved maintenance windows.
- R-2 Maturity (Failure Frequency): Critical incidents that fully block appointment booking or clinical record access shall not exceed 1 incident per 30 days per production environment.
- R-3 Recoverability (MTTR): Mean time to restore service for Severity-1 incidents shall be less than 60 minutes.
- R-4 Integrity Any mismatch shall trigger incident response within 5 minutes of detection.
- R-5 Defect Leakage: Critical defects in production shall be less than 0.2 per 1,000 function points per release, or an approved equivalent internal quality metric.

c. Performance

- P-1 Time Behavior (Read APIs): API read endpoints shall respond at p95 under 200 ms under nominal load.
- P-2 Time Behavior (Write APIs): API write endpoints shall respond at p95 under 500 ms under nominal load, excluding third-party callback delays.
- P-3 Payment Workflow Latency: Payment initialization and callback confirmation processing shall complete at p95 under 3 seconds, excluding user time on payment gateway pages.
- P-4 Capacity
 - o P-5.1 The platform shall support at least 1,000 concurrent active users.
 - o P-5.2 The platform shall support at least 100 booking transactions per minute in the baseline production sizing profile.

d. Dependability

d1. Security

- S-1 Access Control: The system shall enforce role-based access control across all protected APIs.
- S-2 Authentication and Session Security: Authentication tokens shall be cryptographically signed JWTs with expiration, refresh control, and revocation or lockout support.
- S-3 Data Confidentiality
 - o S-3.1 Sensitive medical data at rest shall be encrypted using industry-accepted cryptography (for example, AES-256 class).
 - o S-3.2 All data in transit shall be protected by TLS (minimum TLS 1.2 in production).

- S-4 Auditability: All access to medical records shall be audit-logged with actor, action, timestamp, and outcome.
- S-5 Compliance Alignment: The solution shall align with HIPAA-ready and GDPR-supporting controls as defined by project policy and deployment configuration.
- S-6 Secure Development: The system shall enforce secure coding, dependency scanning, and vulnerability management practices aligned with OWASP ASVS and OWASP Top 10 guidance.

d2. Safety

- SAF-1 Clinical Data Completeness: The software shall prevent unsafe clinical actions caused by incomplete required information (for example, missing patient identifiers or invalid treatment context).
- SAF-2 High-Risk Action Control: High-risk operations (record finalization, prescription issuance, consent revocation) shall require explicit user confirmation and complete audit trails.
- SAF-3 Fail-Safe Behavior: When downstream services are unavailable, the system shall not confirm clinical operations that did not persist successfully.
- SAF-4 Degradation Guidance: In case of system degradation, the platform shall provide clear operator guidance, visible status, and rollback or retry paths.

e. Supportability

- *SUP-1 Observability: Services shall expose health and metrics endpoints for operational monitoring.*
- *SUP-2 Diagnostic Logging: Logs shall be structured and correlated by request or trace identifiers to support distributed troubleshooting.*
- *SUP-3 API Maintainability: API contracts shall be documented and versioned; breaking changes shall require compatibility review.*
- *SUP-4 Delivery Quality Gates: CI/CD pipelines shall include automated tests and quality gates before deployment.*

f. Design Constraints

- DC-1 Architecture and Technology: Backend services shall remain microservice-based and containerized, using TypeScript/NestJS for core services and Go/Python for specialized services.
- DC-2 Data Boundary Constraint: Data persistence shall follow database-per-service boundaries.
- DC-3 Payment Provider Constraint: The payment integration for Vietnamese market operations shall use VNPay unless an approved architecture decision supersedes it.

g. Support Documents

- DOC-1 Central Documentation: The project shall maintain centralized system and API documentation for implementation and operations.
- DOC-2 Operational Documentation: User-facing help and operator runbooks shall be maintained for onboarding and incident handling.
- DOC-3 Traceability: Non-functional and compliance requirements shall be traceable to implementation and evidence documents.

h. Purchased Components

- PC-1 Third-Party Components: The system uses third-party and managed components including VNPay, Google OAuth, Gmail SMTP, and cloud object storage integrations.
- PC-2 Legal and Privacy Terms: Usage of purchased components shall comply with provider licensing terms, data-processing terms, and jurisdictional privacy constraints.
- PC-3 Upgrade Control: Component upgrades shall follow compatibility and regression testing before production rollout.

5. Requirements Appendix

- OR-1 Legal and Regulatory
 - o OR-1.1 Deployments handling PHI/PII shall implement controls consistent with applicable local regulations and healthcare privacy frameworks.
 - o OR-1.2 Consent lifecycle, audit logging, and retention/deletion policies shall be enforceable and reviewable.
- OR-2 Installation and Operations
 - o OR-2.1 The platform shall support reproducible deployment via Docker Compose for development and controlled deployment workflows for production.
 - o OR-2.2 Environment-specific configuration and secrets shall be managed outside source code.
- OR-3 Logging, Monitoring, and Audit
 - o OR-3.1 Security, access, and clinical data events shall be logged with immutable or tamper-evident strategy where applicable.
 - o OR-3.2 Monitoring shall include service health, latency, error rates, queue status, and storage capacity.
 - o OR-3.3 Audit evidence shall be exportable for compliance and post-incident review.

5.1 Business Rules

ID	Rule Definition
BR-01	Users must register a valid account and log in to access system functions.
BR-02	Passwords must be stored as secure hashes and never in plain text.
BR-03	Each account must use unique email, phone number, and username.
BR-04	Account roles are controlled by authorized admins only; users cannot self-change roles.
BR-05	Expert/doctor account creation requires valid professional certification.
BR-06	Users can only view or update data they own, unless their role grants broader access.
BR-07	Finalized medical records are immutable; only draft records can be edited.
BR-08	Schedule and shift assignments must not overlap for the same doctor.
BR-09	Appointments must be booked in valid time slots and follow clinic availability rules.
BR-10	Payment callbacks must pass signature verification before updating payment status.
BR-11	Sensitive actions must be recorded in audit logs for traceability.
BR-12	Only content that passes AI moderation can be published or displayed.
BR-13	OTP verification codes are time-limited and can be used only once.
BR-14	Accounts are temporarily locked after repeated failed login attempts.
BR-15	A doctor can be assigned only to active clinics and active specialties.
BR-16	A treatment room cannot be deleted while active schedules or appointments depend on it.
BR-17	Personal schedule changes require approval when they impact existing bookings.
BR-18	Appointment updates are allowed only for appointments in modifiable statuses.

BR-19	Appointment cancellations must follow cutoff-time and fee policies.
BR-20	Payment transaction IDs must be unique across the system.
BR-21	Refunds are allowed only for eligible paid appointments under policy conditions.
BR-22	Treatment plans requiring cost increase above threshold require renewed patient consent.
BR-23	Electronic prescriptions must include medication, dosage, frequency, and duration.
BR-24	Medical and imaging files must meet allowed format and size constraints.
BR-25	Access to financial reports is restricted to authorized management roles.
BR-26	A nurse may record pre-examination data only into a draft examination session.
BR-27	A nurse has read-only access to the patient directory and medical history.
BR-28	A nurse may upload and update clinical/dental images, but archiving or deleting an image is restricted to Doctor/Admin.
BR-29	A nurse must never issue prescriptions, approve treatment plans, or finalize/amend medical records.

5.1 Appendix1 - Messages List

#	Message code	Message Type	Context	Content
A. Generic — validation and common states				
1	MSG01	In line	There is not any search result	No search result.
2	MSG02	In red, under the text box	Input-required fields are empty	The {field} field is required.
3	MSG03	In red, under the text box	Input value length > max length	Exceed max length of {max_length}.
4	MSG04	In red, under the text box	Email format is invalid	Please enter a valid email address.
5	MSG05	In red, under the text box	Phone number format is invalid	Please enter a valid phone number.
6	MSG06	In line	A list has no records yet	No data to display.
7	MSG07	Toast message	Any unexpected server failure	Something went wrong. Please try again later.
8	MSG08	Toast message	Network request fails or times out	Cannot connect to the server. Please check your connection.
9	MSG09	Toast message	User lacks permission for the action	You do not have permission to perform this action.
10	MSG10	Confirmation dialog	Any delete action	Are you sure you want to delete {item_name}? This action cannot be undone.

B. Authentication and account				
11	MSG11	In line	Username or password is not correct when clicking sign-in	Incorrect username or password. Please check again.
12	MSG12	Toast message	Registration completed successfully	Registration successful. Please verify your email.
13	MSG13	In red, under the text box	Email or phone already registered	This {field} is already in use.
14	MSG14	In red, under the text box	Password does not meet policy	Password must be at least 8 characters and include letters and numbers.
15	MSG15	In red, under the text box	Confirm password does not match	Passwords do not match.
16	MSG16	Toast message	Google sign-in fails	Google login failed. Please try again.
17	MSG17	Toast message	OTP code sent to the user	A verification code has been sent to {email_address}.
18	MSG18	In red, under the text box	OTP code is wrong or expired	The verification code is invalid or has expired.
19	MSG19	Toast message	Password reset completed	Your password has been reset successfully.
20	MSG20	Toast message	Email verified	Email verified.
21	MSG21	In line	Account locked after repeated failed logins (BR-14)	Your account has been temporarily locked. Please try again later.
22	MSG22	Toast message	Session expired, user redirected to login	Your session has expired. Please sign in again.
C. User and role management				
23	MSG23	Toast message	Profile updated successfully	Profile updated successfully.
24	MSG24	Toast message	Admin locks a user account	Account locked.
25	MSG25	Toast message	Admin unlocks a user account	Account unlocked.
26	MSG26	Toast message	Role created, updated or deleted	Role {action} successfully.
27	MSG27	Toast message	Role assigned to a user	Role assigned successfully.

28	MSG28	Toast message	KYC documents submitted for review	Your identity documents have been submitted for review.
29	MSG29	In line	A KYC request is already pending review	A KYC request is already pending review.
30	MSG30	In red, under the text box	KYC consent checkbox not accepted	You must accept the {consent_type} to continue.
31	MSG31	Toast message	KYC file exceeds size or wrong format	{document_kind} must be a jpg, png or webp image under {max_size}.
32	MSG32	Toast message	KYC approved by admin	Identity verification approved.
33	MSG33	In line	KYC rejected by admin	Identity verification was rejected. Reason: {rejection_reason}
D. Clinic and treatment room				
34	MSG34	Toast message	Clinic created, updated or deleted	Clinic {action} successfully.
35	MSG35	Toast message	Treatment room created, updated or deleted	Treatment room {action} successfully.
36	MSG36	In red, under the text box	Room code duplicated within the clinic	This room code already exists in the clinic.
37	MSG37	Toast message	Deleting a room still referenced by schedules or appointments (BR-16)	This room cannot be deleted because it has active schedules or appointments.
38	MSG38	Toast message	Specialty created, updated or deleted	Specialty {action} successfully.
E. Schedule and shift				
39	MSG39	Toast message	Work schedule created or updated	Schedule saved successfully.
40	MSG40	In line	Overlapping shift for the same doctor (BR-08)	This doctor already has a shift assigned in that time slot.
41	MSG41	In red, under the text box	Schedule created without a room	A treatment room is required for this schedule.
42	MSG42	In red, under the text box	Leave end date is before start date	End date must be after start date.

43	MSG43	Toast message	Leave request submitted	Leave request submitted and awaiting approval.
44	MSG44	Toast message	Shift change or transfer notified to affected staff	{staff_name} has been notified of the schedule change.
F. Patient and medical record				
45	MSG45	Toast message	Patient profile added or updated	Patient profile saved successfully.
46	MSG46	In red, under the text box	Patient code already exists	This patient code is already in use.
47	MSG47	Toast message	Medical record saved as draft	Medical records must start as draft.
48	MSG48	Toast message	Medical record finalized	Medical record finalized successfully.
49	MSG49	In line	Editing a finalized medical record (BR-07)	Medical record is already finalized. Create an amendment instead.
50	MSG50	Toast message	Amendment added to a finalized record	Amendment added.
51	MSG51	Toast message	Medical record exported	Medical record exported successfully.
G. Appointment and payment				
52	MSG52	Toast message	Appointment booked	Appointment booked.
53	MSG53	Toast message	Appointment updated	Appointment updated.
54	MSG54	Toast message	Appointment confirmed	Appointment confirmed.
55	MSG55	Toast message	Appointment cancelled	Appointment cancelled.
56	MSG56	Confirmation dialog	Cancelling within the cutoff window (BR-19)	Cancelling now may incur a fee of {cancellation_fee}. Do you want to continue?
57	MSG57	In line	Selected slot no longer available	This time slot is no longer available. Please choose another.
58	MSG58	In line	Service requires a room type the schedule does not provide	The selected service is not available in this room.
59	MSG59	In line	Appointment already in a terminal status	This appointment is already {status} and can no longer be changed.

60	MSG60	Toast message	Confirmation email or SMS sent	Confirmation sent.
61	MSG61	Toast message	Payment initiated, redirecting to gateway	Redirecting to {payment_provider}...
62	MSG62	Toast message	Payment gateway returns no URL	No payment URL returned. Please try again.
63	MSG63	Toast message	Payment completed successfully	Payment completed successfully.
64	MSG64	In line	Payment failed or was cancelled at the gateway	Payment was not completed. Please try again.
65	MSG65	Toast message	Refund requested	Refund request submitted for review.
H. Clinical examination and prescription				
66	MSG66	Toast message	Examination session created	Examination session started.
67	MSG67	Toast message	Symptom or diagnosis added, updated or deleted	{item_type} {action} successfully.
68	MSG68	Toast message	Clinical notes saved	Clinical notes saved.
69	MSG69	In line	Editing a finalized examination session	Finalized examination sessions are locked.
70	MSG70	Toast message	Treatment plan saved as draft	New treatment plans must start as draft.
71	MSG71	Toast message	Treatment plan sent to the patient	Treatment plan sent to the patient for approval.
72	MSG72	In line	Proposing a plan that is not a draft	Only draft treatment plans can be proposed.
73	MSG73	Confirmation dialog	Cost increase above threshold requires renewed consent (BR-22)	The estimated cost has increased to {estimated_cost}. Patient consent must be renewed.
74	MSG74	Toast message	Treatment plan approved by the patient	Treatment plan approved.
75	MSG75	Toast message	Prescription issued	Prescription issued successfully.
76	MSG76	In red, under the text box	Prescription item missing dosage, frequency or duration (BR-23)	Dosage, frequency and duration are required for each medication.

77	MSG77	Toast message	Clinical or diagnostic order created	Clinical order created.
I. Dental imaging				
78	MSG78	Toast message	Image uploaded successfully	Image uploaded successfully.
79	MSG79	Toast message	Batch upload completed	Batch upload successful.
80	MSG80	In red, under the text box	File exceeds allowed size or format (BR-24)	{document_kind} exceeds the maximum file size of {max_size}.
81	MSG81	Toast message	Image annotation added or deleted	Annotation {action}.
82	MSG82	Toast message	Image attached to a treatment record	Image attached to the treatment record.
83	MSG83	Toast message	Image archived	Image archived.
84	MSG84	Toast message	AI analysis request submitted	AI analysis request sent.